Quantifying the Welfare Effects of Gentrification on Incumbent Low-Income Renters*

Robert French, Loyola Marymount University
Ashvin Gandhi, UCLA & NBER
Valentine Gilbert, University of Houston

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This paper investigates how gentrification affects the welfare of low-income incumbent renters. To do so, we construct an annual panel from US Census and IRS data tracking the earnings, workplaces, and residential addresses of over 3 million low-income urban renter households. We use this panel to conduct a comprehensive descriptive analysis and to estimate a dynamic structural model of residential and workplace choice. We find that, because low-income renters are highly mobile, the welfare effects of gentrification differ only modestly among renters initially living in different census tracts within the same metropolitan area. Our findings shift attention beyond changes in incumbent renters' initial neighborhoods to the metropolitan area choice sets that shape their welfare.

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1. Introduction

Since the 1990s, gentrification has transformed the socioeconomic composition of many American inner-city neighborhoods (Hwang and Lin 2016; Baum-Snow and Hartley 2020). Between 2000 and 2017, the share of residents with a college degree in census tracts near their metro areas' central business districts rose by an average of 19 percentage points from a baseline of 26 percent, more than twice the 7-percentage-point increase in suburban neighborhoods (Couture and Handbury 2023). How gentrification affects low-income renters has been the focus of considerable debate. In practice, policymakers and the public have tended to focus on the welfare of low-income incumbent renters—renters already living in these neighborhoods as they gentrified—motivating policies such as rent control (e.g., Zaveri 2026) that may come at the expense of prospective renters or renters living in other neighborhoods throughout the metro area (Glaeser and Luttmer 2003; Diamond, McQuade, and Qian 2019). Although protecting low-income incumbent renters is an important motivation for such policies, it remains unclear whether, or to what extent, they are harmed when their own neighborhood gentrifies.

The extent to which gentrification benefits or harms incumbent renters depends on several factors. First, it depends on how much renters value the associated changes in neighborhood characteristics.[1] Those who value these changes more than any accompanying rent increases benefit from gentrification, while those who do not are harmed. The magnitude of that harm, however, depends on the cost of moving and the availability of other desirable neighborhoods. When moving is costly—for example because renters are strongly attached to their current neighborhood—they may remain in their neighborhood even when the increase in rent exceeds the value they place on these neighborhood changes. When moving costs are low, however, renters can more readily reoptimize their location choice, so their welfare then depends less on what happens to the neighborhood where they initially live and more on the set of alternative neighborhoods available to them.

This study uses detailed administrative US Census, IRS, and proprietary real estate data to assess the effects of gentrification on incumbent low-income renters, measuring gentrification using the change in a census tract's college-graduate share between 2010 and 2019. We follow more than 3 million low-income urban renter households within large U.S. metro areas, linking person-level records from the Census Bureau's Master Address File (MAF), the American Community Survey (ACS), and the Longitudinal Employer-Household Dynamics (LEHD) database with real estate data from CoreLogic and other sources. We first document how residential mobility, neighborhood characteristics, and employment outcomes vary with gentrification. Then, to distinguish the welfare implications of these descriptive patterns, we estimate a dynamic structural model of residential and workplace choice and use it to construct neighborhood-level measures of gentrification's welfare effects on incumbent renters.

Our central finding is that, because low-income renters are highly mobile within cities, their welfare depends less on whether the neighborhood where they initially live gentrifies than on how affordability, amenities, and employment opportunities evolve throughout the metropolitan area. Renters value some

[1] Gentrification has been associated with changes in, for example, private consumption amenities (Glaeser, Luca, and Moszkowski 2023; Hoelzlein 2023; Almagro and Domínguez-Iino 2025), public amenities like public safety (Autor, Palmer, and Pathak 2017; Ellen, Horn, and Reed 2019), and employment opportunities (Lester and Hartley 2014; Meltzer and Ghorbani 2017). For a broader discussion of the welfare implications of gentrification for low-income households, see Vigdor (2002).

of the changes associated with gentrification, offsetting part of the cost of rising rents, while residential mobility gives them substantial scope to avoid further welfare losses when remaining in their initial neighborhood becomes less attractive. These findings suggest that policy attention should not focus narrowly on protecting incumbents in gentrifying neighborhoods, but more broadly consider the availability of accessible and desirable neighborhoods for low-income renters. Throughout the analysis, we report results separately for Black and non-Black households, motivated by longstanding concerns that gentrification has disproportionately harmed Black residents (Freeman 2006; Pattillo 2010).

We begin by showing that low-income renters are highly mobile, with over two thirds of households moving census tracts at least once between 2010 and 2019. Relative to this high baseline mobility rate, gentrification is associated with only modest differences in the likelihood of moving. Gentrification is instead most clearly associated with differences in the neighborhood characteristics experienced by renters who remain in their initial neighborhoods. In particular, stayers in gentrifying neighborhoods experience substantially larger increases in tract-level rents than stayers in non-gentrifying neighborhoods. Movers from gentrifying tracts, by contrast, generally reach neighborhoods that are observably similar to those reached by movers from non-gentrifying tracts, with only slightly higher rents and college-graduate shares. One exception is that Black renters leaving gentrifying tracts move to neighborhoods with lower white population shares than Black renters leaving non-gentrifying tracts, consistent with racial mobility networks shaping migration patterns in gentrifying neighborhoods (Ferreira, Kenney, and Smith 2024; Sharkey 2012). For both movers and stayers, gentrification is associated with relatively limited changes in a commute-time-discounted measure of nearby jobs, as well as in realized employment and earnings outcomes.

Our descriptive findings admit two broad interpretations of gentrification's welfare effects for low-income incumbent renters, depending on the cost of moving. If moving costs are low, high residential mobility may reflect renters' ability to readily reoptimize their location choice, while similar destination choices suggest that renters in gentrifying and non-gentrifying tracts can access similar neighborhoods. Incumbent renters' welfare is then likely to differ little across initial tracts within the same metro area. By contrast, if moving costs are sizeable, high mobility may instead reflect frequent renter-specific shocks arising from employment, financial, or family instability rather than deliberate responses to changing tract characteristics (Collinson et al. 2024; Desmond 2017; DeLuca, Wood, and Rosenblatt 2019; Skobba and Goetz 2013). High moving costs would also help explain why incumbent renters' residential mobility responds only weakly to gentrification. In this case, the welfare effects for renters who remain in their initial neighborhoods are ambiguous because they depend on how much renters value the changes in neighborhood characteristics that accompany gentrification, as emphasized in Vigdor (2010).

To quantify the welfare effects of gentrification on incumbent renters and distinguish between these competing interpretations, we estimate a dynamic discrete neighborhood choice model tailored to low-income renters (Bayer et al. 2016; Davis et al. 2021; Almagro and Domínguez-Iino 2025; Davis, Gregory, and Hartley 2026). Each period, forward-looking renters choose a neighborhood to maximize expected present-discounted utility, subject to rich moving costs. Flow utility depends on expected housing and non-housing consumption, neighborhood amenities, access to employment opportunities, and neighborhood-specific capital that accumulates with tenure, capturing the value of local familiarity and social ties (Kan 2007;

Koenen and Johnston 2024). To model access to employment opportunities, we embed a repeated static workplace choice problem within renters' dynamic neighborhood choice problem. This feature extends existing dynamic neighborhood choice frameworks, yielding time-varying, neighborhood-level measures of job market access that enter renters' flow utilities (Tsivanidis 2026). Importantly, accounting for job market access allows us to distinguish renters' valuation of neighborhood amenities from their valuation of access to employment opportunities.

Empirically, we separately identify renters' preferences over tract rents, job market access, amenities causally tied to neighborhoods' college-graduate shares, and potentially correlated unobserved amenities (Lee and Lin 2018; Ellen, Horn, and Reed 2019; Hoelzlein and Miller 2024). These preferences are especially important for understanding gentrification's welfare effects on renters who stay in their initial neighborhood. We estimate them by relating changes in neighborhood choices across cohorts of low-income renters to changes in neighborhood characteristics between 2010 and 2019. Because we allow changes in observed neighborhood characteristics to be correlated with changes in unobserved amenities, we instrument for changes in observed characteristics over this period.

We instrument for changes in job market access using variation in national employment growth across fine-grained industries interacted with the employment composition of each tract's initial job market access level. Following Severen (2023), Baum-Snow, Hartley, and Lee (2019), and Baum-Snow and Han (2024), this instrument isolates variation in job market access driven by broader industry trends rather than changes in local consumer demand (Couture and Handbury 2020). We instrument for changes in a neighborhood's college-graduate share by interacting tracts' proximity to already gentrified tracts with changes in metro-wide demand for college-educated workers. This instrument is motivated by evidence that gentrification tends to spread to nearby lower-income neighborhoods during periods of strong metro-wide housing demand (Guerrieri, Hartley, and Hurst 2013; Malone and Redfearn 2018). Finally, like Diamond (2016), we interact these predicted changes in college-graduate shares with local housing supply constraints to generate variation in rents (Baum-Snow and Han 2024).

We use the estimated model to quantify the welfare impacts of gentrification from the perspective of incumbent low-income renters in 2010. For each initial tract, we compare expected welfare under the observed evolution of tract characteristics with expected welfare under a counterfactual steady state in which those characteristics remain fixed at their 2010 levels. Within metro areas, differences in welfare effects across tracts are modest and statistically unrelated to measured gentrification between 2010 and 2019 for both Black and non-Black incumbent renters. These null effects reflect two broad forces. First, low-income renters value neighborhoods with higher college-graduate shares and greater job market access, although these valuations differ somewhat across racial groups. While gentrification is associated with only small changes in job market access, renters' positive valuation of higher college-graduate shares offsets some of the cost of rising rents, reducing the welfare loss from remaining in a gentrifying tract.

Second, even though the value from staying in a gentrifying tract typically declines meaningfully on net, the option value of moving tends to increase with gentrification. The option value of moving captures the value of an incumbent renter's feasible alternatives to remaining in their initial tract. It rises when other tracts in the renter's choice set become more attractive relative to the initial tract, net of moving costs. We find that declines in the payoff from staying in gentrifying tracts were generally offset by

increases in the option value of moving, in part because of only moderate estimated moving costs. Notably, the role of the option value of moving varies across metro areas. For example, it is particularly pronounced in Chicago and Dallas but notably weaker in New York and Los Angeles, consistent with renters in New York and Los Angeles facing fewer affordable alternative neighborhoods and greater moving frictions.

We conclude by extending the main welfare analysis in two ways. First, under a restrictive additional assumption, we compare welfare changes for incumbent renters across metro areas. We find that measured gentrification remains largely unrelated to welfare changes, while rent increases are associated with larger welfare losses across metro areas, consistent with broad-based increases in housing costs reducing renters' option value of moving. Second, we approximate expected welfare by combining our model estimates with the observed neighborhood choices households made from 2010 to 2019, rather than using model-implied mobility choices. We again find little relationship between gentrification and changes in incumbent renters' welfare across tracts within the same metro area. Overall, our analyses consistently point to limited welfare differences within metro areas between incumbent renters from gentrifying and non-gentrifying neighborhoods, while suggesting that broader changes in metro-area characteristics can have much larger welfare effects.

To our knowledge, we are the first to provide neighborhood-level estimates of gentrification's welfare effects on low-income incumbent renters across large U.S. metropolitan areas. We further decompose these effects into the value of remaining in one's initial neighborhood and the option value of moving elsewhere in the metro area. This decomposition helps bridge two strands of the gentrification literature. Static models of residential choice and research on changing urban affordability and amenities point to potentially large welfare effects for prospective renters (e.g., Newman and Wyly 2006; Diamond 2016; Su 2022; Hoelzlein 2023; Couture et al. 2024; Baum-Snow and Duranton 2025). In contrast, reduced-form studies generally find small effects of neighborhood gentrification on incumbent residents' mobility and other realized outcomes (e.g., Freeman and Braconi 2004; McKinnish, Walsh, and White 2010; Ellen and O'Regan 2011; Brummet and Reed 2021), which, taken at face value, may suggest modest welfare effects on incumbents. Together, our estimates of high baseline mobility and renters' positive valuation of the amenity changes that accompany gentrification help explain why gentrification induces little additional residential mobility among incumbent renters. Yet we also show that incumbent renters' welfare depends importantly on metro-area characteristics through the option value of moving. Thus, small mobility responses to neighborhood-level gentrification are entirely consistent with large welfare effects from broader changes in metro-area characteristics.

We make two primary contributions to the reduced-form literature finding that gentrification is associated with only small changes in residential mobility (Brummet and Reed 2021; Ding, Hwang, and Divringi 2016; Dragan, Ellen, and Glied 2020; Baum-Snow, Hartley, and Lee 2019; Fransham 2020; Beauregard 2024; Pennington 2021).[2] First, gentrification measures typically rely on noisy five-year ACS tract estimates, so classical measurement error may substantially attenuate estimated mobility responses. We instead link the MAF to the Census Bureau's demographic surveys to construct markedly more precise measures of tract-level sociodemographic change and document the resulting attenuation bias

[2] Baum-Snow, Hartley, and Lee (2019) also examine the effects of neighborhood change on children's long-run outcomes, while Beauregard (2024) use rich Canadian administrative panel data and find that the effects of gentrification on residential mobility are small overall but differ somewhat across cities.

in regressions using five-year ACS aggregates. Second, our large sample allows us to test whether small average mobility responses in the literature mask meaningful heterogeneity across neighborhood and renter characteristics. Even with our more precise measures, we find small responses across a wide range of settings, including tracts with inelastic housing supply or without rent control ordinances and among older renters and renters with long prior residential tenure.

Beyond residential mobility, we contribute to research examining how gentrification affects incumbent residents' economic outcomes (Meltzer and Ghorbani 2017; Brummet and Reed 2021). Using linked individual-level administrative data, we follow incumbent renters over time and measure changes in their employment, earnings, commuting, and job market access. Consistent with prior work, we find small average changes in realized employment and earnings outcomes, while also documenting meaningful racial differences and changes in renters' access to employment opportunities.

Our welfare analysis naturally relates to the dynamic neighborhood choice literature. Among others, Davis, Gregory, and Hartley (2026) estimate preferences over neighborhood racial composition, Diamond, McQuade, and Qian (2018) evaluate the insurance value of rent control, Bishop and Murphy (2019) estimate willingness to pay to avoid violent crime and air pollution, Almagro and Domínguez-Iino (2025) study preferences over horizontally differentiated consumption amenities and how these shape the welfare effects of tourism-induced gentrification in Amsterdam, and Qian and Tan (2021) use a myopic model of neighborhood choice to estimate the welfare effects of high-skill firm entry on incumbent households. We instead use a dynamic model to retrospectively evaluate how gentrification affected incumbent renters across large U.S. metro areas, separating the welfare effects of changes in renters' initial neighborhoods from changes in the option value of moving elsewhere in the metro area.

Finally, our findings connect to a growing literature showing that the trajectories of places and the people who pass through them can diverge sharply. Bilal and Rossi-Hansberg (2021) conceptualize location as an asset, emphasizing that households dynamically adjust their location choices in response to income shocks and differences in locations' costs and future returns. Garin et al. (2026) relatedly show that residents of poor tracts are highly mobile and tend to leave as their earnings rise, while the tracts they leave remain poor. Staiger, Palloni, and Voorheis (2026) show that the HOPE VI Revitalization program reduced poverty rates in targeted neighborhoods primarily by changing the composition of in-migrants rather than by displacing incumbent residents, many of whom would have left even absent redevelopment. In closely related work, Baum-Snow, Farhoodi, and Han (2025) study how Texas renters respond to rent growth by substituting across neighborhoods and cities. Together with this recent literature, our findings shift attention away from changes in incumbent renters' initial neighborhoods alone and toward the broader metropolitan choice sets that shape their welfare.

2. Data and Sample Construction

Data. We use newly available administrative U.S. Census data to track the earnings, workplaces, and residential addresses of more than 3 million low-income urban renter households across large U.S. metro areas—which we define using Core-Based Statistical Areas (CBSAs)—throughout 2010-2019. We link person-level records using Census-provided identifiers from the Census Bureau's Master Address File

(MAF), the American Community Survey (ACS), and the Longitudinal Employer-Household Dynamics (LEHD) database. These data provide annual address-level residential locations, annual earnings and workplace locations, as well as detailed sociodemographic characteristics for our samples of low-income renters. To help identify rental units, we link our person-level panel to property-level characteristics recorded in commercial real estate data provided by CoreLogic and Black Knight, which include information on homeownership status. Given our focus on low-income renter households, we additionally merge our panel to publicly available information on low-income housing and rental support, including details from hand-collected rent control ordinances covering all incorporated and unincorporated places in our analysis. Lastly, we use establishment-level business characteristics from the LEHD and the Longitudinal Business Database (LBD) to construct sample-specific and tract-level measures of job market access. We postpone a detailed discussion of the raw data to Appendix A.1.

Sample Construction. We use the LEHD to define annual cohorts of low-income households from 2010 to 2019. We define our samples primarily using the LEHD because it provides detailed earnings records for roughly 95 percent of U.S. workers, yielding much larger annual samples than we could construct using the ACS. We limit each annual cohort to household heads, whom we define as the highest earner in a given housing unit in that year. We then classify a household head as low-income if the household's combined earnings fall in the bottom tercile of household earnings among households in the same CBSA and decadal age band, after adjusting for household size and composition. We exclude households without any reported LEHD earnings because we cannot reliably distinguish nonworkers from workers not covered by the LEHD.

We classify workers as renters if we identify them as living in a rental unit in the analysis year. We identify rental units using information from the household-level ACS, the MAF, CoreLogic, and Black Knight. Our selection criteria yield a sample of roughly 3 million low-income renters per year across all tracts within the CBSAs in our sample, which we use for our structural analysis. For our descriptive analyses, we further limit the sample to low-income renters who, in 2010, lived in census tracts closer to their metropolitan division's central business district (CBD) than at least 50 percent of the division's population, in tracts within the bottom quartile of the CBSA's 2010 college-graduate share distribution, and in tracts outside the bottom 1 percent of the national tract population distribution.[3] We consider these tracts as "gentrifiable" as of 2010. Table 1 presents summary statistics for our sample and their 2010 census tracts, which we refer to as their origin tracts.

We focus on low-income renters because these households are frequently cited as especially susceptible to gentrification's potential harms (Newman and Wyly 2006).[4] Throughout the paper, we further split

[3]Our CBD definitions come from Fee and Hartley (2013) and are based on 2008 CBSA delineations. To maintain consistency with these CBD definitions, we use 2008 CBSA delineations throughout the analysis. While most CBSAs are associated with a single primary urban center, some large CBSAs are subdivided into metropolitan divisions, each of which may have its own CBD. Although we construct our low-income cutoffs from the earnings distribution of the entire CBSA, our urban-core cutoffs are specific to each urban center's population, including metropolitan divisions. We believe these choices best capture our target population of low-income urban residents.

[4]Low-income homeowners may also be vulnerable to gentrification's potential harms if rising home prices increase property tax burdens in gentrifying neighborhoods (Ding and Hwang 2020; Martin and Beck 2018). We believe, however, that studying homeowners is better suited to a framework that captures intertemporal consumption-savings decisions in response to home price appreciation in gentrifying neighborhoods.

the sample into Black and non-Black household heads given longstanding concerns that gentrification has disproportionately harmed Black residents (Freeman 2006; Pattillo 2010) and evidence that recent gentrification produced larger changes in initially majority Black neighborhoods than in initially majority Hispanic neighborhoods (Ferreira, Kenney, and Smith 2024). We detail our sample selection criteria, validate our samples against the ACS, and address potential data-quality concerns arising from the novelty of our linkages in Appendix A.2. We believe these data enable the most comprehensive analysis to date of residential mobility and employment outcomes among low-income renters in the United States.

TABLE 1. Sample Summary Statistics

A. Household Head Characteristics, 2010

	Low-Income Renters	
	Black	Non-Black
Adj. HH Income	25,050	26,590
	(18,350)	(17,920)
College Degree	0.0698	0.0681
	(0.255)	(0.252)
Age	41.14	41.78
	(10.50)	(10.59)
Parent	0.4406	0.4719
	(0.4965)	(0.4992)
Female	0.6300	0.5183
	(0.4828)	(0.4997)
White	—	0.8555
	(—)	(0.3516)
Immigrant	0.1335	0.4475
	(0.3401)	(0.4972)
Hispanic	0.09036	0.5948
	(0.2867)	(0.4909)
Tenure Length (years)	4.975	5.200
	(3.355)	(3.371)
Commute Distance (miles)	9.61	9.78
	(9.19)	(9.55)
Commute Time (minutes)	30.1	25.1
	(24.2)	(21.1)

B. Household Tract Characteristics, 2010

	Low-Income Renters	
	Black	Non-Black
CBD Distance (miles)	8.406	8.481
	(5.208)	(5.260)
CBD Distance (pop. density)	0.2010	0.2083
	(0.1381)	(0.1404)
Hedonically Adjusted Rents	502.5	554.2
	(141.7)	(147.1)
Gross Median Rents	762.3	812.8
	(205.8)	(213.0)
College Share	0.0954	0.0947
	(0.0328)	(0.0354)
White Share	0.2651	0.5520
	(0.2317)	(0.2327)
Black Share	0.5315	0.1439
	(0.3196)	(0.1869)
Hispanic Share	0.2631	0.5153
	(0.2808)	(0.3032)
Immigrant Share	0.2752	0.4368
	(0.2430)	(0.2456)
Total Adult Population	2,129	2,379
	(912.7)	(922.8)

Number Black Households: 351,000
Number Non-Black Households: 517,000

Notes: Summary statistics are for incumbent low-income urban renters living in tracts among the bottom quartile of their CBSA's college-graduate share distribution in 2010 (the "Baseline mobility sample" in Table A1). Panel A reports characteristics of household heads, while Panel B reports characteristics of households' 2010 census tracts. We report sample means, with standard deviations in parentheses. Dollar amounts are in 2010 USD. Parental status, commute time, and commute distance are computed only for the subsample that we can link to the ACS. CBD distance is measured in miles to the metropolitan division's CBD and, separately, by the population share residing closer to the metropolitan division's CBD than the tract in 2010. We detail hedonically adjusted rents in Appendix A.3 and adjusted household income in Footnote 59. Tract sociodemographic shares and total adult population are constructed using the de-biased tract aggregates detailed in Appendix A.3. All values throughout the paper are rounded according to Census guidelines, which depend on underlying sample sizes and variable type.

Tract Sociodemographic Characteristics. Our descriptive analyses rely on short-term variation in neighborhoods' sociodemographic characteristics. Researchers typically use pooled five-year ACS aggregates to define U.S. neighborhoods' sociodemographic characteristics and changes thereof (e.g., Ding, Hwang, and Divringi 2016; Baum-Snow and Hartley 2020; Brummet and Reed 2021).[5] These five-year ACS aggregates

[5]Researchers also rely on decadal changes in tract-level measures from Decennial Census Summary Files, but since 2010 these measures have come primarily from the short-form census, which includes only a limited set of sociodemographic

are noisily estimated, however, which can exacerbate classical measurement error in descriptive analyses. We instead combine potentially unbiased but imprecise five-year ACS tract-level estimates with biased but more precise tract-level estimates derived from a residential panel covering all persons in the MAF whom we can link to the ACS or the LEHD at some point between 2000 and 2021. For these individuals, we track their annual residential tracts and assign them their time-invariant sociodemographic characteristics recorded in the ACS or LEHD.[6] We then aggregate annual snapshots of the panel to construct measures of census tracts' sociodemographic composition. Finally, we use the ACS to correct for potential bias in these tract-level measures, given selection into the MAF and selective attrition from the residential panel (Sullivan and Genadek 2024). The estimates in Figure A3 imply that the elasticity of tract-level rent growth with respect to our gentrification measure is approximately three times larger with our more precise tract-level estimates than with the corresponding ACS-based estimates. Appendix A.3 details our approach to constructing and validating the tract-level estimates.

3. Descriptive Analysis

Our goal in this paper is to answer a distinctly retrospective question: how did low-income renters in neighborhoods that actually gentrified fare relative to low-income renters in comparable neighborhoods that did not? Research designs that isolate a specific source of variation in gentrification can provide credible estimates for the neighborhoods affected by that variation. Yet different sources of variation may induce different neighborhoods to gentrify, and the contexts of those neighborhoods may mediate gentrification's effects on incumbent renters.[7] Here, we instead carefully select our sample and set of controls so that the comparison between incumbent renters in gentrifying and non-gentrifying neighborhoods is informative beyond any single source of variation. Because our comparison draws on variation across a wide set of neighborhood environments, we also examine whether average patterns mask meaningful heterogeneity across renter or neighborhood characteristics. In this section, after defining our gentrification measure, we first examine baseline residential mobility and show that it is unrelated to subsequent gentrification. We then study the neighborhoods where incumbent renters lived in 2019 and the economic outcomes of those renters.

Defining Gentrification. Economists typically measure gentrification with a unidimensional measure that is often a function of a neighborhood's share of adult college-graduates (e.g., Baum-Snow, Hartley, and Lee 2019; Brummet and Reed 2021; Glaeser, Luca, and Moszkowski 2023). We follow this literature and measure gentrification as the log change between 2010 and 2019 in a tract's share of adult residents over

characteristics and excludes college attainment, the basis of most gentrification measures.

[6] We do not assign time-invariant characteristics that were imputed by the LEHD.

[7] For instance, Baum-Snow, Hartley, and Lee (2019) compare children of incumbent households in tracts differentially exposed to high-skill labor demand shocks; Qian and Tan (2021) compare incumbent households living closer to versus farther from high-skilled firm entries in the U.S. in 1990-2010; Timmins and Wang (2026) compare incumbent households across Los Angeles neighborhoods experiencing different reductions in local toxic pollution; Hong and Macek (2025) compare long-term incumbent residents located either side of the boundaries of two Chicago neighborhoods that imposed a demolition surcharge on teardowns to curb gentrification; Friday and Smith (2023) examine evictions on either side of school attendance boundaries before and after schools receive federal School Improvement Grants; and Agarwal et al. (2024) compare incumbent households in urban and suburban zip codes before and after staggered Uber and Lyft entry to those in MSAs without rideshare entry.

25 with at least a Bachelor's degree, using our bias-corrected tract estimates:

$$\text{Gent} = \ln\left(\frac{\text{Coll}_{n2019}}{\text{Pop}_{n2019}}\right) - \ln\left(\frac{\text{Coll}_{n2010}}{\text{Pop}_{n2010}}\right).$$

While our focus on changes in the share of college-graduate adults is also consistent with a prominent strand of quantitative sociology research on gentrification (e.g., Freeman 2005; Ding, Hwang, and Divringi 2016), other influential sociological studies emphasize gentrification's multifaceted nature, including racialized patterns of neighborhood change as well as reinvestment and renewal often reflected in changes to the housing stock and housing prices (e.g., Hwang and Sampson 2014; Zuk et al. 2018). To partially address the multifaceted nature of gentrification, we separately examine relationships between rent growth and incumbent renter outcomes in Appendix D.4 and include robustness checks using changes in the share of white college-graduate adults throughout the paper.[8]

Much of our analysis uses incumbent residents' migration choices as the dependent variable. To avoid a mechanical relationship between these migration choices and our gentrification measure, we isolate variation in gentrification driven by changes in the number of adult college graduates in a neighborhood. Specifically, throughout our descriptive analysis, we predict changes in our gentrification measure using tract-level changes in the number of residents over age 25 with at least a bachelor's degree, normalized by the tract's total adult population in 2010: $(\text{Coll}_{n2019} - \text{Coll}_{n2010})/\text{Pop}_{n2010}$.[9]

3.1. Residential Mobility

Figure 1 shows baseline mobility rates in the MAF.[10] The figure illustrates how highly mobile low-income renters are. Between 2010 and 2011, roughly 18.5% of low-income renter household heads moved census tracts, nearly twice the 10.3% rate for all household heads. These one-year move rates are very similar in the ACS. By 2019, 70% of non-Black and 73% of Black low-income renter household heads had moved census tracts in the MAF, compared with 52% of all household heads, implying high baseline mobility was not concentrated among a small subset of low-income renters.

Most low-income renters remained within their origin CBSA, however. Between 2010 and 2011, only 3.4% of non-Black and 2.2% of Black low-income renters left their origin CBSA, similar to the rate among all household heads. Their moves are also quite local, and significantly more so than the full population. The median distance of low-income renters' first within-CBSA move was about 4.7 miles, compared with 6.8 miles for all household heads. By 2019, low-income renters who had moved to a new tract within their CBSA lived a median of 5.2 miles from their 2010 origin tract. Although low-income renters' moves are often quite local, the median first-move distances suggest that many moves are not between nearby blocks or adjacent tracts. Instead, many movers likely end up in areas that would reasonably be viewed as

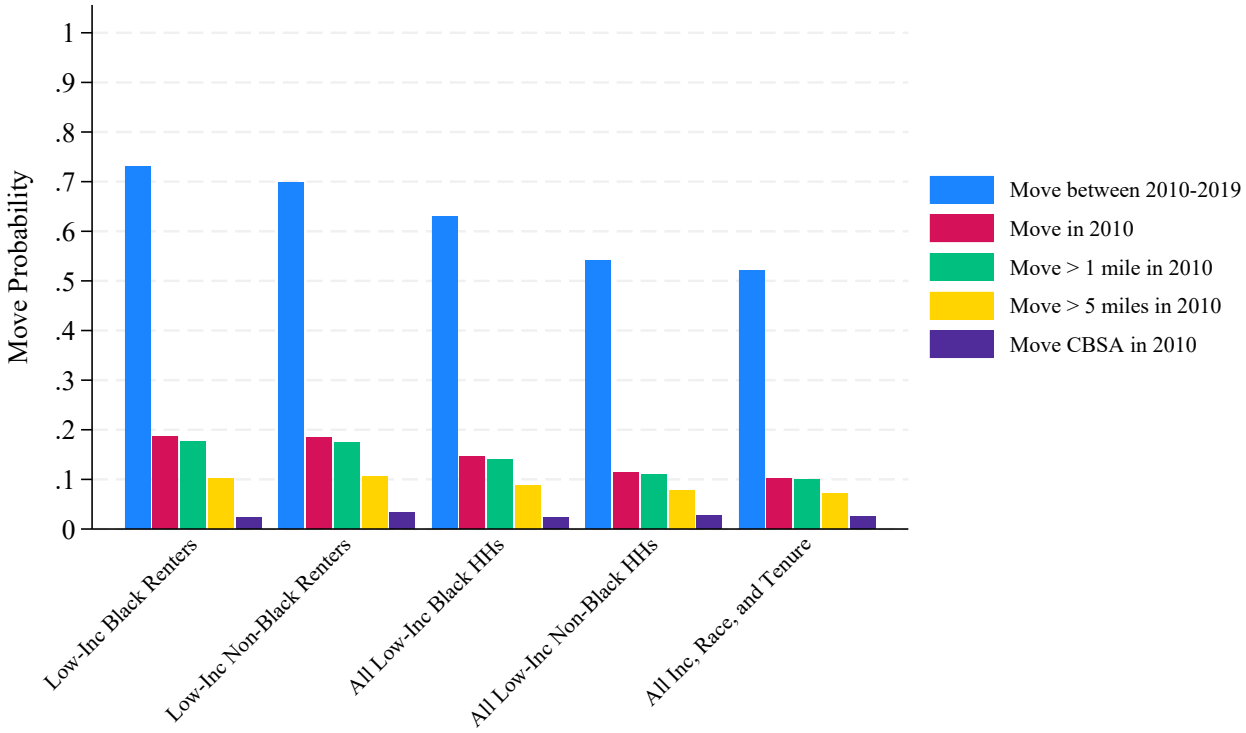
[8] While our examination of rental growth and incumbent renter outcomes addresses frequent comments on earlier drafts of the paper, we additionally refer readers to ongoing work in Baum-Snow, Farhoodi, and Han (2025) for a causal analysis of the impact of rent growth on incumbent renters.

[9] A small minority of incumbent renters in our descriptive analysis sample have college degrees. In Appendix D.2, we show that our primary results are quantitatively similar when excluding these household heads from the descriptive sample.

[10] Table A15 reports additional mobility statistics and sample sizes, and Figure A1 compares baseline move probabilities in the MAF with those in the ACS.

distinct neighborhoods.[11]

FIGURE 1. Baseline Residential Mobility



Notes: Figure 1 summarizes residential mobility for households observed in 2010. The leftmost bars report the share of household heads who lived in a different census tract in 2019 than in 2010; the remaining bars report shares who lived in a different census tract in 2011 than in 2010. "Low-Inc Black (Non-Black) Renters" are low-income urban renters living in tracts in the bottom quartile of their CBSA's college-graduate share distribution in 2010. "All Low-Inc Black (Non-Black) HHs" are all low-income household heads in 2010, regardless of homeownership status. "All Inc, Race, and Tenure" includes all 2010 household heads, irrespective of income, race, or homeownership status. These samples correspond to those defined after applying restrictions (h.i), (f), and (d) in Table A1, respectively. We compute residential mobility statistics only when the household head's census tract is observed in both years relevant to a given bar (2010 and 2019 for the leftmost bars; 2010 and 2011 for the others). We report sample sizes and additional baseline mobility statistics in Table A15.

3.2. Gentrification and Residential Mobility

The high mobility rates reported in Section 3.1 are largely insensitive to whether an incumbent renter's origin neighborhood gentrified. To show this, we regress several measures of incumbent renters' migration choices on our gentrification measure:

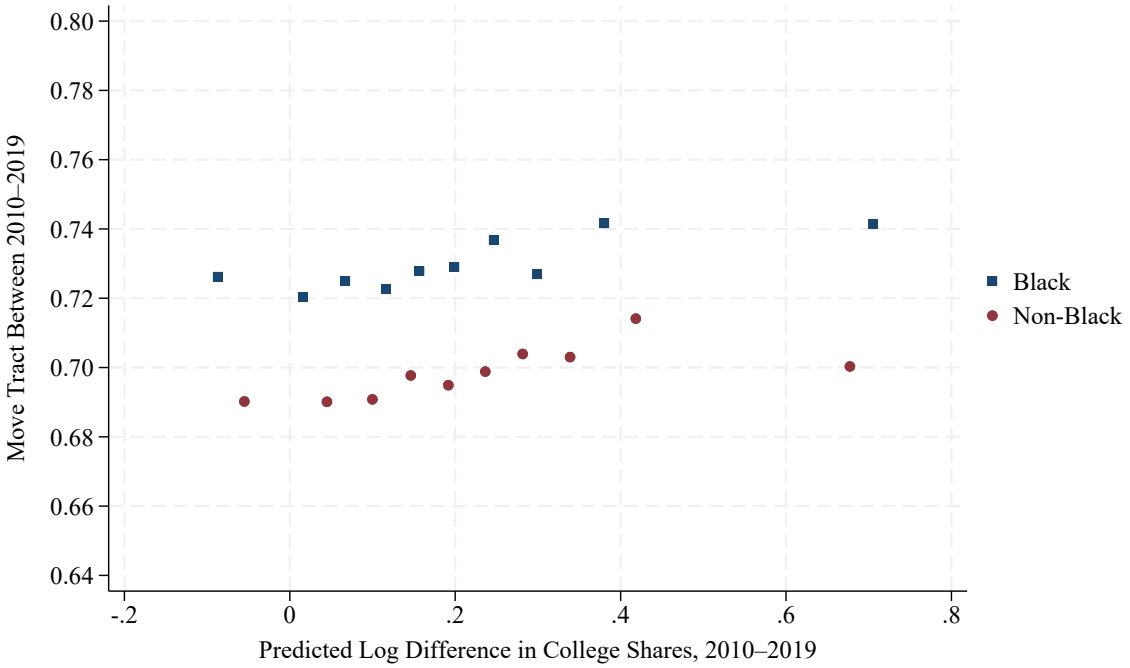
$$Y_i = \beta_0 + \beta_1 \text{Gent}_{n(i)} + \boldsymbol{X}_i'\delta + \boldsymbol{W}_{n(i)}'\lambda + \alpha_{\text{CBSA}} + \varepsilon_i, \tag{1}$$

[11] Mast and Barca (2026) is useful for assessing whether these distances correspond to moves within or across broader neighborhood boundaries. Mast and Barca (2026) construct broader neighborhood units, called "districts," by grouping census tracts with strong observed migration flows. They report that across all movers in the Census Bureau's Environmental Impacts Frame (EIF), 72.6% of the average tract's within-county out-migrants leave their district. Districts in the top quartile of Census Place population density in their sample have a median land area of 4.3 square miles.

where Y_i is a migration outcome for incumbent renter i. These outcomes include an indicator for leaving the origin tract before 2019, an indicator for leaving the origin CBSA before 2019, and measures of the physical distance between the renter's origin tract or CBD and their 2019 tract. $\boldsymbol{X}_i$ is a vector of renter-level controls for age, household income, and residential tenure, while $\boldsymbol{W}_{n(i)}$ is a vector of origin-tract controls for location, baseline sociodemographic composition, job market access, and the prevalence of low-income housing support. α_{CBSA} denotes fixed effects for the renter's origin CBSA. We list these controls in Table A16 and provide a more in-depth discussion of our descriptive estimates in Appendix D.

Figure 2 plots conditional means from Equation 1 for the probability that an incumbent renter moves census tracts between 2010 and 2019, separately for Black and non-Black household heads (Cattaneo et al. 2024). Ten-year mobility rates are close to the baseline mobility rates across all gentrification deciles. Black low-income renters in the top decile of gentrifying tracts were only 1.16 percentage points more likely to move tracts between 2010 and 2019 than the average Black low-income renter household head. For non-Black renters, the corresponding difference is just 0.19 percentage points.

FIGURE 2. Gentrification and Residential Mobility



Notes: Figure 2 shows the probability incumbent renters move census tracts between 2010-2019 as a function of their origin tract's predicted log change in the share of adults with at least a bachelor's degree. The points plot conditional means estimated from Equation 1 (Cattaneo et al. 2024). The main text describes how we predict log changes in college shares. The sample comprises urban low-income renter households observed in both 2010 and 2019, defined after applying restriction (h.ii) in Table A1.

Figure 3 reports the estimated coefficients and standard errors on our gentrification measure from Equation 1. Panel A shows whether an incumbent renter leaves their origin census tract before 2019. Panel B shows whether an incumbent renter leaves their origin CBSA before 2019. The first set of estimates in Panel A corresponds to the conditional means plotted in Figure 2; a 10 percent increase in a census tract's college-graduate share is associated with only a 0.14 percentage-point increase in tract out-migration

among incumbent non-Black low-income renters and a 0.26 percentage-point increase among incumbent Black low-income renters. The corresponding estimates for CBSA out-migration in Panel B are 0.12 and 0.06 percentage points, respectively.

To test whether these small average mobility responses mask meaningful heterogeneity, we separately restrict the sample to census tracts that experienced growth in their share of adult college graduates, to census tracts not covered by any rent control ordinance, and to incumbent renters who had lived in their 2010 origin census tract since at least 2005. We find no economically meaningful association between residential mobility and gentrification across these subsamples. In Appendix D, we consider possible heterogeneous mobility responses across a wider set of neighborhood and household-head characteristics, including characteristics that could plausibly mediate the effects of gentrification, such as whether a tract has inelastic housing supply or is located near its CBD. We also test the sensitivity of our estimates to alternative measures of gentrification and to substantively motivated additional controls. Our results consistently point to small associations between gentrification and residential mobility, even when using tract-level gentrification measures that are more precise than those typically used in the literature.[12]

3.3. Gentrification and Experienced Neighborhood Characteristics

Despite the small association between gentrification and incumbent renter's residential mobility, gentrification may still have been associated with changes in the types of neighborhoods where incumbent renters lived by 2019. This is especially true for renters who remained in their 2010 census tract throughout the decade, since gentrification mechanically entails changes in neighborhood characteristics. To examine these changes, we re-estimate Equation 1 with dependent variables measuring log changes between the characteristics of each household's 2010 and 2019 tract.

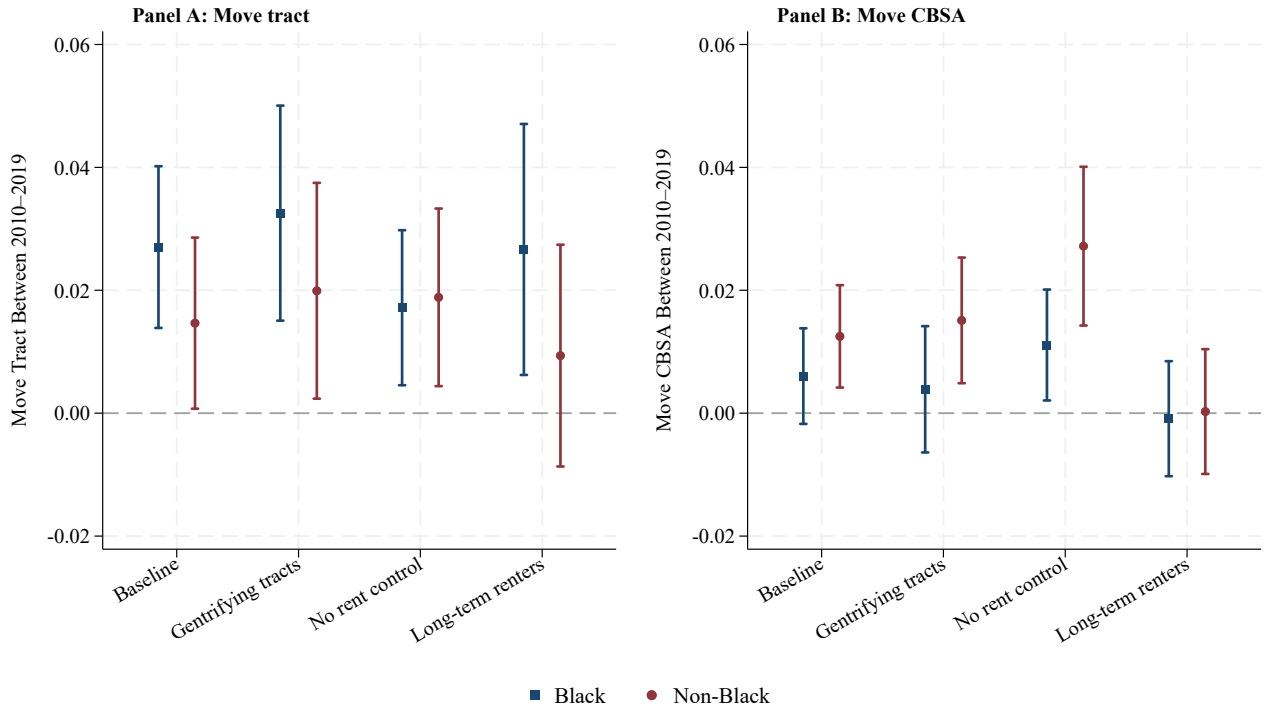
Figure 4 and Table A19 show that incumbent low-income renters who remained in gentrifying origin tracts experienced larger increases in hedonically adjusted tract rents than comparable renters who remained in less-gentrifying origin tracts. A 10 percent increase in a tract's college-graduate share between 2010 and 2019 is associated with roughly 2.5 percent larger rent increases for non-Black household heads and 2.3 percent for Black household heads. Figure A7 suggests that this relationship is not substantially attenuated by low-income renters moving out of gentrifying neighborhoods with the largest rent growth either; it shows a strong relationship between tract rent growth and gentrification, regardless of incumbent households' subsequent residential mobility. This is despite evidence that residential mobility is somewhat more closely associated with tract-level rent increases, which we examine separately in Appendix Section D.4.

Incumbent renters who remained in their origin tract throughout 2010–2019 experienced at most modest increases in measured job market access, a commute-time-weighted measure of access to employment opportunities relevant to each group that we derive from the model in Section 4.2. For these households, a 10 percent increase in their tract's share of adult college graduates is associated with a 0.5 percent average increase in job market access for non-Black households and a 0.2 percent increase for Black households.[13]

[12] Appendix D.2 compares our preferred estimates to those obtained when measuring gentrification using unadjusted 5-year ACS tract-level estimates.

[13] These estimates correspond to 0.018 and 0.008 standard deviations for non-Black and Black households, respectively. For

FIGURE 3. Gentrification and Residential Mobility Coefficient Estimates



Notes: Figure 3 plots coefficient estimates for our gentrification measure from regression Equation 1. In Panel A, the dependent variable is an indicator equal to one if the incumbent household head moved to a different census tract by 2019, whereas in Panel B it equals one if the household head moved to a different CBSA by 2019. We define gentrifying tracts as those whose college share rose from 2010 to 2019, tracts without rent control as those not covered by any type of rent-control ordinance in 2010, and long-term renters as households residing in the same tract since at least 2005. We display 95% confidence intervals based on robust standard errors clustered at the origin census-tract level (Abadie et al. 2023). The sample comprises urban low-income renter households observed in both 2010 and 2019, defined after applying restriction (h.ii) in Table A1.

Figure A9 shows that the modest increase for non-Black households appears to be driven almost entirely by tracts in the top decile of our gentrification measure. And Figure A10 suggests that the difference between Black and non-Black households does not reflect differences in their initial residential locations, but rather differences in how their sample-specific measures of job market access changed with gentrification.

Incumbent renters in gentrifying tracts experienced larger increases in their tracts' white population shares than incumbent renters in less-gentrifying tracts. These increases are largest for Black-headed households who remained in their origin tracts, reflecting the lower baseline white population shares of those tracts. For these households, a 10 percent increase in the tract's share of adult college graduates is associated with a 4.7 percent increase in the tract's white population share, compared with 1.7 percent for non-Black households who remained in their origin tracts. These results are consistent with gentrification since 2000 being driven primarily by young, white college graduates (Couture and Handbury 2023).

In contrast to incumbent renters who stayed in their origin tracts, households that moved lived in

each racial group, we compute the standard deviation of the difference between log residential market access in households' 2019 and 2010 tracts, regardless of whether they remained in their origin tract throughout 2010–2019. We residualize these changes on CBSA fixed effects and weight tracts by the number of households in the corresponding sample.

broadly similar neighborhoods by 2019, regardless of how much their origin tract gentrified. A 10 percent increase in an origin tract's share of adult college graduates is associated with movers ending up in tracts whose adult college-graduate shares are only 0.9 percent higher for Black households and 1.5 percent higher for non-Black households, relative to comparable movers from less-gentrifying tracts. The corresponding increases are roughly 0.3 percent for hedonically adjusted tract rents for both groups, and 0.2 and 0.3 percent for job market access for Black and non-Black movers, respectively.[14] Interestingly, a 10 percent increase in an origin tract's share of adult college graduates is associated with Black movers ending up in tracts with 0.9 percent lower white adult population shares than comparable Black movers from less-gentrifying tracts. This result is consistent with racial mobility networks shaping migration patterns out of gentrifying neighborhoods (Ferreira, Kenney, and Smith 2024).

Overall, gentrification substantially reshaped the neighborhoods of incumbent renters who remained in their origin tracts. Low-income renters who moved, however, relocated to tracts that were largely observably similar to the destinations of comparable movers from non-gentrifying neighborhoods. As Figure 4 shows, we also find little evidence of meaningful heterogeneity across subsamples, including among long-term incumbent renters and those from tracts not covered by rent control ordinances.

3.4. Gentrification and Economic Outcomes

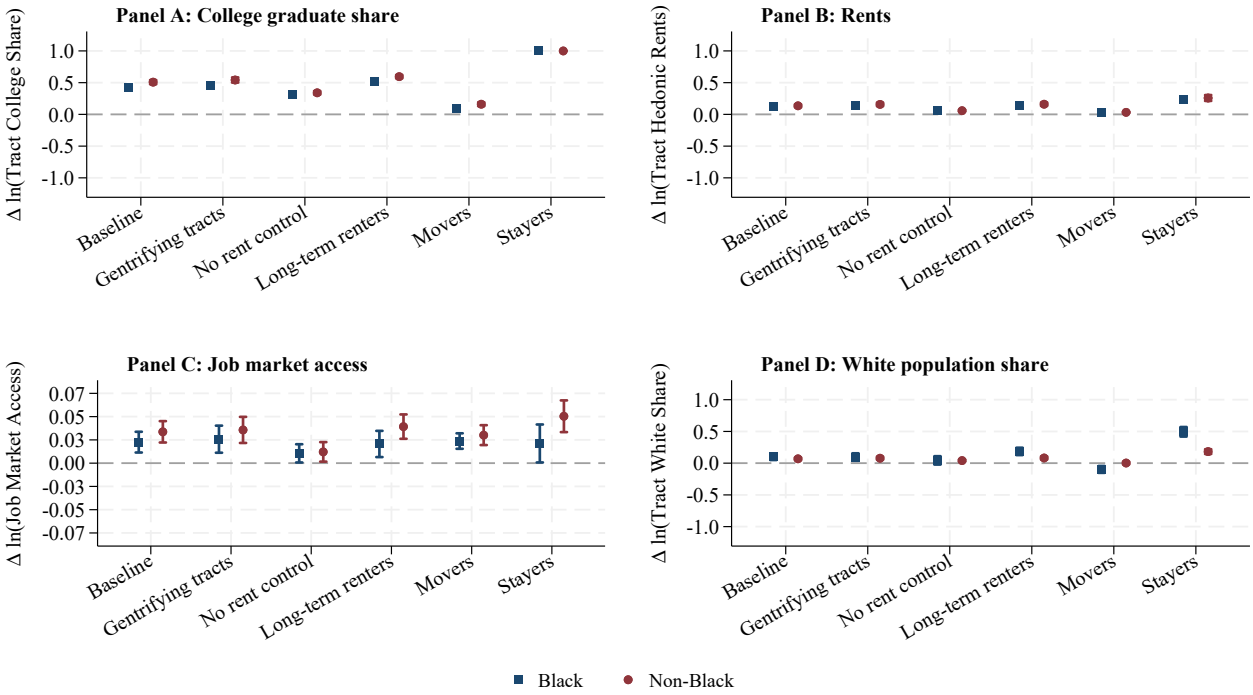
The relatively small association between gentrification and changes in incumbent renters' job market access is reflected in their realized economic outcomes. Table A20 reports estimates from Equation 1 for labor market participation, income, and commute time. Baseline job mobility is high for incumbent renters, as reported in Table A21, so realized economic outcomes are mediated by substantial labor market churn in addition to any changes in job market access associated with gentrification.

While the association between gentrification and realized economic outcomes is generally small, we find racial heterogeneity consistent with the heterogeneity in experienced job market access shown in Figure 4. A 10 percent increase in the origin tract's adult college-graduate share is associated with a 0.07 percentage-point decrease in the probability that Black incumbent renters are observed in the LEHD in 2019, with no corresponding change for non-Black incumbent renters. Given a baseline non-participation rate of 9.8 percent for Black incumbent renters in 2019, this estimate implies a roughly 0.7 percent increase in non-participation. The racial heterogeneity is even more pronounced among incumbent renters who remain in their origin census tract, again consistent with the racial heterogeneity in experienced job market access among stayers.

The income results broadly mirror the labor market participation results because non-participants are coded as having zero income. We find no consistent patterns in commute time, and all commute-time point estimates are economically small. Overall, the realized economic outcomes align with the modest differences in experienced job market access, with somewhat stronger evidence of adverse labor market changes for Black incumbent renters. These findings are broadly consistent with prior evidence that gentrification had limited effects on incumbent adults' employment and income (Meltzer and Ghorbani

[14]These positive associations could partly reflect incumbent renters moving to nearby tracts whose characteristics are likely spatially correlated with those of their origin tracts (Mast and Barca 2026). This is especially relevant for job market access, which is spatially diffuse by construction.

FIGURE 4. Gentrification and Experienced Neighborhood Characteristics Coefficient Estimates



Notes: Figure 4 plots coefficient estimates for our gentrification measure from regression equation 1. In Panel A, the dependent variable is the log difference in the share of adults with a college degree between each household's 2019 and 2010 census tracts. In Panel B, the dependent variable is the log difference in hedonic rents between each household's 2019 and 2010 census tracts. In Panel C, the dependent variable is the log difference in job market access, measured using the sample-specific residential market access index defined in Section 4.2. In Panel D, the dependent variable is the log difference in the share of white adults between each household's 2019 and 2010 census tracts. Movers are household heads observed in a different census tract in 2019 than in 2010. Stayers are household heads observed in the same census tract in 2019 as in 2010. Coefficient estimates and their standard errors are reported in Table A19. Otherwise, the plots are analogous to those in Figure 3.

2017; Brummet and Reed 2021). We discuss the relationship between gentrification and economic outcomes in more detail in Section D.5.

4. A Dynamic Model of Neighborhood and Workplace Choice

Our descriptive findings admit two broad interpretations of the welfare effects of gentrification for incumbent low-income renters. One interpretation is that high residential mobility and similar destination choices imply that renters can readily access comparable neighborhoods, limiting welfare differences between renters from gentrifying and non-gentrifying tracts. Another interpretation is that high mobility reflects large idiosyncratic shocks, perhaps stemming from unstable employment, household disruption, or other financial shocks, rather than deliberate responses to neighborhood change. In this case, high mobility may reveal little about renters' ability to respond to neighborhood change, leaving the welfare effects of gentrification, especially for incumbents who remain in their initial neighborhood, ambiguous. To quantify these welfare effects and distinguish between these competing interpretations, we estimate a single-agent

dynamic discrete neighborhood choice model tailored to low-income working renters, building on dynamic location-choice models with moving frictions and forward-looking behavior (Bayer et al. 2016; Diamond, McQuade, and Qian 2018; Davis et al. 2021; Coate and Mangum 2022; Almagro and Domínguez-Iino 2025; Davis, Gregory, and Hartley 2026).

We extend the standard dynamic discrete neighborhood choice framework by incorporating a repeated-static workplace choice problem (Tsivanidis 2026).[15] Conditional on their neighborhood choice, households choose their workplace location and industry to maximize income net of commuting time costs. Using households' observed commutes in the LEHD to estimate the workplace choice model, we obtain time-varying, tract- and sample-specific measures of job market access that enter households' neighborhood flow utilities. Incorporating these measures allows us to distinguish households' preferences over neighborhood amenities that would otherwise be confounded with preferences over job market access, as our results show in Section 7. This extension also allows us to assess a longstanding concern that gentrification may harm incumbent renters by limiting access to inner-city neighborhoods with better public transportation and greater proximity to employment opportunities (LeRoy and Sonstelie 1983; Glaeser, Kahn, and Rappaport 2008).

In Section 8, we use the estimated model to quantify the welfare impacts of gentrification from the perspective of incumbent low-income renters in 2010.

4.1. Households and Timing of Choices

Low-income renter households are indexed by i and differ ex-ante by race, $k \in \{\text{Black}, \text{Non-Black}\}$. Access to employment opportunities and preferences over neighborhood characteristics can vary by household race. Households also differ by their current residential neighborhood, $n_{i,t-1}$, and the length of residential tenure in this neighborhood, $\tau_{i,t-1}$. These controlled state variables, $x_{i,t} \equiv (n_{i,t-1}, \tau_{i,t-1})$, influence households' choices by affecting the utility cost of moving from their current neighborhood.[16] We consider households choosing among neighborhoods in their current city, c, and an outside option of leaving their city, OO^c. We denote these city-specific choice sets as $\mathcal{N}^c \equiv \{OO^c, 1^c, \ldots, N^c\}$ and abstract from neighborhood choices between cities.

Time is discrete and indexed by t. In each period, households receive idiosyncratic preference shocks ε_{int} for residential neighborhoods within their city and for their outside option of leaving their city. They also receive idiosyncratic productivity shocks $z_{im\iota t}$ across workplace locations, $m \in \mathcal{N}^c$, and industries, $\iota \in \mathrm{I}$, where I denotes the industry choice set. After first observing their residential preference shocks, households choose their residential neighborhood n_{it} and then decide on an optimal level of housing consumption H^k_{nt}. Following this, they observe their idiosyncratic workplace productivity shocks and choose their workplace location and industry.[17] The timing of these choices is summarized in Figure 5.

[15] Warnes (2024) similarly incorporates a repeated-static workplace choice problem into a dynamic neighborhood choice model to evaluate the welfare impact of transportation infrastructure improvements in Buenos Aires.

[16] The individual-level state variables, x_{it}, are "controlled" in that their evolution is fully determined by the household's choices.

[17] We separate households' residential location and workplace choices to ease exposition. Our framework is isomorphic to a framework where households jointly choose residential and workplace locations as in Ahlfeldt et al. (2015).

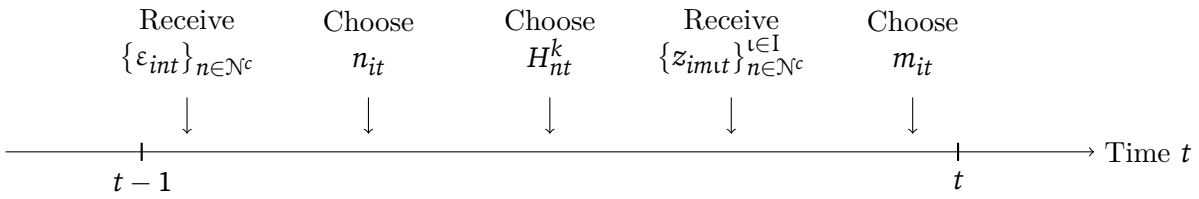


FIGURE 5. Timeline of Household Choices t

In the remainder of this section, we detail the periodic household problem working backwards from their workplace choice.

4.2. Workplace Choice

Conditional on living in neighborhood n, households observe these productivity shocks and choose their workplace location and industry to maximize their commute time–discounted income:

$$I_{int} \equiv \max_{m,\iota} \frac{z_{im\iota t}}{d^k_{nmt}} w^k_{m\iota t},$$

where $w^k_{m\iota t}$ is the wage offered in workplace tract m, industry ι, and period t to type-k households measured in efficiency units. $d^k_{nmt} > 0$ is the cost for a type-k household to commute between neighborhoods n and m in period t, also measured in efficiency units.[18] We assume that households spend a fixed amount of time each day working or commuting, so d^k_{nmt} effectively discounts the total wage offered to household i in workplace m and industry ι: $z_{im\iota t} \cdot w^k_{m\iota t}$. We assume that $z_{im\iota t}$ is drawn independently from a Frechet distribution with shape parameter ϵ^{kc}_t. These shape parameters are specific to each city, year, and racial group, which we make explicit with the superscripts k, c and subscript t.[19] The commute-adjusted earnings index for a type-k household living in tract n at time t is therefore

$$\bar{I}^k_{nt} = \Gamma\left(1 - \frac{1}{\epsilon^{kc}_t}\right) \cdot \left(\text{RMA}^k_{nt}\right)^{1/\epsilon^{kc}_t}$$

where $\Gamma(\cdot)$ is the gamma function and $\text{RMA}^k_{nt} \equiv \sum_{\iota\in I} \sum_{m\in\mathcal{N}^c} \left(\frac{w^k_{m\iota t}}{d^k_{nmt}}\right)^{\epsilon^{kc}_t}$ is the structural index we use to measure job market access, which we follow the literature in terming residential market access. We derive these expressions and describe how we construct their empirical analogs in Appendix B.1.

[18] We allow commute costs to differ by race because Black and non-Black low-income renters exhibit systematically different commuting patterns. Black renters are substantially more likely to commute by public transit and have lower estimated commute-time elasticities, as reported in Tables A5 and A6. These differences may reflect variation in transportation mode, transit access, and other constraints rather than intrinsic differences in preferences by race.

[19] Assuming shocks are independent across years effectively rules out job-switching costs. We do not view this abstraction as particularly restrictive given the high rates of job mobility among low-income renters in our sample (cf. Table A21).

4.3. Neighborhood Choice

Households' Neighborhood Choice Problem. Households choose their residential locations to maximize the sum of their expected discounted utilities,

$$\max_{\{n \in \mathcal{N}^c\}_t^\infty} \quad \mathbb{E}\left[\sum_{t'=t}^{\infty} \delta^{t'-t} \cdot u_n^k\left(s_{it'}^c\right) \,\middle|\, \mathcal{I}_{it}\right] \tag{2}$$

where δ is a known discount factor and s_{it}^c is a vector of state variables that determine household i's flow utility u_n^k from choosing neighborhood n. s_{it}^c includes the commute-adjusted earnings index, $\bar{I}_{nt}^k$, derived in Section 4.2. $\mathbb{E}\left[\cdot|\mathcal{I}_{it}\right]$ denotes the expectation operator conditioned on household i's information set at time t. In each period, households observe the state variables s_{it}^c before choosing their residential location. Flow utilities are then realized, and states evolve. Each household's information set $\mathcal{I}_{it}$ therefore includes all current and past state variables that households may use to form expectations over their future evolution. We specify households' belief formation in Section 5.1.

State Variables. Households' flow utilities depend on the vector of state variables $s_{it}^c \equiv \left(x_{it}, \varepsilon_{it}, \omega_t^{kc}, \xi_t^{kc}\right)$, where $(x_{it}, \varepsilon_{it})$ are household-level observable and unobservable state variables, respectively. By contrast, $\left(\omega_t^{kc}, \xi_t^{kc}\right)$ are city-specific observable and unobservable state variables, respectively. The household-level observable state variables, x_{it}, include the length of households' residential tenure and their neighborhood choice in the previous period, $x_{it} = (n_{it-1}, \tau_{it-1})$. We assume the household's unobservable state, $\varepsilon_{it} \equiv \{\varepsilon_{int}\}_{n \in \mathcal{N}^c}$, is i.i.d. across households, neighborhoods, and time. We conceptualize ε_{it} as a vector of unobserved-to-the-econometrician time-varying household and neighborhood-specific preference shocks. As is common, we assume that ε_{it} is distributed according to a type I extreme value distribution with scale parameter σ^k, which we normalize to 1.

ω_t^{kc} denotes observable city- and type-specific state variables. It includes vectors of housing costs that can vary by the length of residents' tenure in a neighborhood, $r_{\tau nt}$, shares of college graduates across neighborhoods, $\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}$, each neighborhoods' commute-adjusted earnings index, $\bar{I}_{nt}^k$, and an index for each period to explicitly incorporate nonstationarity:

$$\omega_t^{kc} = \left(\{r_{\tau nt}\}, \left\{\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right\}, \left\{\bar{I}_{nt}^k\right\}, t\right).$$

Finally, ξ_t^{kc} is a city-specific vector of unobservable time-varying neighborhood-level amenity valuations among type-k households. For example, ξ_t^{kc} could include time-varying valuations among type-k households for suburban life or proximity to natural amenities, independent of gentrification. To facilitate exposition, we define $\bar{\omega}_t^{kc} \equiv \left(\omega_t^{kc}, \xi_t^{kc}\right)$ as the vector containing both observable and unobservable city-specific state variables.

Flow Utility. Flow utility net of moving costs for a type-k household with residential tenure length τ that chooses neighborhood n can be represented as

$$A_{nt}^k \cdot Q_{\tau nt}^k \cdot \tau^{\beta_\tau^k} \cdot \exp(\varepsilon_{int}), \tag{3}$$

where A_{nt}^k is a type-specific valuation of amenities in neighborhood n and $Q_{\tau nt}^k$ is a consumption composite that is Cobb–Douglas over nonhousing consumption, $C_{\tau nt}^k$, and housing consumption, $H_{\tau nt}^k$:

$$Q_{\tau nt}^k \equiv \left(C_{\tau nt}^k\right)^{\beta_C^k} \left(H_{\tau nt}^k\right)^{1-\beta_C^k}. \tag{4}$$

Households' expected period- and neighborhood-specific budget constraint is given by

$$C_{\tau nt}^k \geq \bar{I}_{nt}^k - r_{\tau nt} \cdot H_{\tau nt}^k,$$

where time-varying neighborhood-level rents can depend on the length of residents' tenure as,

$$r_{\tau nt} = r_{nt} \cdot h_n\left(\tau, \{\bar{\omega}_t^{kc}\}_{t-\tau}^t\right). \tag{5}$$

Equation 5 decomposes rents charged to residents in neighborhood n in period t into current market-level rents, r_{nt}, and a component that depends on the length of residents' tenure and a vector of state variables in neighborhood n: $h_n\left(\tau, \{\bar{\omega}_t^{kc}\}_{t-\tau}^t\right)$.[20] $\{\bar{\omega}_t^{kc}\}_{t-\tau}^t$ can contain information on the prevalence of any local rent control ordinances, changes in historical market rents, and other non-market forces influencing how charged rents vary with the length of neighborhood tenure for low-income renters. Given gross rents may rise more slowly for incumbent renters, incorporating tenure-dependent rents helps us distinguish between households' disutility from paying higher rents and utility from remaining in their home neighborhood all else equal, $\tau^{\beta_\tau^k}$. This feature of our setup is feasible because we can jointly observe rents and the length of incumbents' residential tenure; we detail the empirical construction of $r_{\tau nt}$ in Appendix A.3.

For each k-type, we parameterize neighborhood amenities as

$$A_{nt}^k \equiv \left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right)^{\beta_A^k} \exp\left(\xi_{nt}^k\right).$$

where we decompose unobserved neighborhood- and period-specific amenities, ξ_{nt}^k, into time-invariant neighborhood-specific components, city-wide components that vary over time but not across neighborhoods within a city, and neighborhood-specific time-varying components:

$$\xi_{nt}^k \equiv \alpha_n^k + \alpha_t^{kc} + \bar{\xi}_{nt}^k.$$

Appendix B.2 shows that by taking logs of equation 3, incorporating moving costs, $MC_t^k(n_t, n_{it-1})$,

[20] We constrain $h_{OO^c}\left(\tau, \{\bar{\omega}_t^{kc}\}_{t-\tau}^t\right) = 1$. This constraint assumes households residing outside their associated urban core always pay market rents, regardless of their neighborhood tenure.

solving for optimal housing consumption, and substituting in the amenity specifications yields the following expected flow utility specification for a type-k household choosing neighborhood n with state s_{it}^c:

$$u_n^k\left(s_{it}^c\right) = \tilde{\alpha}_n^k + \tilde{\alpha}_t^{kc} + \tilde{\beta}_I^k \ln\left(\bar{I}_{nt}^k\right) + \tilde{\beta}_r^k \log\left(r_{\tau nt}\right) + \tilde{\beta}_A^k \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) + \tilde{\beta}_\tau^k \ln(\tau_{it}) + MC_t^k(n_t, n_{it-1}) + \tilde{\xi}_{nt}^k + \varepsilon_{int} \tag{6}$$

Flow utility is expected in that it is the value households expect to obtain before they realize their period-specific workplace-tract productivity shocks. It is this flow utility specification that is relevant for households' neighborhood choice and thus for our structural estimation of households' preferences over neighborhood characteristics.

Moving Costs. If a type-k household decides to leave its current neighborhood for another neighborhood in the same city, it incurs a nonmonetary moving cost, $MC_t^k(n_{it}, n_{it-1})$. This nonmonetary moving cost depends on a city-specific fixed disutility of moving and the physical and social distances between households' origin and destination neighborhoods. Conversely, if a type-k household decides to leave its city entirely, it incurs a single, city-specific fixed cost. Specifically,

$$MC_t^k(n_t, n_{t-1}) = \begin{cases} 0 & \text{if } n_t = n_{t-1} \\ MC^{kc} + \beta_d^{k\prime}\boldsymbol{d}(n_t, n_{t-1}) + \beta_s^{k\prime}\boldsymbol{s}(n_t, n_{t-1}) & \text{if } n_t \neq n_{t-1} \text{ and } n_t, n_{t-1} \neq OO^c \\ MC_{OO}^{kc} & \text{if } n_{it} \neq n_{t-1} \text{ and } n_t \text{ or } n_{t-1} = OO^c \end{cases}$$

where MC^{kc} and MC_{OO}^{kc} are the city-specific fixed intensive- and extensive-margin moving costs. $\boldsymbol{d}(n_t, n_{t-1})$ is a vector describing the physical distance between n_t, n_{t-1} and $\boldsymbol{s}(n_t, n_{t-1})$ is a vector describing the social distance between n_t, n_{t-1} in period t:

$$\boldsymbol{d}(n_t, n_{t-1}) \equiv \begin{bmatrix} Dist(n_t, n_{t-1}) \\ \vdots \\ Dist(n_t, n_{t-1})^3 \end{bmatrix} \quad \boldsymbol{s}_t(n_t, n_{t-1}) \equiv \begin{bmatrix} \ln\left(\frac{S(n_t)}{S(n_{t-1})}\right) \mathbb{1}\left\{\ln\left(\frac{S(n_t)}{S(n_{t-1})}\right) \geq 0\right\} \\ \ln\left(\frac{S(n_t)}{S(n_{t-1})}\right) \mathbb{1}\left\{\ln\left(\frac{S(n_t)}{S(n_{t-1})}\right) < 0\right\} \\ \vdots \\ \ln\left(\frac{S(n_t)}{S(n_{t-1})}\right)^3 \mathbb{1}\left\{\ln\left(\frac{S(n_t)}{S(n_{t-1})}\right) \geq 0\right\} \\ \ln\left(\frac{S(n_t)}{S(n_{t-1})}\right)^3 \mathbb{1}\left\{\ln\left(\frac{S(n_t)}{S(n_{t-1})}\right) < 0\right\} \end{bmatrix}$$

where $Dist(n_t, n_{t-1})$ is the straight-line distance between the centroids of neighborhood n_t and n_{t-1} and $\ln\left(S(n_t)/S(n_{t-1})\right)$ is the log difference in the share of college graduates in neighborhoods n_t and n_{t-1}, with both shares measured at time t. We introduce the social distance measure to capture the idea that, while low-income renters may value living in higher-socioeconomic-status neighborhoods, they may face non-financial frictions that make such moves unattractive (Gans 1982; Jargowsky 2009; Bergman et al. 2024). We allow this cost to differ depending on the "direction" of the move as it may be, for example, more costly for low-income renters to move up the neighborhood college-share distribution than down it.

Value Functions, Choice Probabilities, and Expectational Errors. We denote $V^k(s_{it}^c)$ as the value function of the dynamic programming problem associated with Equation 2. By Bellman's principle of optimality,[21]

$$V^k\left(s_{it}^c\right) = \max_{n\in\{OO^c,1^c,\ldots,N^c\}} \left\{\mathbb{E}_{x'|xn}\left[u_n^k\left(s_{it}^c\right)\right] + \delta\mathbb{E}_t\left[V^k\left(s_{it+1}^c\right)|n, s_{it}^c\right]\right\}$$

We define household i's ex-ante continuation value function as the expectation of the value function with respect to ε_{it}:

$$\bar{V}^k\left(x_{it}, \bar{\omega}_t^{kc}\right) \equiv \int V^k\left(s_{it}^c\right) dF^{\varepsilon}(\varepsilon_{it}) \tag{7}$$

and define household i's conditional value function as

$$\begin{aligned} v_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right) &\equiv \mathbb{E}_{x'|xn}\left[u_n^k\left(s_{it}^c\right)\right] - \varepsilon_{int} + \delta\mathbb{E}_t\left[\bar{V}^k\left(x_{it+1}, \bar{\omega}_{t+1}^{kc}\right)|n, x_{it}, \bar{\omega}_t^{kc}\right] \\ &\equiv \bar{u}_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right) + \delta\mathbb{E}_t\left[\bar{V}^k\left(x_{it+1}, \bar{\omega}_{t+1}^{kc}\right)|n, x_{it}, \bar{\omega}_t^{kc}\right] \end{aligned} \tag{8}$$

Then, given our assumption that ε_{it} are distributed i.i.d type I extreme value, the probability that a type-k household with state variables $\left(x_{it}, \bar{\omega}_t^{kc}\right)$ chooses neighborhood n in period t is given by

$$p_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right) = \frac{\exp\left(v_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right)\right)}{\sum_{n'\in\mathcal{N}^c}\exp\left(v_{n'}^k\left(x_{it}, \bar{\omega}_t^{kc}\right)\right)}, \tag{9}$$

and the ex ante value function in 7 has the value

$$\bar{V}^k\left(x_{it}, \bar{\omega}_t^{kc}\right) = \ln\left(\sum_{n\in\mathcal{N}^c}\exp\left(v_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right)\right)\right) + \gamma,$$

where γ is Euler's constant. Combining these two expressions yields the following well-known result, which is critical to deriving our estimating equations (Hotz and Miller 1993):

$$\bar{V}^k\left(x_{it}, \bar{\omega}_t^{kc}\right) = v_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right) - \ln\left(p_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right)\right) + \gamma, \qquad \forall n \in \mathcal{N}^c. \tag{10}$$

Another expression critical for deriving our estimating equations is the difference between households' expected ex-ante continuation values and their realized counterparts:

$$e^{\bar{V}}(x', \bar{\omega}_t^{kc}, \bar{\omega}_{t+1}^{kc}) \equiv \underbrace{\bar{V}(x', \bar{\omega}_{t+1}^{kc})}_{\text{realized}} - \mathbb{E}_{\bar{\omega}'|\bar{\omega}_t^{kc}}\left[\bar{V}(x', \bar{\omega}')|\bar{\omega}_t^{kc}\right] \tag{11}$$

We follow Kalouptsidi, Scott, and Souza-Rodrigues (2021) and term the differences *expectational errors*. These expectational errors allow us to discard households' actual expectations in estimation. Solving for

[21] The expectation operator $\mathbb{E}_{x'|xn}[\cdot]$ is with respect to the future value of households' observed household-level state variables, x', conditional on households' current state and their neighborhood choice. While the current deterministic setup renders this operator redundant, we include it here to be consistent with our empirical application that models the evolution of households' residential tenure stochastically conditional on their neighborhood choice.

households' expectations would be prohibitively costly given the high-dimensional nature of a household's state space, with some urban cores containing over a thousand 2010-delineated census tracts.

Now that the function dependencies are clear, going forward, we suppress their arguments and remove the city superscripts unless we require them for explicative purposes: $\bar{V}^k_{xnt} \equiv \bar{V}^k\left(x_{it}, \bar{\omega}^{kc}_t\right)$, $V^k_{int} \equiv V^k(s^c_{it})$, $\bar{u}^k_{xnt} \equiv \bar{u}^k_n\left(x_{it}, \bar{\omega}^{kc}_t\right)$, $\nu^k_{xnt} \equiv \nu^k_n\left(x_{it}, \bar{\omega}^{kc}_t\right)$, and $p^k_{xnt} \equiv p^k_n\left(x_{it}, \bar{\omega}^{kc}_t\right)$.

5. Structural Estimation

We estimate our neighborhood demand parameters with what is termed in the dynamic discrete choice literature an Euler equations in conditional choice probabilities (ECCP) estimator (Scott 2013; Kalouptsidi, Scott, and Souza-Rodrigues 2021).[22] Similarly to other conditional choice probability estimators, ECCP estimation involves two steps. In the first step, we estimate households' conditional choice probabilities, the soon-to-be-introduced transition distributions for the household-level state variables, and households' variable moving costs, $\boldsymbol{\beta}^{k\prime}_d$ and $\boldsymbol{\beta}^{k\prime}_s$. We then estimate the remaining model parameters in a second step, conditional on our first-step estimates. We estimate the second step using moment restrictions implied by the dynamic optimization of households, which we derive and explain in Section 5.2. With flow utilities linear in the model's parameters, we can evaluate these moment restrictions in a standard 2SLS framework.[23]

The ECCP estimator has many advantages in our setting. First, the ECCP estimator is computationally light. Since our analysis covers the residential history of low-income households for ten years across many large US metropolitan areas at the census tract level, traditional dynamic discrete choice estimation procedures like Rust (1987) that explicitly solve for households' value functions are infeasible. Second, our focus on gentrification implies an inherently nonstationary environment, making modeling the evolution of neighborhood change conceptually and computationally challenging. As we demonstrate when deriving our moment restrictions, ECCP estimation requires neither the complete specification of households' information sets nor a description of how the city-specific state variables evolve. Third, by yielding moment conditions we can evaluate in a 2SLS framework, we can relate our instrumental variables detailed in Section 6 to the recent literature on quasi-experimental shift-share instruments (Goldsmith-Pinkham, Sorkin, and Swift 2020; Borusyak, Hull, and Jaravel 2022).

5.1. Estimation Assumptions

To identify our neighborhood demand parameters, we must make the following set of assumptions:

[22] ECCP estimators are so named given their likeness to Euler equations in models with continuous choice variables (Aguirregabiria and Magesan 2013).

[23] ECCP estimation has been used in a variety of applied settings, from choices over agricultural land use (Scott 2013) and new technology adoption (Groote and Verboven 2019) to occupational choice (Traiberman 2019; Gendron-Carrier 2023) and, most relevant to our setting, residential neighborhoods (Diamond, McQuade, and Qian 2018; Davis et al. 2021; Almagro and Domínguez-Iino 2025). See Kalouptsidi, Scott, and Souza-Rodrigues (2021) for a comprehensive econometric treatment of linear regression techniques with ECCP estimators.

(a) *State Transitions*: The state variables s_{it}^c evolve according to a first-order Markov process with a transition distribution that factors as

$$f\left(s_{it+1}^c | n_{it}, s_{it}^c\right) = f^x(x_{it+1}|n_{it}, x_{it}) \cdot f^{\bar{\omega}}\left(\bar{\omega}_{t+1}^{kc}|\bar{\omega}_t^{kc}\right) \cdot f^{\varepsilon}(\varepsilon_{it+1})$$

(b) *Utility Normalization*: We normalize the utility offered by the outside option for residents with $\tau = 1$, separately for each city:

$$\tilde{\alpha}_{OO^c}^k + \tilde{\beta}_I^k \ln\left(\bar{I}_{OO^c t}^k\right) + \tilde{\beta}_r^k \log\left(r_{OO^c t}\right) + \tilde{\beta}_A^k \ln\left(\frac{\text{Coll}_{OO^c t}}{\text{Pop}_{OO^c t}}\right) + \tilde{\xi}_{OO^c t}^k = \alpha^{kc} \quad \forall\, t$$

(c) *Rational Expectations*: Households' expectations over the evolution of the CBSA-level state variables conditional on their information set $\mathfrak{I}_{it}$ correspond to the conditional expectations of the true data generating process given $\mathfrak{I}_{it}$:

$$\mathbb{E}\left[e^{\bar{V}}(x', \bar{\omega}_t^{kc}, \bar{\omega}_{t+1}^{kc})|\mathfrak{I}_{it}\right] = 0$$

where $e^{\bar{V}}(x', \bar{\omega}_t^{kc}, \bar{\omega}_{t+1}^{kc})$ are the expectational errors defined in Equation 11.

An important implication of assumption (a) is that the market-level state variables $\bar{\omega}_t^{kc}$ are *perceived* as exogenous by individual households; a household cannot expect to individually affect the evolution of $\bar{\omega}_t^{kc}$ with its own neighborhood choice (cf. Assumption (1) in Kalouptsidi, Scott, and Souza-Rodrigues (2021)). Given that the typical 2010-delineated US census tract contains around 4,000 residents, we believe that this assumption is plausible.[24] Note that Assumption (a) does not require the observed and unobserved city-specific state variables to evolve independently. We highlight this to foreshadow the econometric challenge we face when attempting to identify preferences over functions of ω_t^{kc}.

Assumption (b) says that residents who choose to reside outside of their respective city's urban core obtain a time-invariant and city-specific mean utility.[25] This assumption normalizes each city's neighborhood mean utilities to a constant and time-invariant level, which is necessary to compare welfare across households within CBSAs given that logit models identify only differences in mean utilities. The assumption facilitates exposition and, because each α^{kc} is unobserved, highlights the incommensurability of expected welfare across different k-types and cities without additional and restrictive assumptions.

Last, Assumption (c) says that, on average, households correctly anticipate the evolution of $\bar{\omega}_t^{kc}$. An important corollary of Assumption (c) is that the contents of households' information sets in time t are mean independent of their expectational errors at time t as well (cf. Lemma 1 in Kalouptsidi, Scott, and Souza-Rodrigues (2021)). The importance of this corollary will become clear in Section 5.3 when we discuss our choice of IVs to estimate households' preferences.

[24] While households' dynamic optimization implies our estimating equations, they are not informed by any equilibrium conditions, allowing us to remain agnostic over how households' individual choices influence the evolution of the city-wide state variables.

[25] Because logit models identify only differences in mean utilities, the city-wide time effect $\tilde{\alpha}_t^{kc}$ in Equation 6 is identified only relative to the outside option. We therefore define $\tilde{\alpha}_t^{kc}$ as the change over time in the mean utility of a city's urban-core tracts relative to that city's outside option, whose mean utility Assumption (b) holds fixed at α^{kc}. Any time-varying shock common to all locations in a city, including the outside option, is not separately identified and is subsumed in this normalization.

5.2. Deriving Our Estimating Equations

We now use the model setup to derive estimating equations that are linear in households' demand parameters. We begin by defining renewal actions.

Renewal Actions. To derive our estimating equations, we use what are termed renewal actions in the dynamic discrete choice literature (Hotz and Miller 1993; Arcidiacono and Miller 2011). Renewal actions are choices that lead to the same distribution of states in the following period regardless of the household's current state. In our setting, simply moving to a new neighborhood is a renewal action; moving to a new neighborhood resets a household's residential tenure to 0 regardless of the household's previous neighborhood or its current residential tenure. Moreover, because the city-specific state variables, $\bar{\omega}_{t+1}^{kc}$, and unobserved idiosyncratic preference shocks, ε_{it+1}, are independent of the household's state in period t, *all* the remaining state variables have the same conditional distribution following a move to a new neighborhood.[26]

Following the ECCP literature, we exploit such renewal actions when deriving our estimating equation (Scott 2013; Diamond, McQuade, and Qian 2018; Davis et al. 2021; Almagro and Domínguez-Iino 2025). To see how, consider Figure 6. The figure shows the residential choices of two hypothetical type-k households (A and B) with initial tenure length τ between periods $t-1$ and $t+1$. In period $t-1$, these households reside in the same neighborhood, $\underline{n}$, which we call the path origin. In period t, however, household A moves to neighborhood n while household B leaves the urban core, exercising their outside option OO^c. We refer to n as the destination tract. In period $t+1$, both households move to the same neighborhood, $\bar{n}$, which we term the renewal tract. Our estimation procedure involves relating the difference in the expected discounted utilities of the two households' hypothetical neighborhood choices to the difference in the probability that the two households actually make these neighborhood choices. Critically, because moving to neighborhood $\bar{n}$ in period $t+1$ constitutes the same renewal action for both households, , they face the same distribution of future states, equalizing their continuation values. Differences in the expected discounted utilities associated with the two sets of neighborhood choices are thus a function only of households' flow utilities. Constructing estimating equations based on these comparisons therefore helps isolate observed variation in current neighborhood characteristics from variation in households' unobserved continuation values. Throughout the paper we term these consecutive residential location choices *residential paths*.

Our Estimating Equation. Given the residential paths outlined in Figure 6, we can derive the following estimating equation by relating differences in expected discounted utilities associated with the residential paths to the probabilities households take these paths:

$$Y_{x\underline{n}n\bar{n}t}^{k} = \tilde{\bar{\alpha}}_{n}^{k} + \tilde{\alpha}_{t}^{kc} + \tilde{\beta}_{I}^{k}\ln(\bar{I}_{nt}^{k}) + \tilde{\beta}_{r}^{k}\ln(r_{\tau nt}) + \tilde{\beta}_{A}^{k}\ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) + \tilde{\beta}_{\tau}^{k}\tilde{\tau}_{xt} + \widetilde{MC}_{t}^{kc} + \nu_{x\underline{n}n\bar{n}t}^{k} \tag{12}$$

[26] Note that renewal actions depend on our construction of households' neighborhood tenure in Section 4.3. This construction assumes that the lengths of households' prior residential tenures do not impact the value of future residential tenure. While this is a strong assumption, it is necessary to keep the dimension of the household-level state space manageable. We outline the steps we take to mitigate the restrictiveness of this assumption in Appendix B.5.

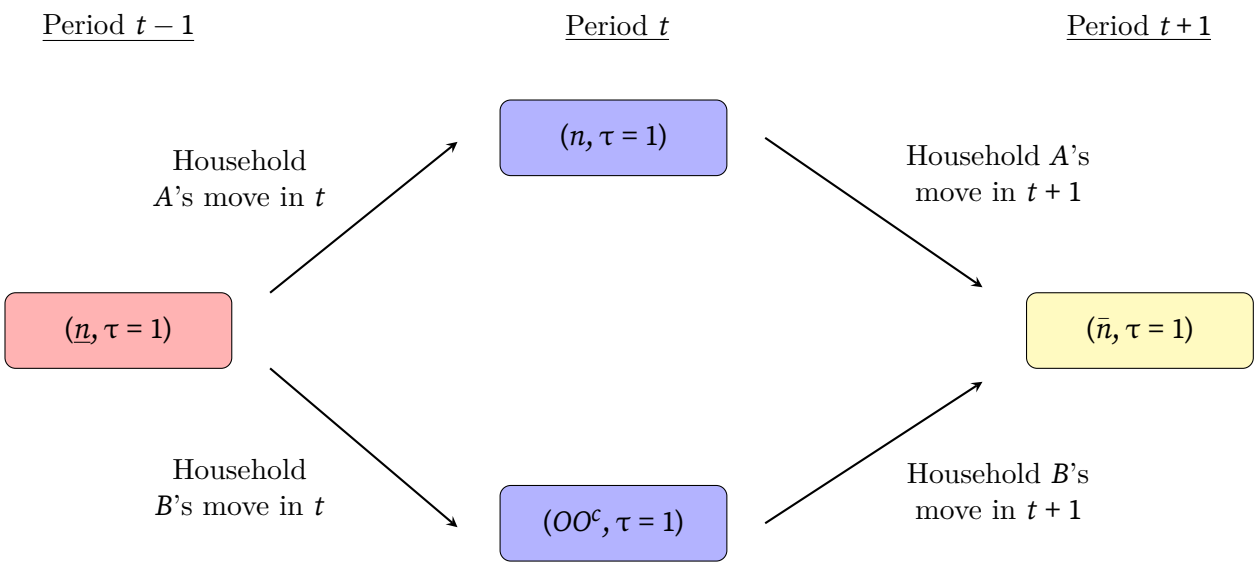


FIGURE 6. Residential Paths for Two Hypothetical Households

Note: Figure 6 shows hypothetical residential paths for two households, A and B. These households both start in neighborhood $\underline{n}$ with tenure $\tau = 1$, move to different neighborhoods for period t, and convene in a common neighborhood in period $t+1$.

where

$$
\begin{aligned}
Y^k_{x\underline{n}n\bar{n}t} &\equiv \ln\left(\frac{p^k_{xnt}}{p^k_{xOO^ct}}\right) + \delta\left(\sum_{x'}\ln\left(p^k_{x'\bar{n}t+1}\right) f^x\left(x'|n, x_t\right) - \sum_{x'}\ln\left(p^k_{x'\bar{n}t+1}\right) f^x\left(x'|OO^c, x_t\right)\right) \\
\tilde{\tau}_{xt} &\equiv \sum_{x'}\ln(\tau_{xt}(x')) f^x\left(x'|n, x_{xt}\right) - \sum_{x'}\ln(\tau_{xt}(x')) f^x\left(x'|OO^c, x_t\right) \\
\widetilde{MC}^{kc}_t &\equiv MC^k_t(n, \underline{n}) - MC^k_t(OO^c, \underline{n}) + \delta\left(MC^k_{t+1}(\bar{n}, n) - MC^k_{t+1}(\bar{n}, OO^c)\right) \\
\nu^k_{x\underline{n}n\bar{n}t} &\equiv \delta \cdot \tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}) + \tilde{\xi}^k_{nt}.
\end{aligned}
$$

Since the full derivation of this estimating equation is becoming well known, we relegate it to Appendix B.4. With estimates of households' conditional choice probabilities, $\hat{p}^k$, and estimates of the household-level transition distributions, $\hat{f}^x$, we can estimate Equation 12 using linear regression techniques. The following subsection details our two-step estimation procedure.[27]

[27]Figure 6 illustrates a representative comparison of two residential paths. In practice, to help identify the components of households' fixed moving costs and their preferences for residential tenure, we further consider alternative comparisons of residential paths between neighborhoods and their city's outside option. In Appendix B.5, we present the full set of residential paths used in estimation, discuss how each pair of residential paths contributes to identification, outline all restrictions imposed on the choices of $\underline{n}$, n, and $\bar{n}$, and detail how we compute migration flows between neighborhoods and their outside options.

5.3. Two-Step Estimation Procedure

In the first step of the estimation procedure, we estimate (i) transition probabilities for the household-level state variables, (ii) households' conditional choice probabilities, and (iii) households' variable moving cost parameters.

Household Transition Distributions. To keep the dimension of the household state space manageable, we follow Almagro and Domínguez-Iino (2025) and discretize our household-level residential tenure measure into two buckets.[28] Specifically, we aggregate tenure into two buckets:

$$\bar{\tau} = \begin{cases} 1 & \text{if } \tau \leq 5 \\ 2 & \text{otherwise} \end{cases}$$

We assume that this aggregated location tenure variable evolves stochastically according to the following distribution function:

$$f^x(x(n_t, \bar{\tau}_t = 2)|n_t, x(n_{t-1}, \bar{\tau}_{t-1})) = \begin{cases} 1, & \text{if } n_t = n_{t-1} \text{ and } \bar{\tau}_{t-1} = 2 \\ g^k_{n_t}, & \text{if } n_t = n_{t-1},\ \bar{\tau}_{t-1} = 1, \text{ and } i \in k \\ 0, & \text{otherwise,} \end{cases}$$

where we estimate $g^k_{n_t}$ directly from the data:

$$\hat{g}^k_{n_t} = \frac{\sum_{i \in n_t, k} \mathbb{1}\{\tau_{xt} = 5\}}{\sum_{i \in n_t, k} \mathbb{1}\{\tau_{xt} \leq 5\}}.$$

$\hat{g}^k_{n_t}$ captures the probability that a household with tenure $\bar{\tau} = 1$ staying in the same neighborhood transitions to $\bar{\tau} = 2$ in the following period. Given that our analysis is at the census tract level, $\hat{g}^k_{n_t}$ is a sparse approximation of $g^k_{n_t}$. This is not problematic for us, however, as we restrict attention to paths originating in tracts with many low-income renters. Appendix B.5 outlines the restrictions we impose on our residential path choices.

Conditional Choice Probabilities. Researchers typically face a trade-off between sparsity and flexibility when estimating first-step conditional choice probabilities, $\hat{p}^k_{xnt}$. Our setting is no different. On the one hand, we may estimate $\hat{p}^k_{xnt}$ directly from the data by calculating the probability that a type-k household with state x_{it} moves to each neighborhood $n_t \in \mathcal{N}^c$.[29] While this approach does not impose any restrictions on the implied data-generating process, it leads to very sparse estimates of $\hat{p}^k_{xnt}$ in our setting given the number of census tracts in our largest CBSAs. On the other hand, we may impose some structure on

[28] The remaining household-level state variable is the household's residential location in the previous year. This state variable evolves deterministically depending on the residential path under consideration. We, therefore, do not need to specify any transition probability for this component of x_t.

[29] We could similarly employ any nonparametric method to compute $\hat{p}^k_{xnt}$ directly from the data.

the implied data-generating process to smooth $\hat{p}^k_{xnt}$. Given that our setting yields particularly sparse empirical choice probabilities, we choose this latter option.

We model the count of k-type households in neighborhood $\underline{n} \in \mathcal{N}^c$ choosing neighborhood $n \in \mathcal{N}^c$ between periods t and $t+1$ as being derived from a Poisson distribution.[30] We parameterize the mean of the Poisson distribution in a way that imposes no additional restrictions on the data-generating process implied by our dynamic model. Specifically, we estimate the following flexible Poisson regression separately for each k-type household:[31]

$$\log\left(\mathbb{E}\left[\left|\underline{n}^k_{\bar{\tau}} \to n^k_{\bar{\tau}}\right|\right]\right) = \underbrace{\gamma^k_{nt\bar{\tau}=1} \cdot \mathbb{1}\{\bar{\tau}=1, n=\underline{n}\} + \gamma^k_{nt\bar{\tau}=2} \cdot \mathbb{1}\{\bar{\tau}=2, n=\underline{n}\}}_{\text{Mean neighborhood utilities and CVs for stayers}} + \underbrace{\mu^k_{nt} \cdot \mathbb{1}\{n \neq \underline{n}\} + \lambda^{kc} \cdot \mathbb{1}\{\underline{n} = OO^c, n \neq \underline{n}\}}_{\text{Mean neighborhood utilities, CVs, and fixed moving costs for movers}} + \underbrace{\left(\beta^{k\prime}_d \boldsymbol{d}(n,\underline{n}) + \beta^{k\prime}_s \boldsymbol{s}(n,\underline{n})\right) \cdot \mathbb{1}\{n \neq OO^c, \underline{n} \neq OO^c\}}_{\text{Variable moving costs}} \tag{13}$$

where $\left|\underline{n}^k_{\bar{\tau}} \to n^k_{\bar{\tau}}\right|$ is the count of k-type households with aggregate tenure status $\bar{\tau}$ in neighborhood $\underline{n}$ choosing neighborhood n in period t. $\gamma^k_{nt\bar{\tau}}$ are tenure- and period-specific fixed effects capturing the value to households of staying in their current neighborhood. μ^k_{nt} is a fixed effect capturing the mean value associated with moving to neighborhood n, and λ^{kc} captures the fixed cost from moving into a city's urban core from its outside option. Finally, $\beta^{k\prime}_d \boldsymbol{d}(n,\underline{n})$ and $\beta^{k\prime}_s \boldsymbol{s}(n,\underline{n})$ are the variable moving costs specified in Section 4.3.

We estimate this Poisson model separately for each k-type, but pool observations across periods, aggregate tenures, and cities. We use the estimates from the model to predict the probability a k-type household with aggregate tenure status $\bar{\tau}$ living in neighborhood $\underline{n}$ chooses neighborhood n in each year: $\hat{p}^k_{xnt}$.[32] Equation 13 additionally identifies our variable moving cost parameters, $\beta^{k\prime}_d$ and $\beta^{k\prime}_s$. Since the cost of moving to neighborhood n differs for each k-type household depending on its current neighborhood, we can separately identify the parameters governing variable moving costs from $\gamma^k_{nt\bar{\tau}}$, μ^k_{nt}, and λ^{kc}. Variation in the distance households move within their urban core, conditional on moving, identifies $\beta^{k\prime}_d$. Similar variation for social distance identifies the cost of moving to neighborhoods socially different to one's origin neighborhood, $\beta^{k\prime}_s$.[33] We report the full set of variable moving cost estimates in Table 3.

[30] We choose to model the data as a Poisson distribution because of its ability to account for sparse data, its relative robustness to the inclusion of many fixed effects Fernández-Val and Weidner (2016), and its computational efficiency (Correia, Guimarães, and Zylkin 2020).

[31] Note that the independence of households' neighborhood moves in any one period implied by the Poisson distribution is embedded in Assumption (a); f^x for household i is independent of all other households' actions. Appendix B.3 details our Poisson regressions and shows how our regression specification does not impose additional restrictions on the neighborhood choice problem of households in our dynamic model.

[32] Many census tracts lack k-type current residents or in-migrants in a given year. Consequently, we cannot compute their conditional choice probabilities and exclude them from type-k households' choice sets. These tracts tend to be suburban or affluent.

[33] Note that our Poisson distribution assumption in Equation 13 does not impact the variable moving cost estimates we obtain here. This is because of the isomorphism between the score of the Poisson distribution and of the conditional logit

Step Two. Given our estimated variable moving cost parameters $\left(\hat{\boldsymbol{\beta}}_s^{k\prime} \text{ and } \hat{\boldsymbol{\beta}}_d^{k\prime}\right)$, our estimated conditional choice probabilities $\left(\hat{p}_{x\underline{n}nt}^k\right)$, and our estimated transition probabilities $\left(\hat{f}^x\right)$ from step 1, we may now construct the empirical analogue of Equation 12:

$$\hat{Y}_{x\underline{n}n\bar{n}t}^k = \bar{\tilde{\alpha}}_n^k + \tilde{\alpha}_t^{kc} + \tilde{\beta}_I^k \ln(\bar{I}_{nt}^k) + \tilde{\beta}_r^k \ln(\hat{r}_{\tau nt}) + \tilde{\beta}_A^k \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) + \tilde{\beta}_\tau^k \hat{\tilde{\tau}}_{xt} + \widehat{\widetilde{MC^F}}_t^{kc} + \nu_{x\underline{n}n\bar{n}t}^k, \tag{14}$$

where $\widehat{\widetilde{MC^F}}_t^{kc}$ is the difference in the fixed portion of moving costs as the variable portion is estimated in the first-step and embedded in $\hat{Y}_{x\underline{n}n\bar{n}t}^k$. We define Equation 14 fully in Appendix B.4.

To be precise about identification, it is worth unpacking the error term, $\nu_{x\underline{n}n\bar{n}t}^k$, in Equation 14. $\nu_{x\underline{n}n\bar{n}t}^k$ is comprised of both unobserved neighborhood-specific amenities, $\tilde{\xi}_{nt}^{kc}$, and expectational errors, $\tilde{e}(x_t, \bar{\omega}_t^{kc}, \bar{\omega}_{t+1}^{kc})$.[34] We consider each of these in turn starting with the unobserved neighborhood-specific amenities, $\tilde{\xi}_{nt}^{kc}$. Since we do not restrict the relationship between $\tilde{\xi}_{nt}^{kc}$ and the remaining time-varying observable neighborhood characteristics, OLS estimates of Equation 14 would be biased. The commute-adjusted earnings index, neighborhood-level housing costs, and the share of college graduates are invariably correlated with many unobserved neighborhood-level factors, such as proximity to natural amenities that we do not observe as econometricians (Lee and Lin 2018).

To distinguish between preferences for observed versus unobserved neighborhood amenities, we start by differencing Equation 14 using residential paths starting in 2017 (i.e., $t-1 = 2017$) and residential paths starting in 2010 (i.e., $t-1 = 2010$), obtaining,

$$\Delta\hat{Y}_{x\underline{n}n\bar{n}}^k = \Delta\tilde{\alpha}_t^{kc} + \tilde{\beta}_I^k \Delta\ln(\bar{I}_{nt}^k) + \tilde{\beta}_r^k \Delta\ln(\hat{r}_{\tau n}) + \tilde{\beta}_A^k \Delta\ln\left(\frac{\text{Coll}_n}{\text{Pop}_n}\right) + \tilde{\beta}_\tau^k \Delta\hat{\tilde{\tau}}_x + \Delta\nu_{x\underline{n}n\bar{n}}^k \tag{15}$$

where Δ corresponds to the difference in the associated variable between $t = 2011$ and $t = 2018$. Differencing Equation 14 removes the time-invariant component of exogenous neighborhood amenities, $\tilde{\alpha}_n^k$, and the time-invariant components of the moving costs variables, MC^{kc}, MC_{OO}^{kc}, and $\hat{\boldsymbol{\beta}}_d^{k\prime}\boldsymbol{d}(n, \underline{n})$.[35] Our main concern now is that *changes* in the observed components of households' flow utilities are correlated with *changes* in unobserved neighborhood amenities and household expectational errors. We must, therefore, construct neighborhood-level instruments, z_n, for our endogenous regressors that are orthogonal to both of these components:

$$\begin{aligned} 0 &= \mathbb{E}\left[z_n \, \Delta\nu_{x\underline{n}n\bar{n}}^k\right] \\ &= \mathbb{E}\left[z_n \left(\Delta\xi_n^k + \Delta\left(\delta \cdot \tilde{e}(x, \bar{\omega}_t^{kc}, \bar{\omega}_{t'}^{kc})\right)\right)\right] \\ &= \mathbb{E}\left[z_n \left(\Delta\xi_n^k + \Delta\left(\delta \cdot e^{\bar{V}}(n, x, \bar{\omega}_t^{kc}, \bar{\omega}_{t'}^{kc}) - \delta \cdot e^{\bar{V}}(OO^c, x, \bar{\omega}_t^{kc}, \bar{\omega}_{t'}^{kc})\right)\right)\right] \end{aligned} \tag{16}$$

(represented in Equation 9) for these continuous variables, yielding identical maximum likelihood estimation (MLE) estimates (Guimarães, Figueiredo, and Woodward 2003).

[34] Specifically, $\nu_{x\underline{n}n\bar{n}t}^k \equiv \delta \cdot \tilde{e}(x_t, \bar{\omega}_t^{kc}, \bar{\omega}_{t+1}^{kc}) + \tilde{\xi}_{nt}^k$.

[35] We estimate the remaining time-invariant parameters in a final stage. Specifically, we estimate Equation 14 conditional on the estimates from our differenced regressions and our first-step Poisson regression. We assume that the effect of residential tenure and moving costs on the likelihood of different residential paths is uncorrelated with unobserved neighborhood amenities.

where $e^{\bar{V}}(n, x, \bar{\omega}_t^{kc}, \bar{\omega}_{t'}^{kc})$ is the difference between the realized k-type ex-ante continuation value and k-type households' expectations of these continuation values, integrated over the potential realizations of the household-level states:

$$e^{\bar{V}}(n, x, \bar{\omega}_t^{kc}, \bar{\omega}_{t'}^{kc}) \equiv \sum_{x'} \left(\bar{V}(x', \bar{\omega}_{t'}^{kc}) - \mathbb{E}_{\bar{\omega}'|\bar{\omega}_t^{kc}} \left[\bar{V}(x', \bar{\omega}') | \bar{\omega}_t^{kc} \right] \right) f^x \left(x' | n, x \right).$$

In addition to being orthogonal to changes in unobserved neighborhood amenities, Equation 16 shows that our instruments must also be mean independent of changes in households' expectational errors, the second component of $\nu_{xnn'\bar{n}t}^k$. Recall Assumption (c) states that households have rational expectations over the evolution of the model's state variables. A corollary of this assumption is that the contents of households' information sets at time t are mean independent of their expectational errors (Kalouptsidi, Scott, and Souza-Rodrigues 2021). Conversely, elements of households' future information sets that cannot be predicted from their period-t information sets will be correlated with their expectational errors. For this reason, our instruments must not predict future values of $\bar{\omega}_t^{kc}$ in a way that cannot simultaneously be predicted from the information in households' period-t information sets.

To see why our instruments must be constructed using elements in households' contemporaneous information sets, consider an instrument that shocks the neighborhood-n elements of $\omega_{t'}^{kc}$ for any $t' \leq 2017$.[36] Assume that this shock cannot be predicted with the information in households' contemporaneous information sets, $\mathcal{I}_{it}$. If it is relevant, the instrument will be mechanically correlated with the realized values of households' time-t ex-ante continuation values, $\bar{V}(x', \bar{\omega}_{t'}^{kc})$, but uncorrelated with households' time-t expectations, $\mathbb{E}_{\bar{\omega}'|\bar{\omega}_t^{kc}} \left[\bar{V}(x', \bar{\omega}') | \bar{\omega}_t^{kc} \right]$, violating the exclusion restriction embodied in 16. For this reason, the instruments we detail immediately below are designed to predict changes in our endogenous variables through 2011–2018 using variation that can be predicted solely from households' 2010 information sets.

6. Identification

We estimate Equation 15 via two-stage least squares (2SLS). We construct three instruments so the parameters $\tilde{\beta}_w^k$, $\tilde{\beta}_r^k$, and $\tilde{\beta}_A^k$ are exactly identified. Our first instrument interacts neighborhood-level measures of proximity to relevant employment opportunities with nationwide trends in tradable-industry employment. This is our "job market access instrument". It helps identify $\tilde{\beta}_w^k$ by shifting neighborhoods' commute-adjusted earnings index, $\bar{I}_{nt}^k$. Our second instrument predicts changes in a neighborhood's share of college graduates by interacting measures of proximity to other neighborhoods' shares of college graduates with growing CBSA-wide demand among this population. This is our "proximity" instrument and it is particularly helpful in identifying households' preferences for endogenous amenities, $\tilde{\beta}_A^k$. By simultaneously shifting access to non-tradable employment opportunities and therefore the commute-adjusted earnings index, $\bar{I}_{nt}^k$, it also helps identify $\tilde{\beta}_w^k$. Our third instrument interacts the proximity instrument with tracts' baseline urban development levels, helping identify households' disutility from paying higher rents, $\tilde{\beta}_r^k$. Neighborhoods' baseline urban development levels proxy for their housing supply elasticities, inducing

[36] If the instrument shocks elements of $\omega_{t'}^{kc}$ for $t' > 2017$, we must also consider how the instrument affects the difference in expectational errors over time. The current example shows that we must construct z_n using variation that can be predicted from households' 2010 information sets.

variation in rent conditional on changes in neighborhood demand.

When estimating Equation 15, we include a similar set of baseline destination-tract controls, notated as $X_{nt=2010}$, as we did for Equation 1.[37] These controls ensure identification comes from how households' migration choices change when endogenous neighborhood characteristics shift among tracts that are observably similar in 2010. The evolution of exogenous neighborhood amenities, $\Delta\xi_n^k$, may therefore vary systematically with the instruments across tracts that were observably different in 2010 without threatening identification. We further include path-origin fixed effects, notated as $\delta_{\underline{n}}$. These fixed effects ensure that identifying variation in households' migration choices comes from residential paths originating from a common neighborhood $\underline{n}$. This is important given the potential for non-random changes in path origins' exogenous amenities, $\Delta\xi_{\underline{n}}^k$, to affect neighborhood exit rates and thus overall move probabilities to nearby tracts, whose characteristics may be correlated with $\Delta\xi_{\underline{n}}^k$. For example, if path origins with declining exogenous amenities are located near neighborhoods with low measures of proximity to other neighborhoods' shares of college graduates—which households in neighborhood $\underline{n}$ had a high ex-ante probability of moving to—our proximity instrument would not be exogenous. Instrument validity thus implies the following augmented moment restriction,

$$0 = \mathbb{E}\left[\tilde{z}_n \, \Delta\nu_{x\underline{n}n\bar{n}}^k\right],$$

where $\tilde{z}_n = z_n - \mathbb{E}\left[z_n \middle| X_{nt=2010}, \delta_{\underline{n}}\right]$.

Job Market Access IV. We predict changes in neighborhoods' commute-adjusted earnings indices $\left(\bar{I}_{nt}^k\right)$ by interacting baseline measures of proximity to relevant employment opportunities with nationwide trends in tradable industry employment. We build our instrumental variable from neighborhood- and industry-specific measures of employment exposure separately for both k-type households,

$$\text{EE}_{m\iota t}^k = \sum_{m \in \mathcal{N}^c \setminus n} e^{-\eta^{ck}\tau_{n,m}^k} l_{m\iota t}^k, \tag{17}$$

where $l_{m\iota t}^k$ is the number of jobs held by low-income workers of race k in workplace tract m, 6-digit NAICS industry ι, and time t; where $\tau_{n,m}^k$ is the type-specific travel time between tracts n and m; and where η^{ck} is a CBSA- and type-specific spatial decay parameter governing the importance of faraway jobs relative to closer ones for quantifying employment exposure.[38] Like $\bar{I}_{nt}^k$, we derive Equation 17 from the household workplace choice problem and describe its empirical construction in Appendix B.1.

We interact our industry-specific measures of job market access with changes in national industry

[37] Table A16 lists the baseline controls. Unlike for Equation 1, we take logs of tract college graduate shares and hedonic rents to match the form of the baseline endogenous variables. We also replace the residential market access controls with a suite of employment exposure measures that support identification through shocks to national industry growth (Borusyak, Hull, and Jaravel 2022). We discuss these controls below and in Appendix C.1.

[38] We use all low-income workers of race k when constructing these measures as Table A21 suggests there is little labor-market segmentation by homeownership status conditional on income. Cities in the model, c, correspond to CBSAs in practice.

employment,

$$\widetilde{\Delta \text{EE}}^{k}_{nt_0t} = \sum_{\iota \in \mathcal{T}} \frac{\text{EE}^{k}_{n\iota t_0}}{\sum_{\iota'} \text{EE}^{k}_{n\iota' t_0}} \cdot \ln\left(\frac{L^{c}_{\iota t}}{L^{c}_{\iota t_0}}\right), \tag{18}$$

where $L^{c}_{\iota t}$ is national industry employment in period t and industry ι excluding that employment in CBSA c which we obtain from the LBD. $\mathcal{T}$ denotes the set of tradable industries we use to predict changes in the commute-adjusted earnings index. We obtain measures of industry trade costs from Gervais and Jensen (2019) and use them to exclude non-tradable industries from $\widetilde{\Delta \text{EE}}^{k}_{nt_0t}$.[39] Recall that, to satisfy the exclusion restriction embodied in 16, households must predict national employment growth using only the information available in period t_0. Choosing t and t_0 in Equation 18 therefore involves a distinct trade-off. Measuring national employment growth over a longer horizon would strengthen the first stage, but it would also require assuming that households in period t_0 could predict industry employment growth farther into the future. We set $t_0 = 2010$ and $t = 2013$ because this is the shortest horizon that delivers sufficient first-stage power. Our instrument construction therefore implicitly exploits serial correlation in industry employment growth between 2010-2013 and 2013-2019.

Our job market access instrument builds on Severen (2023), Baum-Snow, Hartley, and Lee (2019), and Baum-Snow and Han (2024). In contrast to these papers, we disaggregate employment exposure at the 6-digit NAICS level rather than at 2-digit sectors reported in the public LEHD Origin-Destination Employment Statistics. This finer disaggregation serves two purposes. First, by focusing on tradable employment growth within each sector, we preserve the logic of the Bartik shift-share and avoid evolving consumer preferences jointly influencing households' location choices and industry employment (Bartik 1991). For example, evolving local demand for nontradable services may simultaneously influence employment in those industries and households' within-CBSA residential location choices (Liu and Yang 2021; Couture and Handbury 2020). Second, substantial variation in 6-digit NAICS industry employment growth during 2010-2013 lets us treat national industry growth—weighted by each tract's 2010 employment mix—as conditionally exogenous while allowing initial establishment locations to correlate with changing exogenous amenities (Borusyak, Hull, and Jaravel 2022).

Intuitively, identification based on our job market access instrument (18) requires that (i) households can predict national industry employment growth through 2010-2013 using only information available to them in 2010 and (ii) establishments in industries with increasing or decreasing nationwide employment are not concentrated near tracts with systematically changing exogenous amenities, conditional on our rich set of tract-level controls and path-origin fixed effects. Following Borusyak, Hull, and Jaravel (2022), we further relax (ii) by conditioning on baseline exposure to 2-digit sector employment, so only tract-weighted within-sector residual employment growth must be orthogonal to changing exogenous neighborhood amenities.

[39] We find that excluding industries among the highest 25% of trade costs in Gervais and Jensen (2019) maximizes first-stage power. We crosswalk Gervais and Jensen (2019)'s 2007 NAICS codes to 2012 NAICS codes using industry employment weights obtained from the 2010 LBD cross section which records both 2007 and 2012 NAICS codes for each establishment. For 2007-defined industries that are assigned to a single 2012-defined NAICS code or for industries that amalgamate, we construct an employment-weighted average of the 2007-defined trade costs.

We discuss these and other identification considerations thoroughly and more formally in Appendix C.1. One point of discussion worth previewing here, however, is how the source of identifying variation for job market access informs how we interpret the corresponding preference estimates. Shocks to employment exposure in tradable industries capture the value households place on both direct access to relevant jobs and on neighborhood characteristics responsive to improved tradable market access, such as consequent non-tradable retail activity, public service investments in transportation infrastructure, and the general economic vitality of a neighborhood. We believe this holistic interpretation most accords with the value low-income renters place on a neighborhood's job market access when choosing where to live.

Proximity IV. We predict changes in a neighborhood's college-graduate share by interacting its distance-weighted exposure to other neighborhoods' college-graduate shares with changes in CBSA-wide demand among this population. The instrument is motivated by research showing that neighborhood change is spatially correlated, and that, among low-income neighborhoods, those closest to high-income neighborhoods experience the greatest home price appreciation during positive labor demand shocks (Guerrieri, Hartley, and Hurst 2013; Malone and Redfearn 2018). We construct the proximity instrument using the geographic-diffusion model in Saiz and Wachter (2011), closely resembling the gentrification instruments of Glaeser, Luca, and Moszkowski (2023) and Brummet and Reed (2021). Specifically, we construct,

$$\widetilde{\Delta \text{Prox}}_{nt_0t} = \sum_{m \in \mathcal{N}^c \setminus n} e^{-\rho \tau_{n,m}} \frac{\text{Coll}_{m,t_0}}{\text{Pop}_{m,t_0}} \cdot \ln\left(\frac{\text{Coll}^c_t}{\text{Coll}^c_{t_0}}\right), \tag{19}$$

where $\tau_{n,m}$ is the straight-line distance between census tracts n and m, and ρ is a spatial decay parameter governing the importance of faraway neighborhoods' college graduate shares relative to closer neighborhoods' shares. As $\rho \to \infty$, only the neighborhoods closest to n matter for determining $\widetilde{\Delta \text{Prox}}_{nt_0t}$. Conversely, as $\rho \to 0$, every neighborhood receives the same weight in $\widetilde{\Delta \text{Prox}}_{nt_0t}$, eliminating any within-CBSA variation and rendering the instrument irrelevant. Since we do not have a good prior for ρ, we calibrate its value via k-fold cross-validation, choosing ρ that minimizes the out-of-sample sum of squared errors in a set of unweighted tract-level first-stage regressions. Multiplying our proximity measures by log-changes in CBSAs' college graduate population, $\ln\left(\frac{\text{Coll}^c_t}{\text{Coll}^c_{t_0}}\right)$, helps ensure instrument monotonicity. We again select $t_0 = 2010$ and $t = 2013$ for our instrument horizon.

Identification based on our proximity instrument (19) requires that (i) households can predict their CBSA's growth in college graduates over 2010-2013 using only information available in 2010 and (ii) a neighborhood's proximity-weighted exposure to other neighborhoods' college-graduate shares is uncorrelated with changes in exogenous amenities once we condition on our rich set of tract-level controls. This latter condition permits changes in neighborhood college shares to alter the local mix of private consumption amenities (Almagro and Domínguez-Iino 2025) or the supply of public amenities like crime (Porreca 2023) or school quality (Pearman 2018). Indeed, β^k_A captures both households' direct preference for a neighborhood's college-graduate share and their preference for the way college graduates reshape local private and public amenities, following Diamond (2016) and Su (2022). We assume, however, that conditional on observable neighborhood characteristics, exogenous amenity shocks that alter college-graduate shares are uncorrelated with $\widetilde{\Delta \text{Prox}}_{nt_0t}$. For instance, although we recompute market-access

measures each year, if public infrastructure investments are both correlated with $\Delta\widetilde{\text{Prox}}_{nt_0t}$ and raise neighborhood college-graduate shares, we cannot disentangle low-income renters' preference for improved transit access from preferences for endogenous amenities $\left(\tilde{\beta}_A^k\right)$.

We predict rent changes by interacting the proximity instrument with baseline tract-level urban development measures from Baum-Snow and Han (2024).[40] Because these development measures also enter Equation 15 directly as controls ($X_{nt=2010}$), identification proceeds analogously to a difference-in-difference estimator. We effectively compare differences in rent growth between heavily and lightly developed tracts when $\Delta\widetilde{\text{Prox}}_{nt_0t}$ is low to corresponding differences when $\Delta\widetilde{\text{Prox}}_{nt_0t}$ is high. This setup mirrors the price instruments in Diamond (2016) and permits changes in exogenous amenities to covary with the development measures (Davidoff 2016).

7. Parameter Estimates

Table 2 presents our preferred estimates from three variations of Equation 15, which alternately include log-changes in neighborhood college shares, job market access, or both characteristics. Across all specifications, the rent coefficients imply that households' preferences for lower rents translate into quantitatively large migration responses to rent changes, all else equal. In the full specification (column 3), a 10 percent increase in a tract's rent reduces the probability that a low-income renter chooses it by roughly 22 percent for Black households and 26 percent for non-Black households. The same coefficients imply that a 10 percent increase in own-tract rents, absent other potentially compensating neighborhood changes, raises the annual probability of leaving the tract by 3.8 and 4.5 percentage points, respectively. This is roughly one-fifth of baseline mobility for Black households and one-quarter for non-Black households. These migration responses incorporate moving costs through baseline choice probabilities, indicating that the frictions low-income renters face do not prevent them from responding to market rents.[41]

Conversely, a 10 percent higher college-graduate share raises the probability that a tract is chosen by about 5 percent for Black households and 13 percent for non-Black households, equivalent to annual rent premiums of roughly \$107 and \$267.[42] The corresponding responses to a 10 percent increase in each racial group's job market access are markedly larger for Black households, who become roughly 50 percent more likely to choose such a neighborhood, implying a premium of \$997 per year, compared with 17 percent and a \$349 premium per year for non-Black households. These estimates are sensitive to our specification

[40] We use the share of urban land development in the census tract in 2011. Baum-Snow and Han (2024) suggest researchers use housing supply elasticities generated from their quadratic finite mixture models which are functions of, among other tract characteristics, urban land development. As such, the two measures are highly correlated within CBSAs. In our data, and after conditioning on a rich set of tract-level controls, the raw urban development shares have a stronger first-stage relationship with subsequent rent growth than the estimated elasticities do, a pattern consistent with attenuation from estimation uncertainty in the latter.

[41] For tract n, $\partial\, p_{xnt}^k / \partial \ln r_{\tau nt} = \tilde{\beta}_r^k\, p_{xnt}^k (1 - p_{xnt}^k)$ and $\partial \ln p_{xnt}^k / \partial \ln r_{xnt} = \tilde{\beta}_r^k (1 - p_{xnt}^k)$. For the sake of these calculations, we assume the probability of moving to any particular alternative census tract is approximately 0 and we obtain the probability of staying in one's current tract from Table A15. These calculations hold continuation values fixed and therefore correspond to transitory rent changes; persistent rent changes will generate larger responses because forward-looking households incorporate the discounted stream of future rents into their neighborhood evaluations, which our welfare simulations in Section 8 capture.

[42] The willingness-to-pay calculations are implied mean values computed using the hedonically adjusted mean tract rents reported in Table 1. Baseline mean college shares are 9.54 percent for the Black sample and 9.47 percent for the non-Black sample.

choice, however. Excluding job market access (column 1) raises the implied college-share premiums to $317 and $415, while excluding college shares (column 2) raises the job market access premiums to $1,627 and $1,042. Accordingly, including job market access reduces the estimated willingness to pay for college shares by about 66 percent for Black households and 36 percent for non-Black households. Consistent with this pattern, Panel C of Figure 4 shows that changes in job market access are positively correlated with changes in college-graduate shares for low-income renters who remain in their origin neighborhoods, conditional on baseline tract characteristics. Thus, omitting job market access from households' flow utility would lead us to overstate low-income renters' valuation of the amenity changes associated with gentrification, especially for Black renters.[43]

Our estimates are similar when we proxy for changes in endogenous amenities using neighborhood shares of white college graduates. Table A12 shows that Black and non-Black low-income renters are willing to pay $24 and $191 more per year in rent, respectively, to live in a neighborhood with a 10% higher white college-graduate share. Our baseline estimates are also broadly unaffected by controlling for changes in neighborhood racial composition. This robustness check is especially useful because our job market access measures are sample-specific, and therefore helps alleviate concerns that racial heterogeneity in preferences over job market access partly reflects preferences over neighborhood racial composition. Appendix C.2 discusses these and other robustness checks in more detail.

Moving Costs. Table 3 summarizes our fixed moving cost and high residential tenure preference estimates. Recall that $\tilde{\beta}_{\tau}^{k}$ captures the additional value that incumbent renters place on remaining in their home tract if they have lived there for at least five years. Consistent with Almagro and Domínguez-Iino (2025), Diamond, McQuade, and Qian (2018), and our baseline mobility statistics in Table A15, we estimate only a moderate attachment to home census tracts among low-income renters.[44] High-tenure ($\bar{\tau}^{k} = 2$) low-income Black and non-Black renters would be willing to pay $584 and $386 more in annual rents, respectively, to remain in their home census tract than their low-tenure ($\bar{\tau}^{k} = 1$) counterparts. These moderate tenure premia mirror our relatively moderate fixed moving cost estimates, at least compared with residential choice models that imply fixed moving costs on the order of annual household income or larger, even when choices are defined over relatively fine geographies (e.g., Galiani, Murphy, and Pantano 2015; Bayer et al. 2016; Diamond, McQuade, and Qian 2018; Cook, Li, and Binder 2026).

Panel A of Table 3 reports the pseudo-median of our CBSA-specific fixed moving cost estimates, which we interpret as model-implied benchmarks rather than literal moving costs. In discrete choice migration models, fixed moving costs absorb residual location attachment needed to match observed residential mobility given the researcher's information set and therefore depend on the model's temporal and geographic partition (Howard 2026). Consistent with this interpretation, Table A14 shows that the

[43]That low-income renters remain willing to pay more to live in neighborhoods with higher college-graduate shares even after conditioning on job market access echoes recent studies (e.g., Su 2022; Almagro, Chyn, and Stuart 2024; Couture et al. 2024) that find weaker homophily by education or income than suggested by the influential estimates in Bayer, Ferreira, and McMillan (2007). We extend this evidence to a dynamic setting with forward-looking households and reveal meaningful racial heterogeneity, at least in the US context.

[44]If low-income sample households faced a constant annual hazard equal to their 2010 tract-level migration rates, the probability that Black and non-Black sample households would leave their home tract before 2019 would be roughly 84.5 and 84.1 percent, respectively. These predicted rates exceed the corresponding empirical estimates of 73.0 and 69.8 percent by 11.5 and 14.3 percentage points. We view these differences as moderate relative to the high overall mobility rates.

TABLE 2. Parameter Estimates

	(1)		(2)		(3)	
	Black	Non-Black	Black	Non-Black	Black	Non-Black
Rent $\left(\tilde{\beta}_r^k\right)$	-4.105 (0.327)	-3.476 (0.152)	-2.074 (0.355)	-2.405 (0.213)	-2.633 (0.383)	-3.133 (0.280)
College share $\left(\tilde{\beta}_A^k\right)$	2.206 (0.102) [\$317]	2.206 (0.127) [\$415]	—	—	0.485 (0.253) [\$107]	1.293 (0.390) [\$267]
Job market access $\left(\tilde{\beta}_w^k\right)$	—	—	5.197 (0.343) [\$1,627]	3.672 (0.314) [\$1,042]	4.226 (0.684) [\$997]	1.681 (0.684) [\$349]
Path-origin fixed effects	✓	✓	✓	✓	✓	✓
Baseline destination-tract controls	✓	✓	✓	✓	✓	✓
BHJ 2022 employment exposure controls	✓	✓	✓	✓	✓	✓
SW F-stat (Rent)	102.9	544.8	134.7	324.3	93.69	42.66
SW F-stat (College share)	238.9	466	—	—	78.31	53.67
SW F-stat (Job market access)	—	—	156	148.2	83.26	45.57
Kleibergen-Paap F-stat	67.84	320.9	90.69	78.96	22.86	9.29
Residential paths (000s)	58,320	71,610	58,320	71,610	58,320	71,610

Notes: Table 2 reports 2SLS estimates of the endogenous variables in Equation 15. Column (1) omits neighborhood job market access; Column (2) omits neighborhood college-graduate shares. We describe the residential paths used in estimation and their weights in Appendix B.5. Standard errors, in parentheses, are clustered by path origin. Values in square brackets translate the coefficients into households' willingness to pay for a 10 percent increase in a tract's commute-adjusted earnings index or college-graduate share, measured in annual hedonically-adjusted 2010-rents. We report the full coefficients for baseline destination-tract controls in Table A10 and describe our preferred employment exposure controls following Borusyak, Hull, and Jaravel (2022) in the main text and Appendix C.1. "SW F-stat" denotes the Sanderson–Windmeijer conditional F-statistic.

number and density of census tracts explain more than 90 percent of the variation in our fixed moving cost estimates across CBSAs. Larger, denser choice sets contain more close-substitute neighborhoods, so the model requires a larger fixed cost to rationalize the observed probability that households remain in their current tract. This sensitivity has little bearing on our welfare analysis, however. While moving costs are critical for mediating the welfare effects of gentrification, their estimated levels are not independently informative; their welfare significance depends on how they shape households' choices and option values within our model.

Panel B of Table 3 reports the variable moving cost estimates, while Figure A5 converts them into implied willingness-to-pay measures for a given physical or social distance. We find physical moving costs are quite substantial for low-income renters. Black and non-Black households would pay roughly \$7,800 and \$8,200, respectively, in annual rents to avoid a 10-mile move relative to a move of negligible distance. Social moving costs also rise with the difference in college-graduate shares between origin and destination tracts. These costs are lower for moves toward higher college-share tracts than for comparable moves toward lower college-share tracts, consistent with Garin et al. (2026), who find substantial neighborhood upgrading among low-income households following positive income shocks. Overall, however, our estimates suggest that physical distance is a more important moving friction than social distance. The implied cost of moving from a tract with a 9 percent college-graduate share, the sample average, to one with a 66 percent share is about \$1,200 for Black households and \$3,300 for non-Black households in annual rents.

TABLE 3. Moving Costs

	Black	Non-Black
Panel A: Fixed moving costs and neighborhood tenure		
High neighborhood tenure $\left(\tilde{\beta}^k_\tau\right)$	0.351 (0.073) [$584]	0.255 (0.0329) [$386]
Fixed tract moving cost $\left(MC^{kc}\right)$	-3.319 [$15,240]	-3.221 [$11,942]
Fixed CBSA moving cost $\left(MC^{kc}_{OO}\right)$	-3.368 [$15,639]	-3.348 [$12,711]
Path-origin fixed effects	✓	✓
Destination-tract fixed effects	✓	✓
Year fixed effects	✓	✓
Residential paths (000s)	423,300	506,400

Panel B: Variable moving costs	Black			Non-Black		
	Linear	Quadratic	Cubic	Linear	Quadratic	Cubic
Physical distance $\left(\beta^k_d\right)$	-.273 (.0000308)	.00589 (1.69e-06)	-.000049 (2.41e-08)	-.319 (0.0000213)	.00741 (1.13e-06)	-.0000611 (1.55e-08)
Upward social distance $\left(\beta^k_{s,\text{Up}}\right)$	.128 (.000518)	-.245 (.000585)	.0289 (.000177)	.0875 (.000358)	-.548 (.000434)	.0951 (.000138)
Downward social distance $\left(\beta^k_{s,\text{Down}}\right)$	-.179 (.000576)	-.218 (.000701)	.029 (.000225)	-.141 (.000373)	-.287 (.000424)	.0479 (.000129)
Pseudo R-squared		.9438			.9555	
PIK-Years (000s)		9,472			25,470	

Notes: Panel A reports the high neighborhood tenure coefficients and pseudo-medians of the city-specific fixed moving costs estimated from Equation 14, conditional on the coefficient estimates reported in column (3) of Table 2. We report additional pseudo-percentiles of the city-specific fixed moving costs in Table A13. We pool all cross-sections from 2010–2019 and use the same residential paths and weights as when estimating Equation 15. Standard errors, in parentheses, are clustered by origin-tract. Values in square brackets translate the coefficients into households' willingness to pay to avoid leaving their current tract, and show how this willingness to pay rises for households with high aggregate neighborhood tenure, measured in annual hedonically-adjusted 2010-rents. Panel B reports the variable moving cost coefficients estimated from Equation 13. We weight observations for each move-origin tract, year, and tenure group by the number of sample households in that cell, while each outside-option cell is assigned the maximum weight of any observation within the same CBSA, year, and tenure group. We smooth raw move counts using a three-year rolling window, as detailed in Appendix B.3. We report the total number of PIK-years included in the estimation sample. We translate the variable moving cost coefficient estimates to willingness to pay measures in Figure A5.

8. Welfare Analysis

We use the model estimates to quantify the welfare impacts of gentrification from the perspective of incumbent low-income renters in 2010. In our main analysis, for each origin tract, we compare expected welfare under the observed evolution of tract characteristics after 2010 with expected welfare under a counterfactual steady state that holds tract characteristics fixed at their 2010 levels. We then relate these welfare changes to changes in census-tract college-graduate shares, comparing tracts within the same CBSA that experienced different degrees of gentrification. This comparison measures how gentrification affected incumbent renters relative to other current or prospective renters in the same CBSA. The analysis combines our model estimates with publicly available tract-level data, allowing us to examine CBSA-level

heterogeneity in welfare effects that we could not otherwise report under Census disclosure guidelines.

As a complementary exercise, Appendix F uses households' observed neighborhood choices in the restricted data to impute measures of experienced welfare. We define experienced welfare as the present-discounted sum of neighborhood flow utilities and expected idiosyncratic utility shocks, conditional on households' observed choices. Both analyses suggest that, on average, incumbent low-income renters in gentrifying neighborhoods were not made meaningfully worse off than otherwise similar renters in less-gentrifying tracts in the same CBSA. When gentrification reduced the value of staying in one's origin neighborhood, the model-implied option value of moving largely offset the reduction.

Expected Welfare Change. We construct counterfactual steady-state welfare, $\bar{V}^k_{SS}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$, by holding tract characteristics fixed at their 2010 levels and solving for the distribution of exogenous amenities consistent with a stationary distribution of low-income renters. Appendix E.2 formally defines the stationary distribution and provides additional model details, including how we implement the welfare analysis using publicly available tract-level data. We construct expected welfare under the observed evolution of tract characteristics, $\bar{V}^k_{Gent}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$, via backward induction from the terminal year 2030, using forecasts of 2022–2030 tract characteristics described in Appendix E.1. At each step in the backward induction process, we calibrate exogenous amenities to match the observed distribution of low-income renters within each CBSA given our parameter estimates. This preserves the retrospective nature of the welfare analysis by grounding incumbent renters' estimated welfare changes in the realized evolution of low-income renters throughout each CBSA. The change in expected welfare for each k-type is then

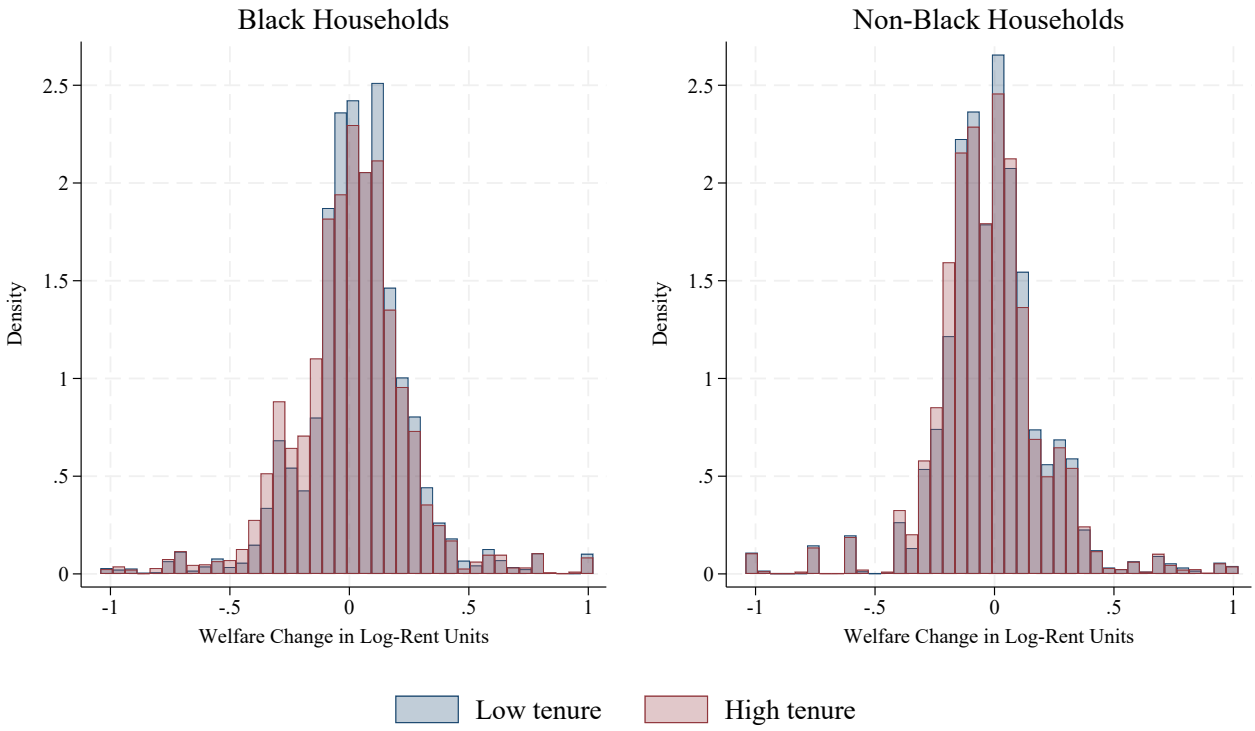
$$\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) \equiv \bar{V}^k_{Gent}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) - \bar{V}^k_{SS}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right).$$

Figure 7 plots the distribution of expected welfare differences, $\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$, demeaned from their respective average CBSA difference. A one-standard-deviation difference in within-CBSA expected welfare changes equals 0.339 and 0.325 log-rent units for Black and non-Black low-income incumbent renters with initially high aggregate tract tenure, respectively. Expressed as total discounted annual-rent-equivalent welfare differences, these values correspond to about \$2,430 and \$2,550, respectively.[45] The corresponding figures are similar for renters with initially low aggregate tract tenure, although their welfare-change distributions are shifted right relative to those of high-tenure renters. Overall, the tract-level welfare change measures are quite modest and symmetric around zero.

Figure 8 shows that expected welfare changes vary little with our gentrification measure. It plots conditional means of expected welfare changes against the log change in tracts' shares of adult college graduates, estimated from regressions that include CBSA fixed effects and controls for baseline neighborhood characteristics. A 10 percent increase in a tract's college-graduate share over 2010–2019 is associated with an approximately 0.008 increase in expected welfare for Black renters with high aggregate baseline tenure and essentially no change for their non-Black counterparts, measured in log-rent units. The same qualitative pattern holds if we exclude controls for baseline neighborhood characteristics or focus on incumbent renters with low initial aggregate tenure. Table A22 reports the corresponding coefficient

[45] Recall that our willingness-to-pay conversions use hedonically-adjusted rents measured in 2010-dollars.

FIGURE 7. Distribution of Expected Welfare Changes



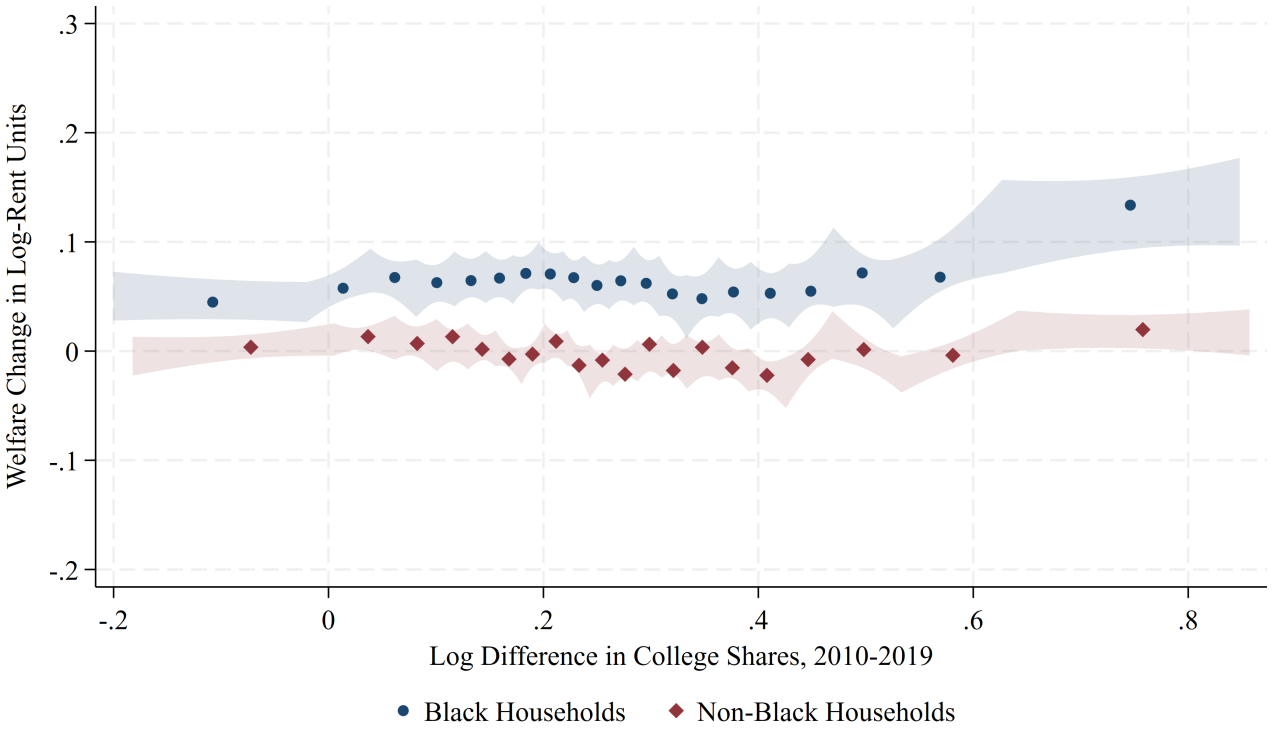
Notes: Figure 7 plots the distribution of expected welfare differences, $\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$, demeaned from their respective CBSA average expected welfare changes and normalized into log-rent units, separately for Black and non-Black renters. For each observation, the plotted value equals the change in expected welfare minus the mean change in expected welfare in the same CBSA. The navy bars show low-aggregate-tenure sample renters ($\bar{\tau}^k = 1$) and the maroon bars show high-aggregate-tenure sample renters ($\bar{\tau}^k = 2$). The sample includes tracts in the bottom quartile of the tract-level college-graduate share distribution in 2010 that are closer to their metropolitan division's CBD than 50 percent of all tracts in the same metropolitan division. We drop tracts below the 1st and above the 99th percentiles of the within-CBSA welfare change distribution. Figure A12 also plots these distributions alongside the corresponding distributions before demeaning by CBSA average welfare changes.

estimates and standard errors.

Option Value of Moving. Why are within-CBSA differences in expected welfare changes so modest, and why do they vary so little with our measure of gentrification? Relative changes in expected welfare within a CBSA depend not only on how a renter's origin tract changed, but also on how the value of moving changed. If a neighborhood changed in a way that an incumbent renter did not value, then the welfare loss is moderated by the renter's ability to move to another neighborhood, net of moving costs. We find that the option value of moving largely explains why expected welfare changes differ so little across tracts in the same CBSA.

Appendix E.2 shows that changes in expected welfare can be formally decomposed into the present discounted value of changes in the payoff from staying in one's origin tract and the present discounted

FIGURE 8. Gentrification and Changes in Expected Welfare



Notes: Figure 8 plots conditional means of expected welfare differences, $\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$, normalized into log-rent units, separately for Black and non-Black renters. The log difference in college shares is measured for low-income renters' 2010 origin tracts. Both series are estimated from regressions that include CBSA fixed effects and pool high- and low-aggregate-tenure groups, and are weighted separately by Black and non-Black low-income tract renter shares in 2010 (Cattaneo et al. 2024). Each series is shifted to match the corresponding unconditional mean welfare difference for that race. Shaded areas show the corresponding 95 percent uniform confidence bands. The sample includes tracts in the bottom quartile of the tract-level college-graduate share distribution in 2010 that are closer to their metropolitan division's CBD than 50 percent of all tracts in the same metropolitan division. We drop tracts below the 1st and above the 99th percentiles of the within-CBSA gentrification distribution. Table A22 reports estimates from the corresponding regressions, separately by aggregate tenure group.

value of changes in the option value of moving,

$$\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)=\underbrace{\text{PDV}\Delta\bar{u}^k_n\left(x(n,\bar{\tau}),\bar{\omega}^{kc}\right)}_{\text{PDV of payoff from staying}}-\underbrace{\text{PDV}\Delta\ln\left(p^k_n\left(x(n,\bar{\tau}),\bar{\omega}^{kc}\right)\right)}_{\text{PDV of option value of moving}}.$$

Here, Δ represents the difference between each welfare component computed under the observed evolution of tract characteristics since 2010 and the corresponding component computed under the counterfactual steady state. The option value of moving in period t can also be written as

$$-\ln\left(p^k_n\left(x(n,\bar{\tau}_t),\bar{\omega}^{kc}_t\right)\right)=\ln\left(1+\sum_{n'\neq n}\exp\left(v^k_{n'}\left(x(n,\bar{\tau}_t),\bar{\omega}^{kc}_t\right)-v^k_n\left(x(n,\bar{\tau}_t),\bar{\omega}^{kc}_t\right)\right)\right).$$

This expression shows that the option value of moving in period t is a function of the value of moving to alternative neighborhoods, n', relative to the value of staying in the origin neighborhood, n, for another

year. It is therefore increasing in the value of alternative tracts in the same CBSA relative to the origin tract. If gentrification lowers the value of staying in one's origin neighborhood more than it lowers the value of moving to alternative neighborhoods, then the option value increases with an origin tract's degree of gentrification. If, instead, conditional values fall at similar rates across tracts in the same CBSA under the observed evolution of tract characteristics relative to the counterfactual steady state, then the option value remains nearly unchanged, and incumbent residents bear the welfare loss. Importantly, a decline in the origin tract's conditional value has a meaningful offsetting welfare effect only when there are attractive alternative neighborhoods available, net of moving costs.[46] The option value of moving can therefore compress within-CBSA differences in welfare changes when renters in adversely affected origin tracts have attractive alternatives net of moving costs.

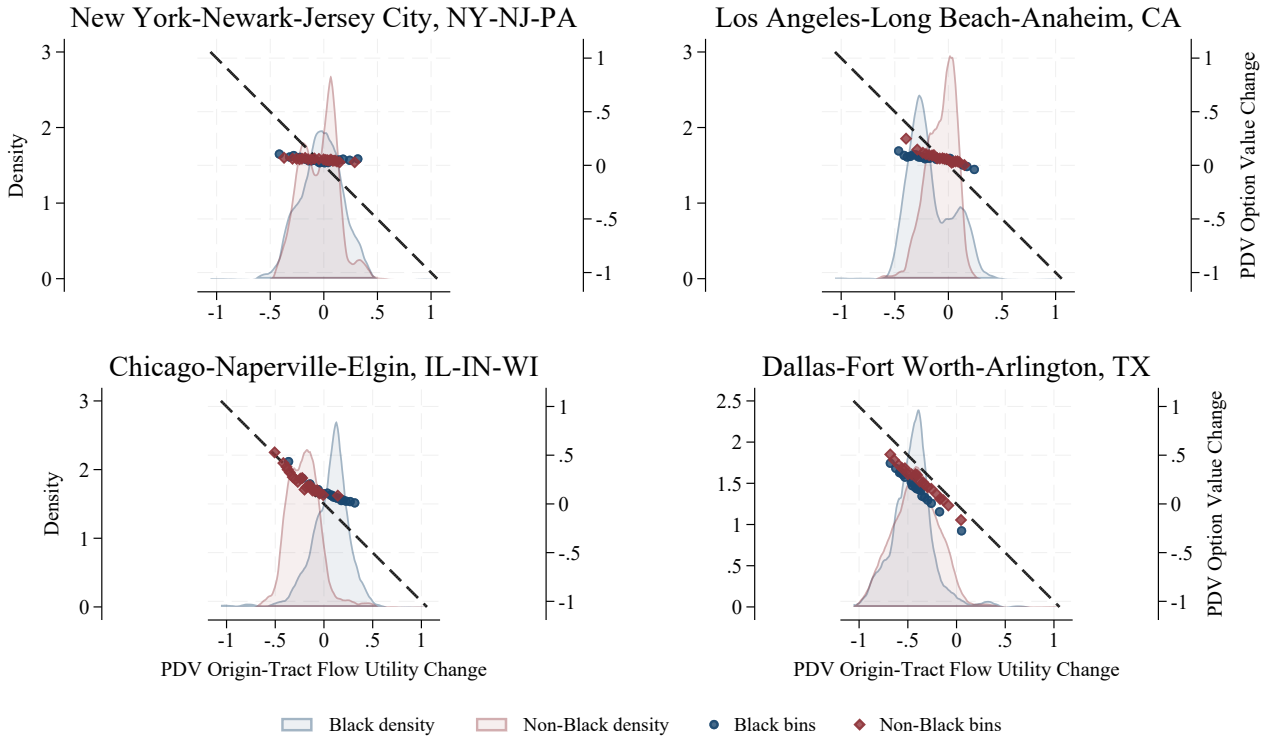
Figure 9 shows how changes in the option value of moving relate to changes in the payoff from staying in the origin tract among highly gentrifying tracts in the four largest CBSAs in the United States. The x-axis reports the present discounted change in the payoff from staying in the origin tract from 2010 onward for a low-income incumbent renter. The y-axis reports the corresponding present discounted change in the option value of moving. Each plot shows conditional means of option-value changes for a given change in the payoff from staying, restricting the sample to tracts in the top quartile of the distribution of log changes in college-graduate shares. The dashed 45-degree line separates tract-level welfare gains from losses relative to the counterfactual steady state that holds tract characteristics fixed at their 2010 levels. Points above the line correspond to tracts where the change in the option value of moving is large enough, relative to the change in the payoff from staying, to leave expected welfare higher than in the steady-state counterfactual. When the payoff from staying falls, these are tracts where the option value of moving fully offsets the loss from remaining in one's origin tract.

Changes in the option value of moving are inversely related to changes in the payoff from staying in one's origin tract, though the strength of this relationship differs meaningfully across CBSAs. In New York-Newark-Jersey City and Los Angeles-Long Beach-Anaheim, changes in the option value of moving are only weakly inversely related to changes the payoff from staying among gentrifying tracts. This weak relationship is consistent with changes in the value of relevant alternative tracts being correlated with changes in the value of incumbents' origin tracts, leaving little change in the relative value of moving. It is also consistent with higher moving frictions. In New York and Los Angeles, slower rent growth for incumbent renters under binding rent-control ordinances acts as an additional cost of moving, further attenuating the option value of moving.

In Chicago-Naperville-Elgin and Dallas-Fort Worth-Arlington, by contrast, changes in the option value of moving are strongly inversely related to changes in the payoff from staying. This suggests that when origin tracts gentrified in ways incumbent renters disliked, relevant alternative neighborhoods remained available and helped offset the declining value of staying. These results imply that the option value of

[46] Holding the values of alternative neighborhoods fixed, $\partial\left[-\ln\left(p_n^k\left(x(n,\bar{\tau}_t),\bar{\omega}_t^{kc}\right)\right)\right]\Big/\partial\left[v_n^k\left(x(n,\bar{\tau}_t),\bar{\omega}_t^{kc}\right)\right] = -\left(1-p_n^k\left(x(n,\bar{\tau}_t),\bar{\omega}_t^{kc}\right)\right)$. A decline in $v_n^k\left(x(n,\bar{\tau}_t),\bar{\omega}_t^{kc}\right)$ raises the option value in proportion to the model-implied probability of moving from the origin tract. The moderating effect of the outside option is therefore largest when alternative neighborhoods are sufficiently attractive net of moving costs that the renter is unlikely to remain in their origin tract irrespective of changes to the payoff from staying.

FIGURE 9. Option Value Changes in Gentrifying Neighborhoods



Notes: Figure 9 plots changes in the option value of moving against changes in the payoff from staying in a gentrifying origin tract in 2010. The x-axis reports the present-discounted change in origin-tract flow utility, and the y-axis reports the present-discounted change in the option value of moving. Both measures are expressed in hedonically adjusted 2010 log-rent units. We plot tracts in New York-Newark-Jersey City, Los Angeles-Long Beach-Anaheim, Chicago-Naperville-Elgin, and Dallas-Fort Worth-Arlington that are in the bottom quartile of the 2010 nationwide college-graduate share distribution, the top quartile of the national distribution of log changes in college-graduate shares over 2010–2019, and that are closer to their metropolitan division's CBD than 50 percent of all tracts in the same metropolitan division. Scatter points are conditional means of the change in option value within 20 quantile bins of origin-tract flow-utility changes, constructed separately by race and weighted by their respective tracts' low-income renter populations. The dashed line plots where the change in the option value of moving exactly offsets the change in the payoff from staying.

moving provided a stronger buffer against the negative welfare consequences of gentrification in Chicago and Dallas than in New York or Los Angeles.

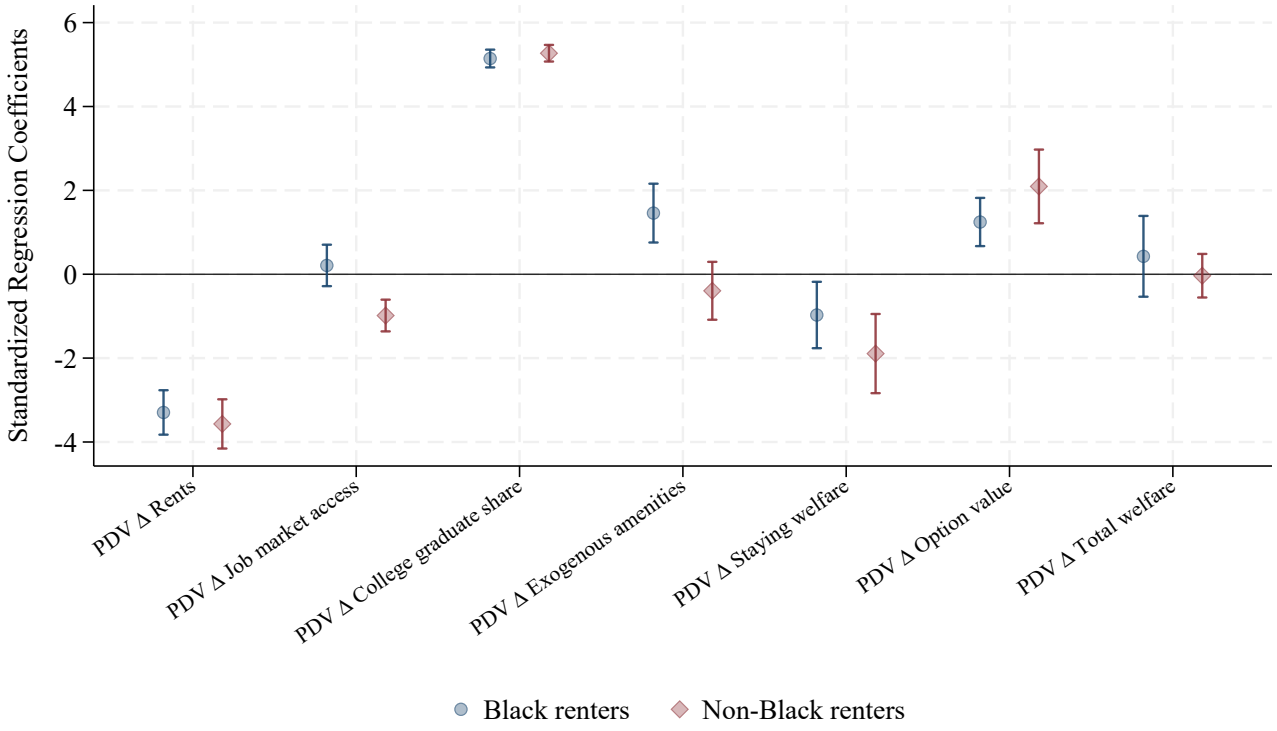
Perhaps surprisingly, Figure 9 also shows only moderate variation in the payoff from staying in one's origin tract. In New York-Newark-Jersey City, the difference in the payoff to staying between the 1st and 99th percentiles of the distribution plotted in the figure is only 0.904 log-rent units for Black incumbent renters and 0.757 log-rent units for non-Black incumbent renters. The corresponding ranges in Chicago-Naperville-Elgin are 1.580 and 1.036 log-rent units. To better understand the moderate variation in the payoff from staying in gentrifying tracts, we decompose its present discounted value into changes in job market access, rents, college-graduate shares, and exogenous amenities:

$$\text{PDV}\Delta\bar{u}_n^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}\right) = \underbrace{\text{PDV}\tilde{\beta}_I^k\Delta\ln\left(\bar{I}_n^k\right)}_{\text{job market access changes}} + \underbrace{\text{PDV}\tilde{\beta}_r^k\Delta\ln\left(r_{\bar{\tau}n}\right)}_{\text{rent changes}} + \underbrace{\text{PDV}\tilde{\beta}_A^k\Delta\ln\left(\frac{\text{Coll}_n}{\text{Pop}_n}\right)}_{\text{amenity changes}} + \underbrace{\text{PDV}\Delta\xi_n^k}_{\text{exogenous changes}}.$$

We then plot in Figure 10 standardized coefficients from regressions of each of these components on our

measure of gentrification, estimated separately for each k-type low-income renter. For each component j, we estimate $C_{nj}^k = \alpha_c^k + \gamma_j^k \Delta \ln(\text{Coll}_n/\text{Pop}_n) + \varepsilon_{nj}^k$, where α_c^k are CBSA fixed effects and C_{nj}^k includes the rent, job market access, college-share, exogenous amenity, payoff-from-staying, and option-value components of total welfare change.[47]

FIGURE 10. Gentrification and Components of Welfare Change



Notes: Figure 10 plots coefficients from component-specific regressions of welfare changes on the 2010–2019 log change in tract college-graduate share. Regressions are estimated separately for Black and non-Black low-income renters in tracts initially in the bottom quartile of the nationwide 2010 college-graduate share distribution and that are also closer to their metropolitan division's CBD than 50 percent of all tracts in the same metropolitan division, excluding tracts outside the 1st and 99th percentiles of the within-CBSA distribution of the gentrification measure. Low- and high-aggregate-tenure types are pooled within each type-k group. Coefficients are weighted by their respective tract-level low-income-renter populations and standardized by the within-CBSA standard deviation of the corresponding component, computed separately by renter type. 95 percent confidence intervals are based on standard errors clustered at the CBSA level. Table A24 reports the corresponding coefficient and standard errors alongside the R-squared values.

The standardized regression coefficients plotted in Figure 10 show that measured gentrification has large counteracting impacts on the components on the payoff from staying in one's origin neighborhood. A one-log-point increase in a tract's college-graduate share between 2010 and 2019 is associated with 5.27 and 5.14 within-CBSA standard deviation increases in the present discounted college-graduate share component for non-Black and Black incumbent renters, respectively. The corresponding rent coefficients

[47] To make coefficient estimates comparable across components, we report $\widehat{\gamma}_j^k / \widehat{\sigma}_{j,c}^k$, where $\widehat{\sigma}_{j,c}^k$ is the within-CBSA standard deviation of component j for type k. The plotted coefficients therefore give the change in component j, in within-CBSA standard-deviation units of that component, associated with a one-unit increase in the log change in college-graduate share. This normalization is especially useful because the market-access measures we construct from public data are on a different scale than the measures we construct with the Census microdata. Since we calibrate exogenous amenities to match the observed distribution of low-income renters, changing the scale of market access reallocates variation between the market-access and exogenous-amenity components. It does not, however, impact tract mean utility levels.

are -3.57 and -3.30, suggesting that higher rents offset much of the welfare gain associated with higher college-graduate shares. The job market access coefficients are smaller, at -0.99 for non-Black renters and 0.21 for Black renters. The coefficients on exogenous amenities are also smaller, at -0.40 and 1.46. The positive exogenous-amenity coefficient for Black renters suggests that, conditional on observed rents, market access, and college-graduate shares, gentrifying tracts also experience unobserved changes that are valued positively by Black incumbent renters.

Together, these components imply a moderate negative impact of gentrification on the present discounted payoff from staying in one's origin tract. A one-log-point increase in a tract's college-graduate share between 2010 and 2019 is associated with 1.89 and 0.97 within-CBSA standard deviation decreases in staying welfare for non-Black and Black incumbent renters, respectively, though the standard deviation of our gentrification measure is only 0.28 log points. These overall negative impacts on staying welfare are nevertheless more than offset by positive impacts on the option value of moving. As a result, the total model-implied welfare impact of gentrification is statistically indistinguishable from zero for both Black and non-Black renters, helping explain the null relationship in Figure 8.

Experienced Welfare. In Appendix F, we complement the main welfare analysis by using the internal census microdata to construct a measure of welfare using households' observed residential histories from 2010 to 2019. The analysis replaces model-implied mobility responses and exposure to tract characteristics with the tracts households actually occupied, while using the estimated preference parameters to impute the value of those tract choices. We term the constructed measure "experienced welfare". This complementary analysis reinforces the findings from the main welfare analysis. Relative differences in imputed welfare along observed residential histories among households in the same CBSA are largely unrelated to measured gentrification. We similarly find that rent increases reduce welfare in gentrifying origin tracts, while changes in college-graduate shares, job market access, and other neighborhood characteristics partly offset those reductions. Omitting neighborhood tenure or moving costs when imputing the value of households' observed residential histories has little effect on the estimates, consistent with the limited mobility responses to gentrification documented in Section 3.2.

Cross-CBSA Comparisons of Welfare Changes. Appendix E.3 extends the main welfare analysis to assess differences in expected welfare changes across CBSAs. Assumption (b) in Section 5.1 normalizes the utility of the outside option to a separate constant for each CBSA, so expected welfare levels are not directly comparable across CBSAs. Conditional on our parameter estimates, however, we can compare expected welfare changes, $\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$, across CBSAs under the additional assumption that the value of each CBSA's outside option changes by the same amount over our analysis period.[48] Imposing this assumption, Figure A12 shows how including between-CBSA variation affects the distribution of expected welfare changes by comparing the unconditional distribution with the corresponding within-CBSA distributions from Figure 7. The distributions are similar for non-Black households, but the unconditional distribution for Black households has a meaningfully larger variance, suggesting that Black households' expected welfare changes also reflect changes in broad, CBSA-wide characteristics.

[48] Using the notation from Section 5.1, this assumption requires $\underset{t\to t'}{\Delta}\alpha^{kc} = \underset{t\to t'}{\Delta}\alpha^{k}$, for all c, t, and t'.

Figure A13 re-estimates the specifications underlying Figure 8 with and without CBSA fixed effects. Expected welfare changes remain unrelated to measured gentrification when we exclude CBSA fixed effects. Figure A14, however, shows a different pattern for rent changes. With CBSA fixed effects, expected welfare changes are very weakly related to changes in log tract rents. Without CBSA fixed effects, higher log tract rents are associated with modest but highly statistically significant decreases in expected welfare, especially for Black households. Among incumbent households with high initial aggregate tenure ($\bar{\tau} = 2$), a one-log-point increase in tract rents is associated with expected welfare decreases of 0.959 and 0.267 log-rent units for Black and non-Black households, respectively. One likely explanation, consistent with CBSA-level variation in the option value of moving shown in Figure 9, is that some CBSAs experience more broad-based changes in tract characteristics, limiting incumbent renters' ability to readily substitute across tracts as their origin tract rents increase.

9. Discussion

Our central finding is that expected welfare changes among incumbent low-income renters vary little with measured gentrification across tracts with initially low college-graduate shares in the same CBSA. The option value of moving largely increases for incumbent low-income renters when their origin tracts change in ways they do not value. This finding suggests that gentrification affects incumbent low-income renters primarily by changing the characteristics of other neighborhoods in their choice sets. It helps reconcile limited mobility responses to within-city differences in gentrification with research pointing to potentially large welfare effects from gentrification for prospective low-income renters. It also suggests that policies focused narrowly on preserving neighborhood incumbency may be poorly aligned with incumbent renters' welfare if they reduce the availability or affordability of alternative neighborhoods elsewhere in the metro area. Because our baseline utility normalization holds the value of the outside option fixed within each CBSA, we interpret comparisons of welfare effects across CBSAs cautiously.

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Appendix A. Sample and Variable Construction

A.1. Data Sources

Residential Locations. We record residential location histories using two complementary sources developed by the U.S. Census Bureau: the Master Address File-Auxiliary Reference File (MAF-ARF) and, post 2011, the Longitudinal Employer-Household Dynamics' Individual Characteristics File (LEHD-ICF). Both rely on the Census Bureau's Master Address File (MAF), a continuously updated inventory of all known U.S. housing units that serves as the sample universe for, among other surveys, the Decennial Census and the American Community Survey (ACS). Each address in the MAF is assigned a stable identifier (MAFID) and includes detailed physical and census-geography attributes. Each year, the Census Bureau commits the live MAF into a research extract known as the Master Address File Extract (MAF-X), which contains a preferred address, the address' MAFID, geographic identifiers, and some property characteristics for every housing unit (Sullivan and Genadek 2024; Graham, Kutzbach, and Sandler 2017).

The MAF-ARF is an annual, person-level panel starting in 2000 that links each individual who can be assigned a Protected Identification Key (PIK) to the MAFID of the housing unit where that person appears in federal administrative records during the reference year.[49] It draws PIK-level address data from sources including IRS Forms 1040 and 1099, the Medicare Enrollment Database, Indian Health Service patient records, Selective Service registrations, and Housing and Urban Development's (HUD) Public and Indian Housing and Tenant Rental Assistance Certification System databases. Prior to 2012, all observed addresses per PIK-year are retained in the MAF-ARF. From 2012 onward, one is selected at random. In 2019, the MAF-ARF covered approximately 305 million individuals, about 93 percent of the U.S. resident population (Sullivan and Genadek 2024).

Post 2011, the LEHD-ICF acquires its residential locations from internal versions of the LEHD Residence Candidates File (RCF).[50] The RCF construction and sources broadly follow those of the MAF-ARF, but it models residence location probabilistically given address conflicts in the source files (Graham, Kutzbach, and Sandler 2017). The LEHD-ICF records only the most preferred location for each year in the RCF (Graham et al. 2022). We show in Appendix A.2 that this choice is associated with higher tract-level location matches with the ACS than for the MAF-ARF.

We construct PIKs' residential location histories by combining the MAF-ARF and LEHD-ICF. To do so, we first link PIKs across the 2000-2019 MAF-ARF and, separately, the 2012-2019 LEHD-ICF. We randomly retain one MAFID for PIKs with multiple MAFIDs in a given pre-2012 MAF-ARF year. We then smooth PIKs' MAFID histories separately in the LEHD-ICF and MAF-ARF by first carrying the nearest available MAFID forward or backward to fill any missing MAFID-year records. When a PIK appears at the same MAFID within a ten-year window, we treat them as residing continuously at that

[49] A PIK is an anonymized identifier assigned by the Census Bureau's Person Identification Validation System, which matches name, date of birth, sex, and, when available, Social Security number to the SSA's master file of person records—the Numident (Wagner and Layne 2014).

[50] Before 2012, the LEHD-ICF acquires its residential location histories from the Composite Person Record (CPR) source, produced by the Center for Administrative Records Research and Administration. We are unable to access address identifiers (MAFIDs) from the CPR, however, and the CPR has been found to differ in important ways from the MAF-ARF (Graham, Kutzbach, and Sandler 2017). We therefore rely solely on the MAF-ARF for residential locations prior to 2012.

MAFID during the intervening years. We truncate PIKs' residential histories whenever the Numident records a person's death before their history's nominal end. Finally, we overlay the smoothed MAF-ARF and LEHD-ICF residential histories, taking the LEHD-ICF as authoritative in any conflict and conduct the same smoothing procedure as we did separately for the MAF-ARF and LEHD-ICF. These choices help ensure the high levels of residential mobility we observe in our panel are not caused by sampling error in the Census residential address infrastructure.

Property Characteristics. We obtain property-level characteristics from the following sources: the 2017 vintage of CoreLogic's Tax History and Multiple Listing Service (MLS) files, which both span years 2000-2016, though coverage is sparse in earlier years; the 2019 and 2020 Black Knight Property Assessment files for these same years; the internal household-level ACS for years 2005-2021; and the 2019 MAF-X. From these sources, we primarily retain information related to each unit's occupancy status (rental vs. owner-occupied).

The Census Bureau has added property-level tax assessment data aggregated from county and township assessment records by CoreLogic and Black Knight into its Integrated Research Environment (IRE). These property assessments have been merged with the MAF where possible (Brummet 2014; Clark et al. 2023). For properties merged to the MAF, we record the reported occupancy status in assessed years, which we understand is determined by comparing property owners' mailing addresses to the assessed property addresses (Brummet et al. 2016). The Census Bureau has similarly integrated CoreLogic's MLS aggregation into the IRE. We record from the MLS data which properties were classified as rental units at listing time.

From the household-level 2005-2021 ACS, we record whether current residents reported owning or renting the surveyed unit. We also record from the ACS gross rents and other property characteristics like home size, which we use to compute tract-level hedonic rents, detailed in Appendix A.3. Finally, in addition to geographic identifiers used to map households' residential locations to 2010-delineated Tabulation Census Tracts, we record from the MAF-X whether each property belonged to a multi-unit building. We discuss in Appendix A.2 how we combine these data to assign annual rental unit indicators to MAFIDs.

Employment, Earnings, and Workplace Establishments. We obtain person-level employment and earnings outcomes, as well as workplace-establishment characteristics for select states and for years 2001-2019 from the S2021_R2022Q4 snapshot of the administrative LEHD database (Graham et al. 2022).[51] We draw person-level employment and earnings data from the Employment and Job History Files (LEHD-EHF and LEHD-JHF). These files contain quarterly earnings for each job covered by state unemployment insurance programs, covering roughly 95% of employment in the US.[52] Earnings and employment are

[51]States must independently approve external Census projects' access to their LEHD employment and jobs data. Our project has access to the following states' data: Arizona, California, Colorado, Connecticut, Delaware, Indiana, Iowa, Kansas, Maine, Maryland, Nebraska, Nevada, New Jersey, New Mexico, New York, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, South Dakota, Tennessee, Texas, Utah, Virginia, Washington, Wisconsin, and Wyoming.

[52]Notable exclusions from state UI coverage include independent contractors, the unincorporated self-employed, and Federal government workers (Graham et al. 2022).

recorded by state UI programs at the firm level. The Census Bureau additionally imputes workers' physical establishments for workers employed at multi-establishment firms. To do so, it uses a probabilistic model to assign workers to specific establishments based on the distance between the worker's residence and candidate workplaces as well as the relative size of each establishment within multi-establishment firms sourced from the Quarterly Census of Employment and Wages (QCEW) (Abowd et al. 2009). The model is trained on data from Minnesota, which reports establishment employment information for workers. The LEHD-JHF provides outcomes of ten probabilistic imputations, which we use to assign multi-establishment firm workers to specific establishments based on the modal imputation, splitting ties randomly. We sum earnings over quarters and job spells to obtain person-level annual earnings among jobs covered by the LEHD. We assign each worker to the establishment where they earned the most in a given year. Lastly, we obtain establishments' 6-digit NAICS industry codes and census tract locations from the Employer Characteristics File (LEHD-ECF), which is sourced primarily from the QCEW.[53]

Non-Economic Person Characteristics. We obtain non-economic characteristics from the LEHD-ICF, the person-level 2005-2021 ACS, and the SSA Numident file. The LEHD-ICF provides demographic information, including race, ethnicity, gender, date and place of birth, and education, compiled from the SSA Numident, the 2000 and 2010 Decennial Censuses, and the ACS. Because many person-characteristics in the LEHD-ICF are missing from the underlying source files, the Census Bureau imputes these values with a classification and regression tree described in Graham et al. (2022). We track which characteristics are imputed and exclude imputed values when constructing our tract-level sociodemographic aggregates, which we show in Appendix A.3 more closely match the ACS as a result. For individuals observed in the ACS, we obtain additional characteristics, including parental status and commute time. We also use the ACS to fill in missing LEHD-ICF values. When the two sources disagree, we give precedence to the ACS.

Low-Income Housing and Rental Support. We collect information on low-income housing and rental support from three main sources: the US Department of Housing and Urban Development's (HUD) Picture of Subsidized Households (POSH) database; HUD's national Low-Income Housing Tax Credit (LIHTC) database; and hand-collected rent control ordinances.

We construct annual counts of units subsidized by HUD for every 2010-delineated census tract from 2009-2019. We use HUD's publicly available Picture of Subsidized Households (POSH), which includes annual tract-level counts of Housing Choice Vouchers, Public Housing, Project-Based Rental Assistance, and properties under the Section 236 Preservation Program, wherein property owners may refinance on favorable terms conditional on maintaining cost-based rents. For years 2009-2013, we sum program-level units within each tract. From 2014 onward, we adopt HUD's tract-level totals. We crosswalk the pre-2012 files from 2000 to 2010 tract boundaries weighting tracts by rental-unit counts. We use these cross-sectional counts as controls in our descriptive analyses and hedonic rent regressions.

We use HUD's national LIHTC database to construct annual tract-level counts of units that are likely

[53]Establishments' tract locations are recorded using 2020 Census tract boundaries. We crosswalk these tracts to 2010 Census tract boundaries randomly using probability weights based on land overlap measures between the 2010 and the 2020 boundaries. We crosswalk establishments to their same 2010 tract boundaries each year.

subject to affordability requirements during 2008-2019. We record each building's allocation year, total number of low-income units, 2010 tract identifier, flags for Qualified Census Tract (QCT) status and whether HUD was still monitoring the property in 2023. We deem a building's units subject to affordability requirements if (i) it was allocated within the past 15 years, (ii) it is HUD-monitored as of 2023, or (iii) it was allocated 16-30 years ago and lies outside a QCT, a proxy for extended-use compliance; pre-1990 projects are excluded unless still monitored.[54] We multiply HUD's reported low-income unit count by this affordability indicator and sum across buildings to obtain tract-level totals for each year in 2008-2019. As with the POSH, we use cross-sectional LIHTC counts as controls in our descriptive analyses and hedonic rent regressions.

We additionally construct annual, tract-level estimates of the share of low-income renter households living in rent-controlled units. First, we hand-collect details on over 600 rent-control ordinances covering all incorporated and unincorporated places in our analysis for years 2000-2019. We record each ordinance's years of operation, maximum allowable annual rent increase, and building-age and size eligibility rules. We then merge these ordinance details with the 2005–2019 ACS household-level microdata, which contain the building characteristics we need to determine whether a unit is subject to rent control. We assign units as rent-controlled when their attributes satisfy the applicable ordinance criteria. We finally compute for each tract and year, the share of low-income renters in our sample occupying such units, smoothing the data over three-year bands.

A.2. Sample Construction

Identifying Rental Units. Our focus on renters is possible given our distinct ability to classify housing-unit tenure through linking the CoreLogic, Black Knight, ACS, and MAF-X databases. To this end, we build a housing tenure indicator panel in the following stages. (i) Using the 2005-2021 household-level ACS, we mark a property as rental or owner-occupied according to the interviewed household's survey response. We also use this data as our "ground truth" to assess our subsequent rental assignment imputations. (ii) We label a unit as rental if its CoreLogic MLS record shows a "for-rent" listing that year. When the ACS and MLS disagree, we treat the unit as owner-occupied in the survey year and rental in the following year, capturing cases in which an owner subsequently lists the home for rent. (iii) To expand our coverage of rental units, we use absentee-owner status from the CoreLogic Tax History files to classify MAFIDs as rental units and owner-occupier status to classify MAFIDs as non-rental units. We find that 34 percent of properties classified by CoreLogic as absentee-owned are reported as owner-occupied in the ACS data. (iv) We further classify units as owner-occupied when the Black Knight assessment files identify them as such.[55] After interpolating the resulting housing tenure flags, first forward and then backward, to fill missing years within MAFIDs, we assign any still-unclassified MAFID-years as rental units in those years if the MAFID is marked as a multi-unit building in the MAF-X. This imputation falsely labels owner-occupied units as rental units in just 9.7 percent of observations in the ACS. Overall, we estimate that 16.4 percent of all properties we classify as rental units are actually owner-occupied. The false-positive rate in our final sample is likely significantly lower, however, because this estimate is based on all MAFIDs

[54] We form these restrictions based on our reading of Khadduri et al. (2012).

[55] We were unable to distinguish absentee-ownership status from missing data in the Black Knight database.

rather than those occupied by low-income households, who are more likely to be renters ex ante.[56]

Annual Samples. We define annual samples for each year between 2010 and 2019. To do so, we merge PIKs' residential histories, employment and earnings, and non-economic characteristics from the aforementioned datasets. For each year between 2010 and 2019, we define samples by implementing the following restrictions:

(a) Persons must be between 25 and 65 years old and not residing in a group quarter, as labeled by the MAF-X;
(b) Persons must be residing in one of the 100 largest CBSAs located within the LEHD states we can access;[57]
(c) Persons must be household heads, defined as PIKs with the highest earnings in the MAFID-year, breaking ties randomly;
(d) Household heads must have positive earnings for the year;[58]
(e) Household heads must be residing in a MAFID with an adjusted household income in the bottom tercile of their CBSA and decadal age-band's income distribution;[59]
(f) Household heads must never earn more than $150,000 (2010 dollars) in a single year during our analysis period.[60]

We finally subset household heads based on their reported race (Black and non-Black) and whether their MAFID is a rental unit in the sample year. We record the outcome of this sample selection procedure in Table A1. The structural estimation uses data formed by stacking the 2010-2019 cross-sectional samples, whereas the descriptive and welfare analyses track each household head from its sample year forward.

Sample Validation. Following Sullivan and Genadek (2024), we validate our residential panel against the ACS. The leftmost columns of Table A2 show the share of sample households whose census tracts match those in the ACS. These tract-level match rates are remarkably high, especially considering households' high mobility rates and point-in-time survey variation. Match rates are higher from 2012 onward, likely

[56]We compute the 16.4 percent rate by applying the estimated false-positive rates to the rounded number of new rental assignments at each assignment step. The ACS directly identifies 500,000 rental properties, which we treat as correctly classified. The MLS data add 473,000 rental assignments with a 37.1 percent false-positive rate, the CoreLogic Tax History files add 13.047 million rental assignments with a 33.9 percent false-positive rate, and and the MAF-X multi-unit imputation adds 34.63 million rental assignments with a 9.7 percent false-positive rate.

[57]We include a CBSA only when at least 95% of its resident workers commute to jobs in states for which we have LEHD data, measured using work- and home-tract locations from the person-level ACS. This threshold ensures our LEHD-based employment counts yield accurate market-access measures for every CBSA in the analysis.

[58]We restrict to workers with positive earnings in a year given the innate difficulty of distinguishing between poor and rich non-workers, self-employed workers, and workers employed by firms not covered by the LEHD.

[59]We consider four equally-sized age-bands for each sample year containing household heads aged 25-34, 35-44, 45-54, and 55-65. We follow the Census Bureau's equivalence-scale guidelines to compute adjusted household income. We calculate adjusted household income for childless households as $\text{THI}/\sqrt{(\#\text{Adults})}$; for a single parent with one child as $\text{THI}/1.8^{.7}$; for single parents with multiple children as $\text{THI}/(1.8 + .5 \times (\#\text{Children} - 1))^{.7}$; and for all other households as $\text{THI}/(\#\text{Adults} + .5 \times \#\text{Children})^{.7}$, where THI is total household income. We approximate households with PIKs residing in the same MAFID and partition adults and children based on whether they are below 18 years of age. We additionally smooth adjusted household income for each household head over the sample year and the previous two years.

[60]We impose this restriction because many high earners in the LEHD experience substantial fluctuations in earnings over time. Although the restriction may introduce some selection bias into our descriptive estimates, we consider it necessary to avoid including high earners whose earnings temporarily fall below our low-income threshold, even after smoothing household income.

TABLE A1. Constructing 2010 Samples

	Restriction	Number MAFIDs	Sample Description
	No restriction	99,980,000	
(a)	Between 25 and 65 and not group quarters	84,410,000	Tract aggregates sample
(b)	100 largest CBSAs	38,880,000	
(c)	Household Heads	38,880,000	
(d)	Earnings > \$0	31,760,000 (3,750,000)	
(e)	Low-income	10,480,000 (1,870,000)	
(f)	Earnings never > \$150,000 (2010 USD)	9,350,000 (1,750,000)	RMA measure sample
(g)	Rental unit	3,270,000 (864,000)	Structural analysis sample
(h.i)	2011-observed, urban, & low-college-share tracts	868,000 (351,000)	Baseline mobility sample
(h.ii)	2019-observed, urban, & low-college-share tracts	796,000 (324,000)	Regression analysis sample
(h.iii)	Gentrifying, urban, & low-college-share tracts	677,000 (269,000)	Welfare analysis sample

Notes: Table A1 reports the number of MAFIDs retained in the 2010 sample after applying each successive restriction. Parenthetical counts show the number of MAFIDs retained for households with Black household heads. Samples for other years are defined analogously. Samples (h.i), (h.ii), and (h.iii) are subsets of sample (g). The 2011- and 2019-observed restrictions require observing the household heads' residential census tract in both 2010 and 2011 or 2019, respectively. The urban restriction requires residence in a tract closer to the metropolitan division CBD than at least 50% of that division's population. The low-college-share tracts restriction requires residence in a tract in the bottom quartile of the CBSA's 2010 college-graduate share distribution. The rightmost column names the samples formed at each cleaning stage.

because of the LEHD's model-driven approach to choosing among candidate residential locations starting in that year. The rightmost columns report households' one-year, tract-level move rates. Like Sullivan and Genadek (2024), we find considerable variation in these rates over time. As this variation likely reflects sampling error, we interpret cross-year move-rate differences cautiously and structure our analyses to circumvent its effects, as we document throughout the text.

Because baseline residential mobility informs moments that are central to our main findings, we compare our tract-level estimates with self-reported mobility in the ACS. Figure A1 reports one-year baseline mobility rates in the MAF-ARF for each sample alongside mobility rates in the ACS for the subset of household heads we can match to the 2010 ACS. Because ACS respondents report where they lived one year earlier, the figure compares 2009 mobility rates in the ACS with 2010 mobility rates in the MAF-ARF.

The estimates are quite similar across samples. The main exception is our primary descriptive sample of urban, low-income renters in tracts in the bottom quartile of the CBSA college-graduate share distribution, for whom the ACS reports slightly lower residential mobility. This gap may reflect nonresponse bias in the

TABLE A2. Sample Validation

	ACS Census Tract Match		One-Year Move Rate	
Year	Non-Black	Black	Non-Black	Black
2010	.791	.775	.181	.185
2011	.790	.772	.303	.308
2012	.864	.846	.220	.228
2013	.862	.848	.208	.221
2014	.863	.847	.234	.257
2015	.873	.860	.151	.162
2016	.880	.874	.227	.229
2017	.875	.866	.173	.182
2018	.878	.868	.173	.183
2019	.887	.867	-	-

Notes: The leftmost columns of Table A2 report the share of sample households whose census tracts match those in the ACS. The rightmost columns of Table A2 report households' one-year, tract-level move rates. The samples correspond to those defined after applying restriction (g) in Table A1, with the ACS Census Tract Match sample defined only for household heads we can merge to the ACS.

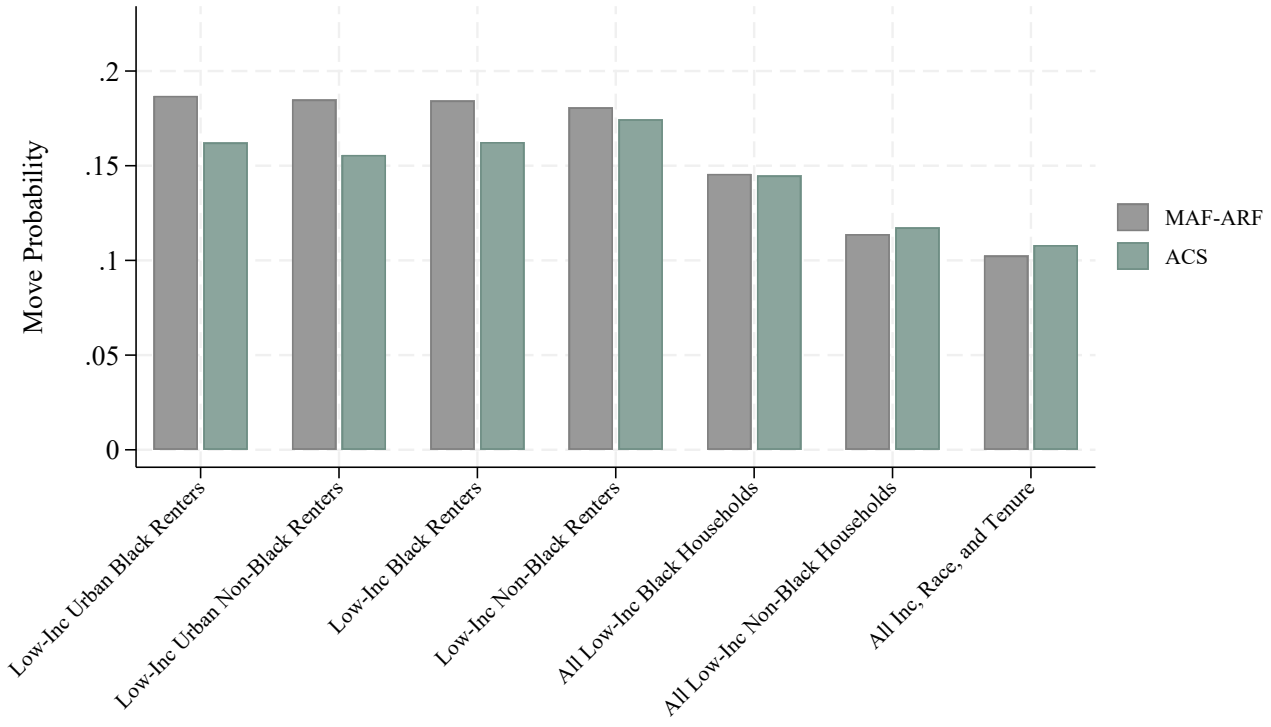
ACS, particularly because these renters live in more disadvantaged tracts than households in the other samples. Consistent with this possibility, Bollinger et al. (2019) document substantial earnings nonresponse among low earners in survey data, raising the possibility of similar underreporting of residential mobility among disadvantaged households. Alternatively, the higher MAF-ARF mobility rates could reflect greater measurement error among disadvantaged households in the MAF-ARF itself. Overall, we view these comparisons as supporting our baseline mobility estimates in Figure 1 and Table A15. We nevertheless test the robustness of our primary welfare analysis by additionally calibrating baseline mobility rates to these ACS estimates, as described in Section E.2.

A.3. Variable Construction

Neighborhood Sociodemographics. Our descriptive analyses rely on short-term variation in neighborhoods' sociodemographic characteristics. Ordinarily, researchers use pooled 5-year ACS aggregates to define US neighborhoods' sociodemographic characteristics and changes thereof. These 5-year ACS aggregates are noisily estimated, however, and exacerbate classical measurement error in descriptive analyses. We instead utilize our administrative data and combine potentially unbiased, but imprecise, 5-year ACS tract-level estimates with biased, but more precise, tract-level estimates derived from our residential panel defined after applying restriction (a) in Table A1.[61] We document below our novel approach to constructing these annual tract-level sociodemographic aggregates.

[61] Bond et al. (2014) and Brummet and Reed (2021) show that children, recent movers, immigrants, low-income, unemployed, male, non-college-educated, and group-quarter residents are less likely to receive a PIK, thus biasing any tract aggregate derived from the residential panel.

FIGURE A1. Baseline Residential Mobility in the ACS and the MAF



Notes: Figure A1 compares one-year baseline tract-level residential mobility for households in the MAF-ARF with the subset of households that can also be linked to the 2010 ACS. We measure residential mobility in the MAF-ARF between 2010 and 2011. Because the ACS asks retrospectively about residential mobility, the corresponding ACS measure is between 2009 and 2010. Column pairs 1 & 2, 3 & 4, 5 & 6, and column 7 correspond to the samples obtained after applying restrictions (h.ii), (h.i), (f), and (d) in Table A1, respectively. The MAF-ARF mobility estimates are the same as those reported in Table A15.

We assume the pooled five-year ACS aggregates are unbiased and use them to bias-correct tract aggregates derived from our residential panel to obtain precise and unbiased annual sociodemographic aggregates. Specifically, we assume that the logit-transformed pooled 5-year ACS sociodemographic neighborhood shares are unbiased measures of the true logit-transformed shares,[62]

$$Y^*_{\text{ACS}} = Y^*_{\text{True}} + \epsilon, \tag{A1}$$

and that the logit-transformed biased measures of the same neighborhood shares constructed from the residential panel are related to the true logit-transformed shares via a linear function of covariates,

$$Y^*_{\text{Panel}} = Y^*_{\text{True}} + \beta X, \tag{A2}$$

where $Y^* = \log\left(\frac{Y}{1-Y}\right)$, ϵ is an i.i.d. mean zero error, Y^*_{True} is the true logit-transformed shares, and X is a vector of covariates including logit-transformed tract-level shares of adults with college degrees, adults who report as female, white, Black, another race, white but not Hispanic, Hispanic, foreign born,

[62] We construct the pooled 5-year ACS measures internally in the Census IRE using person-level weights and cross-walking to 2010-delineated census tracts using block-to-block crosswalks and population weights.

white with a college degree, white with a college degree but not Hispanic, with positive earnings in the LEHD, log-transformed tract-level measures of mean earnings, median earnings, adult population size, and tract-level housing supply elasticity and occupancy rate estimates from Baum-Snow and Han (2024), a quadratic in tracts' distance to their metropolitan division's CBD, a cubic in the calendar year, and CBSA-level fixed effects.

Substituting A1 in to A2 we obtain

$$Y^*_{\text{Panel}} - Y^*_{\text{ACS}} = \beta X + \epsilon$$

which we estimate via OLS and use to predict $\widehat{Y^*} = Y^*_{\text{Panel}} - \hat{\beta}X$. We finally apply the inverse logit transformation to $\widehat{Y^*}$ to obtain our unbiased and precise annual tract-level measures of neighborhood sociodemographic shares. We obtain analogous unbiased and precise estimates for tract-level measures of mean earnings, median earnings, total adult population, and mean adult age substituting in log-transformations to equations A1 and A2. We obtain measures of total populations in each sociodemographic group referenced by multiplying our unbiased and precise estimates of tract-level shares with our estimates for total tract adult population.

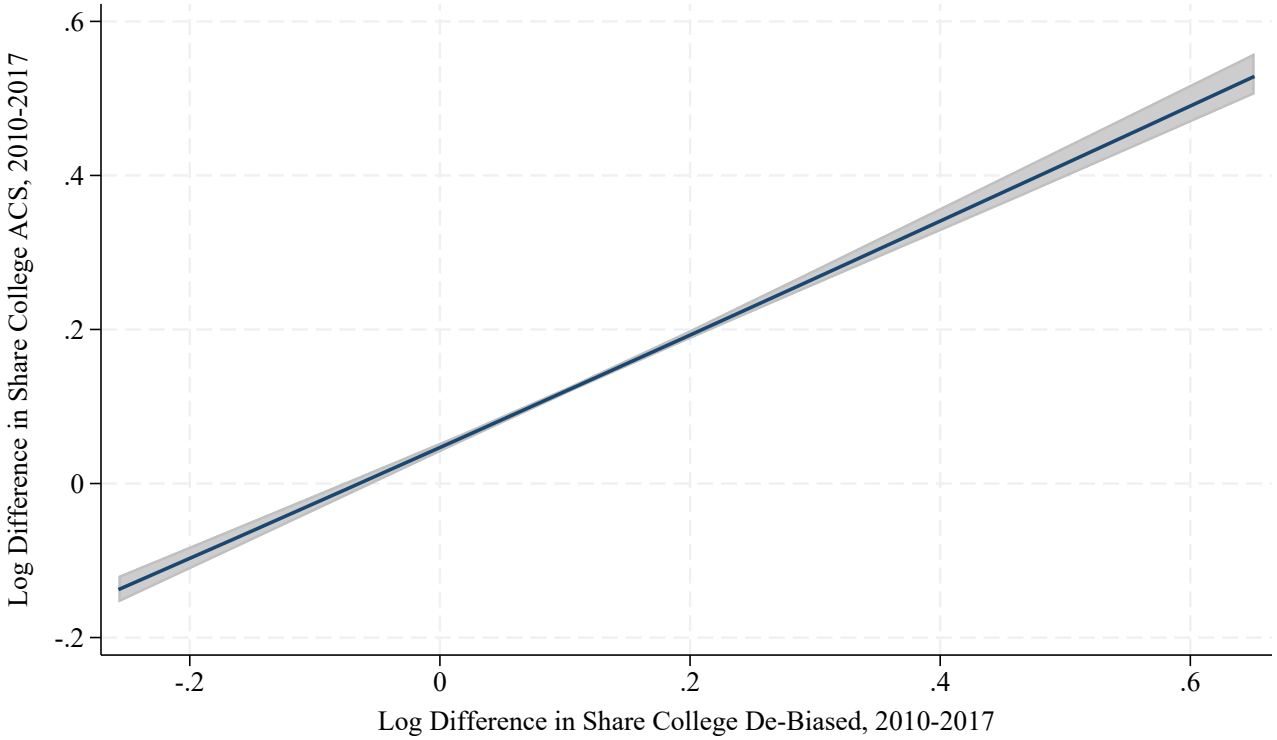
Panel A in Table A3 presents summary statistics comparing estimates of tracts' adult college-graduate shares computed using the ACS aggregates against these same shares computed using three estimates derived from our residential panel. These include estimates derived from all PIKs in our residential sample after applying restriction (a) in Table A1, estimates derived from these same PIKs but excluding those for whom the LEHD imputes educational attainment, and our de-biased tract estimates described above. Table A3 shows that using the full sample of PIKs in our residential panel to estimate the tract aggregates underestimates tracts' college-graduate shares, especially in later years. By contrast, we initially overestimate tracts' college-graduate shares when excluding from our residential panel PIKs whose educational attainment was imputed in the LEHD. These tract aggregates are significantly closer to their ACS counterparts than when we include PIKs with imputed educational attainment, however, as demonstrated by the markedly smaller Kullback-Leibler divergence measures. For this reason, we construct our de-biased college-graduate share tract aggregate estimates excluding PIKs with imputed educational attainment. As expected, our de-biased tract aggregates closely align with their ACS counterparts in both 2010 and 2017 according to our Kullback-Leibler divergence measures, the tract pseudo-medians, and the tract standard deviations.[63] Panel A additionally reports rounded pseudo-median counts of the number of PIKs used to compute each tract aggregate, rounded to the nearest 50 PIKs. The median de-biased tract aggregate is constructed using between 2.7 to 4.6 times more PIKs than its ACS counterpart, whose PIKs are further dispersed across five-year intervals.

Panel B in Table A3 presents similar summary statistics for log-differences in tract college-graduate shares between 2010 and 2017 for the pooled 5-year ACS aggregates and for our de-biased tract aggregates. While the pseudo-median tract aggregates are very similar across the two sets of estimates, their

[63]We present cross-section results for 2010 and 2017 so that the pooled 5-year ACS estimates centered at these years exclude years 2020 and 2021, since 5-year aggregates centered at 2018 and 2019 may be distorted by neighborhood re-sorting in response to COVID-19. In practice, we find very similar results when centering our pooled 5-year ACS estimates around 2018 and 2019. We nevertheless, estimate β in Equation A2 excluding years 2018 and 2019, and conduct out-of-sample predictions for these same years.

distribution—as indicated by the Kullback-Leibler divergence and standard deviation estimates—differ markedly. Smaller standard deviations among college-graduate tract shares derived from the de-biased panel implies meaningful sampling error in the ACS-based measures. While we believe our de-biased tract-level aggregates are a significant improvement over the pooled 5-year ACS estimates, they are not perfect; Figure A2 plots the results from a non-parametric kernel regression of log-changes in neighborhood college-graduate shares between 2010-2017 using the pooled ACS 5-year aggregates on the same measure constructed using the de-biased panel aggregates. That the kernel regression line has slope less than one suggests some measurement error remains in the de-biased panel aggregates. We view these estimates as a significant improvement over the raw ACS estimates, especially given the much stronger relationship between log changes in tract college-graduate shares and tract rents when using the de-biased estimates rather than the ACS estimates, shown in Figure A3.

FIGURE A2. Assessing Measurement Error in De-Biased Panel Aggregates



Notes: Figure A2 plots results from the non-parameteric regression of log-changes in tracts' shares of adults holding college degrees between 2010-2017 using pooled ACS 5-year aggregates on the same measure constructed using the de-biased panel aggregates. We use epanechnikov kernals and select bandwidths via cross validation. 95% confidence interval bands are in gray.

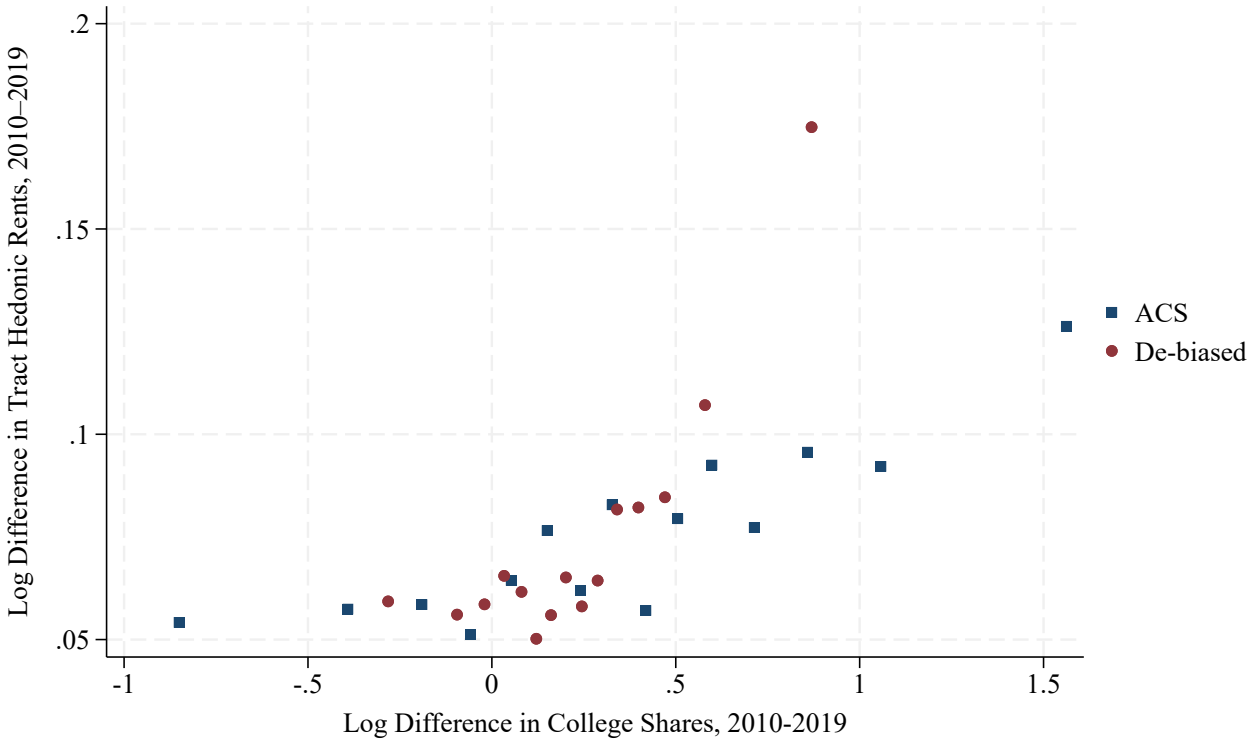
Neighborhood Rents. We estimate two series of tract-level hedonic rents. The first series we estimate is tailored for our descriptive analyses to capture average changes in tract rents over time. The other series is tailored for our structural and welfare analyses to capture changes in rents that incumbent renters actually pay, and therefore vary by the length of renters' neighborhood tenure. To construct the former series, we estimate tract-level hedonic rents by regressing real-2010 log gross rents from the household-level

TABLE A3. Tract Aggregate Comparisons

Statistic	Measure	Estimate	
		2010	2017
Panel A: Cross-Section Aggregates			
Kullback-Leibler divergence	College share all	.1428	0.2127
	College share not imputed	.0191	0.0031
	College share de-biased	.0039	0.0007
Tract pseudo-median	College share ACS	.2770	.3182
	College share all	.2670	.2583
	College share not imputed	.3038	.3125
	College share de-biased	.2744	.3127
Tract standard deviation	College share ACS	.2053	.2138
	College share all	.1444	.1338
	College share not imputed	.2033	.2134
	College share de-biased	.1968	.2137
Tract pseudo-median sample size	ACS	150	150
	All	2100	2200
	Not imputed	550	500
	De-biased	550	500
		2010-2017	
Panel B: Neighborhood Change Measures			
Kullback-Leibler divergence	Share college de-biased	.2714	
Tract pseudo-median	College share ACS	.1049	
	College share de-biased	.1039	
Tract standard deviation	College share ACS	.3530	
	College share de-biased	.1961	

Notes: Table A3 reports sample statistics for tract-level estimates of the share of residents aged between 25 and 65 who hold at least a Bachelors degree. These estimates are computed using (i) pooled 5-year ACS aggregates centered at 2010 and 2017 ("ACS"); (ii) all adults in our residential panel after applying restriction (a) of Table A1 ("All"); (iii) the "All" sample but excluding those for whom the LEHD imputes educational attainment ("Not Imputed"); and (iv) our de-biased tract-level estimates described in the appendix main text ("De-Biased"). Kullback-Leibler divergences are computed as $\sum_{x\in\mathcal{B}} P(x)\log\left(P(x)/Q(x)\right)$, where $P(x)$ are the comparison probability masses and $Q(x)$ the corresponding ACS reference probability masses, and $\mathcal{B}$ the set of 20 bins we approximate these distributions with. Pseudo-median values are the mean across the 11 tracts surrounding the median tract, where tracts are ranked by college-educated share for the estimates and by PIK count for the sample sizes. Sample sizes correspond to the underlying number of PIKs used to compute each tract aggregate and are rounded to the nearest 50 per Census guidelines. Neighborhood change is measured in log differences.

FIGURE A3. Validating De-Biased Panel Aggregates



Notes: Figure A3 shows the log change in tract-level hedonic rents as a function of the log change in the share of college-educated adults in sample households' 2010 residential tracts. The points plot conditional means estimated from a regression that includes CBSA fixed effects (Cattaneo et al. 2024). Tracts are weighted by the number of sample households residing in each tract in 2010. The sample comprises urban low-income renter households in 2010, defined after applying restriction (h.i) in Table A1. In the ACS series, we derive log differences in college shares only from the ACS tract estimates. In the de-biased series, we derive log differences in college shares from our de-biased tract estimates.

2005-2021 ACS on lot size, indicators for the number of bedrooms, rooms, and the building type, as well as tract-by-year fixed effects. We then predict rents for a representative property in each tract-year before smoothing these predictions over three-year bands, linearly interpolating them over missing values within census tracts, and winsorizing them at the first and ninety-ninth percentiles within each CBSA-year hedonic-rent distribution.[64]

We estimate the latter series tailored for our structural and welfare analyses similarly, but incorporate rental discounts stemming from the differential propensity for sample households to reside in LIHTC or otherwise subsidized housing units within tracts, as well as the share of sample households residing in rental units with binding rent control ordinances.[65] To do so, we additionally include in our hedonic rent regressions interactions between a sample household indicator and the tract counts of LIHTC units and,

[64] We predict hedonic rents using the approximation $\widehat{\text{Rent}_{nt}} \approx \exp\left(\ln\left(\widehat{\text{Rent}_{nt}}\right) + \frac{\text{Var}(\widehat{u_{nt}})}{2}\right)$, where $\widehat{u_{nt}}$ are residuals from the hedonic rent regressions.

[65] We define binding rent control ordinances as ordinances with maximum rent increases lower than their MSA's change in traditional, all transaction, Housing Price Index (HPI), retrieved from the Federal Housing Finance Agency given we do not observe counterfactual rent increases in the absence of rent control. When rent ordinances define maximum rent increases relative to the Consumer Price Index (CPI), we use MSA-wide CPIs to compute these maximum increases when possible, and regional CPIs when not, both obtained from the US Bureau of Labor Statistics.

separately, other HUD-subsidized units detailed in A.1, as well as a four-way interaction between the sample indicator, an indicator denoting whether the unit has a binding rent control ordinance, the tract's share of low-income renters in rent-controlled units, and the household's tenure length. We estimate these hedonic regressions separately for each CBSA to accommodate heterogeneous impacts of LIHTC units, HUD subsidies, and rent control ordinances across metro areas.

We use these hedonic regressions to predict the gross rent sample households with a given tenure length would pay in each tract-year for a representative property. The length of residential tenure only influences hedonic rents through our binding rent control measure, helping prevent reverse causality in which lower rents encourage households to remain longer in their units (Gallin et al. 2025), while capturing the partial insulation from market rent growth that incumbent renters enjoy in rent-controlled units (Diamond, McQuade, and Qian 2019; Chen, Jiang, and Quintero 2023). We similarly smooth predictions over three-year bands, interpolate missing values within census tracts, and winsorize values at the fifth and ninety-fifth percentiles within each CBSA-year hedonic rent distribution (there is a higher share of outliers in this latter series than in the former hedonic rent series). For each tract-year, we finally aggregate predicted rents for households with tenure length less than or equal to five years ($\bar{\tau} = 1$) and for households with tenure length greater than five years ($\bar{\tau} = 2$).

Tract and Industry Sample Employment Counts. Our measures of residential and job market access depend on counts of low-income workers employed in each 6-digit NAICS industry in every 2010-delineated census tract. We construct these counts by recording workplace-tract locations and industries for low-income households defined after applying restriction (f) in Table A1 from the LEHD-ECF. We crosswalk industry definitions to their 2012-NAICS code using Census-provided concordance files and employment weights we construct from the LBD.[66] Before conducting any analyses, we manually removed employment counts for industries in the LEHD-ECF whose establishments appeared to be improperly geocoded or assigned to headquarters that do not reflect employees' physical place of work.[67]

[66]We follow establishments' changing industry classifications over time in the LBD to infer the share of employment in each 2007- and 2017-delineated NAICS code that falls under each 2012-delineated NAICS code whenever industry definitions change. We ensure establishments' 2012-NAICS code assignments do not vary across years.

[67]These industries include: 112130—Dual-purpose cattle ranching and farming; 237310—Highway, street, and bridge construction; 491110, 488111, 485113, 482111, 48212, 485111, 492110—Postal services, Air traffic control, Bus and other motor vehicle transit systems, Railroads, Mixed-mode transit systems, Couriers and express delivery services; 561330, 541612, 541214, 541120—Professional employer organizations, Human resources consulting services, Payroll services, Offices of notaries; 611110—Junior and elementary schools; 813910, 813110, 814110, 813219—Business associations, Religious organizations, Private households, Other grantmaking and giving services; 92—Public services.

Appendix B. Microfoundations and Structural Estimation Details

B.1. Residential Market Access

Setup. In this section we derive our measure of residential market access introduced in Section 4.2 under the assumption that the labor market is segmented for each k-type worker.[68] Our derivation nests workers' choices across industries, which facilitates microfounding our job market access instruments described in Section 6. The workplace choice problem reported in Section 4.2 involves households choosing over both workplace, m, and industry, ι, conditional on living in neighborhood n:

$$I_{int} \equiv \max_{m,\iota} \frac{z_{im\iota t}}{d^k_{nmt}} w^k_{m\iota t}.$$

We assume z_{imt} is distributed iid Frechet over workplace locations and industries, $F(z_{im\iota t}) = \exp\left(-\left(z_{im\iota t}\right)^{-\epsilon^{kc}_t}\right)$ with shape parameter $\epsilon^{kc}_t > 1$ for each city c, period t, and for each type-k worker. The probability of a type-k worker living in tract n taking a job in location m is then given by,

$$\begin{aligned} \pi^k_{m|nt} &= \frac{\sum_\iota (w^k_{m\iota t}/d^k_{nmt})^{\epsilon^{kc}_t}}{\sum_\iota \sum_{m'} (w^k_{m'\iota t}/d^k_{nm't})^{\epsilon^{kc}_t}}, \\ &= \frac{\sum_\iota (w^k_{m\iota t}/d^k_{nmt})^{\epsilon^{kc}_t}}{\text{RMA}^k_{nt}}, \end{aligned} \tag{A3}$$

where $\text{RMA}^k_{nt} \equiv \sum_\iota \text{RMA}^k_{n\iota t} \equiv \sum_\iota \sum_{m'} (w^k_{m'\iota t}/d^k_{nm't})^{\epsilon^{kc}_t}$.

Define labor supply to tract m in time t by $\ell^k_{mt} = \sum_\iota \left[\left(w^k_{m\iota t}\right)^{\epsilon^{kc}_t}\right] \text{FMA}^k_{mt}$, where FMA^k_{mt}, represents the access firms in tract m have to k-type workers in period t. Equating labor supply of type-k workers to tract m in period t to workers' choice probabilities yields an expression for FMA in terms of RMA:[69]

$$\begin{aligned} \ell^k_{mt} &= \sum_n \pi^k_{m|nt} \cdot \pi^k_{nt} \cdot N^{kc}_t \\ &= \sum_n \frac{\sum_\iota (w^k_{m\iota t}/d^k_{nmt})^{\epsilon^{kc}_t}}{\text{RMA}^k_{nt}} \cdot \pi^k_{nt} \cdot N^{kc}_t \\ &= \sum_\iota \left[\left(w^k_{m\iota t}\right)^{\epsilon^{kc}_t}\right] \cdot N^{kc}_t \cdot \sum_n \frac{\left(\left(\pi^k_{nt}\right)^{1/\epsilon^{kc}_t} / d^k_{nmt}\right)^{\epsilon^{kc}_t}}{\text{RMA}^k_{nt}} \end{aligned}$$

[68] Baum-Snow, Hartley, and Lee (2019) show how to extend the model to an integrated labor market with multiple types. The resulting expressions are identical to those we derive when assuming labor markets are segmented. The derivations in this section follow an established literature microfounding measures of market access in economic fundamentals (Donaldson and Hornbeck 2016; Tsivanidis 2026; Baum-Snow and Han 2024).

[69] N^{kc}_t is the total mass of type-k workers in city c and period t, and π^k_{nt} denotes the share of these workers choosing neighborhood n in period t.

$$\text{(A4)} \qquad \equiv \sum_{\iota} \left[\left(w_{m\iota t}^{k} \right)^{\epsilon_t^{kc}} \right] \cdot \text{FMA}_{mt}^{k}$$

The penultimate equality yields,

$$\text{FMA}_{mt}^{k} = N_t^{kc} \sum_{n} \frac{\left(\left(\pi_{nt}^{k} \right)^{1/\epsilon_t^{kc}} / d_{nmt}^{k} \right)^{\epsilon_t^{kc}}}{\text{RMA}_{nt}^{k}}.$$

Furthermore, dividing both sides of A4 by $\left(d_{nmt}^{k} \right)^{\epsilon_t^{kc}}$ and summing over m yields an expression for RMA_{nt}^{k} in terms of FMA_{mt}^{k}. We subsequently obtain the following system of equations for RMA and FMA:

$$\text{(A5)} \qquad \begin{aligned} \text{FMA}_{mt}^{k} &= N_t^{kc} \sum_{n} \frac{e^{-\kappa^k \epsilon_t^{kc} \tau_{nmt}^{k}} \pi_{nt}^{k}}{\text{RMA}_{nt}^{k}} \\ \text{RMA}_{nt}^{k} &= \sum_{m} \frac{e^{-\kappa^k \epsilon_t^{kc} \tau_{nmt}^{k}} \ell_{mt}^{k}}{\text{FMA}_{mt}^{k}}, \end{aligned}$$

where we have defined $d_{nmt}^{k} \equiv e^{\kappa^k \tau_{nmt}^{k}}$ with τ_{nmt}^{k} denoting the commute time between tracts n and m for a type-k worker in period t and κ^k governing the magnitude of commute costs for type-k workers. [70] We solve for the unique (to scale) fixed point in A5 to obtain values for RMA_{nt}^{k} in each neighborhood, sample, and year (Tsivanidis 2026). We form measures of $N_t^{kc} \cdot \pi_{nt}^{k}$ and ℓ_{mt}^{k} directly from our samples defined after applying restriction (f) in Table A1. We describe how we obtain estimates of commute times (τ_{nmt}^{k}) and commuting elasticities ($\eta_t^{kc} \equiv \kappa^k \cdot \epsilon_t^{kc}$) below.

Commute Times. Period and k-type specific commute times relating all tract pairs within a CBSA (τ_{nmt}^{k}) play a crucial role in estimating RMA_{nt}^{k}. We obtain measures of τ_{nmt}^{k} from the ACS microdata for each n, m, t, and k combination. To do so, we match our samples to the ACS microdata and record their home census tract, n, their workplace census tract, m, and their reported commute time in minutes. We then compute the median reported commute time for each tract-pair with positive commute flows in each CBSA, separately for each sample and each five-year interval centered at every year between 2007 and 2017.[71]

Since there are far more potential tract-pair commuting routes within CBSAs than ACS survey respondents we can match to each sample, we are missing many observed period and k-type commute times. We therefore construct an empirical forecasting model to predict missing commute times between all neighborhood pairs within each CBSA for each sample and every year between 2001 and 2019. We follow Baum-Snow, Hartley, and Lee (2019) and predict these commute times using a CBSA fixed effect, the log-distance between the tract-pair centroids, and the log-distances between the home- and workplace-tract centroids and their respective metropolitan division CBDs. We estimate the following forecasting model

[70] $1 - e^{-\kappa^k \tau_{nmt}^{k}}$ thus represents the portion of time that type-k workers in tract n spend commuting to tract m in period t.
[71] We avoid using reported commute times from 2020 onward given the increased prevalence of working from home.

separately for each sample and each year,

$$\text{(A6)} \qquad \ln\left(\tau^k_{nmt}\right) = \alpha^k_{d,t} \ln\left(\text{Dist}\right)_{nm} + \alpha^k_{h,t} \ln(\text{Home CBD Dist})_n + \alpha^k_{w,t} \ln(\text{Work CBD Dist})_m + \alpha^{kc}_t + u^k_{nmt},$$

and construct $\widehat{\tau^k_{nmt}}$ using the approximation $\widehat{\tau^k_{nmt}} \approx \exp\left(\widehat{\ln\left(\tau^k_{nmt}\right)} + \frac{\widehat{\text{Var}(u^k_{nmt})}}{2}\right)$.

TABLE A4. Forecasting Commute Times

	Low-Income Renters		All Low-Income	
	Black	Non-Black	Black	Non-Black
Commute Distance $\left(\alpha^k_{d,t}\right)$	0.3565	0.4211	0.3717	0.4601
	(0.00449)	(0.002158)	(0.003202)	(0.00117)
Home-CBD Distance $\left(\alpha^k_{h,t}\right)$	-0.06569	-0.06897	-0.06462	-0.06631
	(0.005334)	(0.002484)	(0.003776)	(0.001379)
Work-CBD Distance $\left(\alpha^k_{w,t}\right)$	-0.06424	-0.07636	-0.06464	-0.07613
	(0.003375)	(0.001826)	(0.002289)	(0.0009395)
Adjusted R-Squared	0.2796	0.3472	0.2877	0.3928
N	27,000	91,000	51,000	286,000

Notes: Table A4 records the pseudo-median of the coefficient estimates of all CBSA-level regressions in the year 2010 from Equation A6, separately for each sample. It also reports the mean of the standard errors and the adjusted R-squared statistics from the regressions forming the pseudo-median estimates. Each CBSA×year×sample regression is weighted by the number of commuters observed in each tract-pair cell in the ACS microdata.

Table A4 shows that commute time responds less to additional travel distance for low-income renters than for the overall low-income population, with the elasticity smallest for Black households. This muted relationship among Black commuters is consistent with their substantially lower propensity to commute by car, as shown in Table A5. Black low-income renters are about twice as likely as non-Black low-income renters to commute by public transit, while commute times are similar across the two samples conditional on mode of transportation. For both samples, commutes by public transit are much longer than those by private motor vehicle. If public transit travel times are less closely related to physical distance because transit quality, availability, and routing vary across home- and workplace-tract pairs, the greater reliance of Black renters on public transit may help explain why commute distance is less predictive of their commute times.[72]

Commute Elasticities. In addition to tract-pair commute times, we must obtain estimates of the commute elasticities $\eta^{kc}_t \equiv \kappa^k \cdot \epsilon^{kc}_t$, which govern how the value of employment opportunities to workers decays with

[72] In addition to providing context for interpreting our type-specific commuting forecasts and elasticities, the heterogeneity in Table A5 suggests that studies of spatial mismatch (e.g., Card, Rothstein, and Yi 2024) may benefit from measuring proximity to employment opportunities using expected commute times in addition to physical distance.

TABLE A5. Commute Modes and Times

	Low-Income Renters			
	Black		Non-Black	
	Share	Time	Share	Time
Car, Motorcycle, or Taxi	0.619	22.63	0.771	21.66
		(17.74)		(15.91)
Public Transit	0.286	49.13	0.143	48.59
		(26.76)		(30.13)
Walk, Bike, or Other	0.071	20.25	0.086	16.24
		(20.54)		(17.34)

Notes: Table A5 reports commute-mode shares and mean commute times in minutes among our low-income renter samples matched to the ACS in 2010, separately for Black and non-Black household heads. Commute times are conditional on using the indicated transportation mode. Standard deviations of commute times are in parentheses.

commute time. As η_t^{kc} increases, the more sensitive RMA_{nt}^k is to the proximity of workplace locations relative to these locations' employment levels, ℓ_{mt}^k (and thus equilibrium wages). We estimate η_t^{kc} through a series of gravity equations derived from the workplace choice model. Taking the (natural) log of equation A3 yields,

$$\begin{aligned} \ln\left(\pi_{m|nt}^k\right) &= \ln\left(\text{RMA}_{nt}^k\right) + \ln\left(\sum_\iota \left(w_{m\iota t}^k\right)^{\epsilon_t^{kc}}\right) - \kappa^k \epsilon_t^{kc} \tau_{nmt}^k \\ &= \alpha_{nt}^k + \rho_{mt}^k - \eta_t^{kc} \tau_{nmt}^k. \end{aligned} \tag{A7}$$

We estimate these gravity equations separately for each CBSA, k-type worker, and period using observed commute flows in the LEHD to approximate $\pi_{m|nt}^k$ for each of our samples.

Table A6 reports estimates of η_t^{kc} for our four low-income samples. As our estimates must include data from at least three of our LEHD-states, we are unable to display the distribution of η_t^{kc} across our sample CBSAs. We instead report three pseudo-quantiles (25th, 50th, 75th) for each sample in 2010 and 2019. The pseudo-percentiles correspond to the mean of eleven estimates of η_t^{kc} centered at the given percentile in their distribution across CBSAs in a given year and sample. The elasticities in Table A6 are within the range of existing estimates, though on the higher end (Ahlfeldt et al. 2015). Like Baum-Snow, Hartley, and Lee (2019), we find substantial heterogeneity in commute elasticities across CBSAs, as revealed by the range of the pseudo-percentiles. We additionally find that low-income renters have higher commute elasticities than the broader low-income population, a pattern that likely reflects their relatively lower frictions to relocating in response to changing job market access. Further, Black households exhibit lower commute elasticities than non-Black households.[73] This heterogeneity again coincides with substantial differences in commuting mode reported in Table A5, with Black low-income renters much more likely

[73]We find similar patterns across Black and non-Black households when using commute distance as our key variable in Equation A7.

to commute by public transit. If commutes on public transit involve larger fixed time costs but lower marginal costs of commute time, the greater reliance of Black renters on public transit could help explain both their longer commute times and their lower estimated commute elasticities.

TABLE A6. Gravity Regressions

		Low-Income Renters		All Low-Income	
		Black	Non-Black	Black	Non-Black
Panel A: 2010					
Commute Time $\left(\eta^{kc}_{2010}\right)$	25th percentile	0.06047	0.08794	0.05685	0.08533
		(0.00375)	(0.000350)	(0.00106)	(0.000184)
	50th percentile	0.08081	0.1021	0.07747	0.09581
		(0.00393)	(0.000288)	(0.00276)	(0.000146)
	75th percentile	0.09981	0.1127	0.09628	0.1058
		(0.00319)	(0.000194)	(0.000858)	(0.0000844)
	N	736,000	1,939,000	1,477,000	6,082,000
Panel B: 2019					
Commute Time $\left(\eta^{kc}_{2019}\right)$	25th percentile	0.06204	0.08289	0.06082	0.07929
		(0.00282)	(0.000491)	(0.00148)	(0.000215)
	50th percentile	0.07518	0.09597	0.07401	0.09208
		(0.00196)	(0.000240)	(0.00111)	(0.000121)
	75th percentile	0.08890	0.1101	0.08893	0.1022
		(0.00159)	(0.000255)	(0.00132)	(0.000121)
	N	967,000	2,527,000	1,837,000	7,079,000

Notes: Table A6 reports the pseudo-percentiles of the distribution of the parameter clusters within each sample separately. Panel A reports these estimates for the year 2010 and Panel B for the year 2019. Means of the standard errors from the regressions forming each pseudo-percentile are in parentheses. We weight each gravity regression by the number of commuters in each origin census tract. We use Poisson pseudo-maximum likelihood to account for sparse tract-pair commute flows (Correia, Guimarães, and Zylkin 2020).

Industry Employment Exposure. To transparently relate our job market access instruments discussed in Section 6 to the recent advances in the quasi-experimental shift-share literature, we also disaggregate employment exposure by industry and take linear approximations of Equation A5 to obtain Equation 17:

$$\text{EE}^{k}_{mit} = \sum_{m \in \mathcal{N}^c \setminus n} e^{-\eta^{ck}\tau^{k}_{n,m}} l^{k}_{mit}, \tag{A8}$$

These linear approximations relate to earlier notions of market potential in international and regional trade theory, which conceptualize the demand for goods in a given region as the sum of demands in surrounding regions, weighted by bilateral transportation costs (Harris 1954; Hanson 2005). It is common in the regional economics literature to take linear approximations of structural measures of market access (e.g., Donaldson and Hornbeck (2016) and Herzog (2021)).

B.2. Neighborhood Flow Utility

In this section we derive our expression for neighborhood expected flow utilities. Conditional on the commute-adjusted earnings indices, households solve for the their optimal level of housing consumption in each neighborhood. Because there is no saving in the model, households' budget constraints bind with equality. Optimal housing consumption in neighborhood n for a household with τ years of tenure under Cobb-Douglas preferences in Equation 4 is therefore

$$H^{k*}_{\tau nt} = \frac{(1-\beta^k_c)\cdot \bar{I}^k_{nt}}{r_{\tau nt}} \tag{A9}$$

Substituting A9 and our amenity parameterizations into Equation 3 and then taking the (natural) log yields Equation 6,

$$\begin{aligned} u^k_n\left(s^c_{it}\right) = \tilde{\alpha}^k_n + \tilde{\alpha}^{kc}_t + \tilde{\beta}^k_I \ln\left(\bar{I}^k_{nt}\right) + \tilde{\beta}^k_r \log\left(r_{\tau nt}\right) + \tilde{\beta}^k_A \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) \\ + \tilde{\beta}^k_\tau \ln(\tau_{it}) + MC^k_t(n_t, n_{it-1}) + \tilde{\xi}^k_{nt} + \varepsilon_{int} \end{aligned}$$

where

$$\begin{aligned} \tilde{\alpha}^k_n &= \left[\alpha^k_n + \beta^k_C \ln\left(\beta^k_C\right) + \left(1-\beta^k_C\right)\ln\left(1-\beta^k_C\right)\right]\frac{1}{\sigma^k} \\ \tilde{\alpha}^{kc}_t &= \alpha^{kc}_t / \sigma^k \\ \tilde{\xi}^k_{nt} &= \bar{\xi}^k_{nt} / \sigma^k \\ \tilde{\beta}^k_I &= 1/\sigma^k \\ \tilde{\beta}^k_A &= \beta^k_A / \sigma^k \\ \tilde{\beta}^k_r &= (\beta^k_C - 1) / \sigma^k \\ \tilde{\beta}^k_\tau &= \beta^k_\tau / \sigma^k \\ r_{\tau nt} &= r_{nt} \cdot h_n\left(\tau_{it}, \{\bar{\omega}^{kc}_t\}^t_{t-\tau}\right). \end{aligned}$$

B.3. First-Step Conditional Choice Probabilities

To see how Equation 13 exactly captures the neighborhood choice problem of households in our dynamic model, start by considering our specification for a household's conditional value function defined in 8,

$$\begin{aligned} v^k_n\left(x_{it}, \bar{\omega}^{kc}_t\right) &\equiv \bar{u}^k_n\left(x_{it}, \bar{\omega}^{kc}_t\right) + \delta \mathbb{E}_t\left[\bar{V}^k\left(x_{it+1}, \bar{\omega}^{kc}_{t+1}\right)|n, x_{it}, \bar{\omega}^{kc}_t\right] \\ &= \tilde{\alpha}^k_n + \tilde{\alpha}^{kc}_t + \tilde{\beta}^k_I \ln\left(\bar{I}^k_{nt}\right) + \tilde{\beta}^k_r \log\left(r_{nt}\right) + \tilde{\beta}^k_A \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) + MC^k_t(n_t, n_{it-1}) + \tilde{\xi}^k_{nt} \\ &\quad + \sum_{\bar{\tau}=1,2}\Bigg(\delta \mathbb{E}_t\left[\bar{V}^k\left(x_{it+1}(n,\bar{\tau}), \bar{\omega}^{kc}_{t+1}\right)|n, x(n_{t-1}, \bar{\tau}_{t-1}), \bar{\omega}^{kc}_t\right] \\ &\qquad + \tilde{\beta}^k_\tau \ln(\bar{\tau}) + \tilde{\beta}^k_r \ln\left(h_n\left(\bar{\tau}, \{\bar{\omega}^{kc}_t\}^t_{t-\bar{\tau}}\right)\right)\Bigg) \cdot f^x(x(n_t,\bar{\tau})|n_t, x(n_{t-1}, \bar{\tau}_{t-1})). \end{aligned} \tag{A10}$$

Let $\tilde{u}^k_{nt} \equiv \tilde{\alpha}^k_n + \tilde{\alpha}^{kc}_t + \tilde{\beta}^k_I \ln\left(\bar{I}^k_{nt}\right) + \tilde{\beta}^k_r \log\left(r_{nt}\right) + \tilde{\beta}^k_A \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) + \tilde{\xi}^k_{nt}$. Then, we can define the components of Equation 13 as,

$$\begin{aligned}
\gamma^k_{nt\bar{\tau}=1} \equiv \tilde{u}^k_{nt} &+ \delta\left[\mathbb{E}_t\left[\bar{V}^k\left(x_{it+1}(n,\bar{\tau}=1),\bar{\omega}^{kc}_{t+1}\right)|\bar{\omega}^{kc}_t\right] + \tilde{\beta}^k_r \ln\left(h_n\left(\bar{\tau}=1,\{\bar{\omega}^{kc}_t\}\right)\right)\right]\cdot\left(1-g^k_{nt}\right) \\
&+ \delta\left[\mathbb{E}_t\left[\bar{V}^k\left(x_{it+1}(n,\bar{\tau}=2),\bar{\omega}^{kc}_{t+1}\right)|\bar{\omega}^{kc}_t\right] + \tilde{\beta}^k_r \ln\left(h_n\left(\bar{\tau}=2,\{\bar{\omega}^{kc}_t\}\right)\right) + \tilde{\beta}^k_\tau \ln(2)\right]\cdot g^k_{nt}, \\
\gamma^k_{nt\bar{\tau}=2} \equiv \tilde{u}^k_{nt} &+ \delta\left[\mathbb{E}_t\left[\bar{V}^k\left(x_{it+1}(n,\bar{\tau}=2),\bar{\omega}^{kc}_{t+1}\right)|\bar{\omega}^{kc}_t\right] + \tilde{\beta}^k_r \ln\left(h_n\left(\bar{\tau}=2,\{\bar{\omega}^{kc}_t\}\right)\right) + \tilde{\beta}^k_\tau \ln(2)\right] \\
\mu^k_{nt} \equiv \tilde{u}^k_{nt} &+ \delta\left[\mathbb{E}_t\left[\bar{V}^k\left(x_{it+1}(n,\bar{\tau}=1),\bar{\omega}^{kc}_{t+1}\right)|\bar{\omega}^{kc}_t\right] + \tilde{\beta}^k_r \ln\left(h_n\left(\bar{\tau}=1,\{\bar{\omega}^{kc}_t\}\right)\right)\right] + MC^{kc} \\
&+ \left(MC^{kc}_{OO} - MC^{kc}\right)\cdot\mathbb{1}\left\{n = OO^c\right\} \\
\lambda^{kc} \equiv\ & MC^{kc}_{OO} - MC^{kc}
\end{aligned}$$

Having defined $\gamma^k_{nt\bar{\tau}=1}$, $\gamma^k_{nt\bar{\tau}=2}$, μ^k_{nt}, and λ^{kc} in this way, we can rewrite Equation A10 as their linear combination plus our variable moving costs specification for movers within cities' urban cores:

$$\begin{aligned}
v^k_n\left(x_{it},\bar{\omega}^{kc}_t\right) = \ & \gamma^k_{nt\bar{\tau}=1}\cdot\mathbb{1}\{\bar{\tau}=1, n=\underline{n}\} + \gamma^k_{nt\bar{\tau}=2}\cdot\mathbb{1}\{\bar{\tau}=2, n=\underline{n}\} \\
&+ \mu^k_{nt}\cdot\mathbb{1}\{n\neq\underline{n}\} + \lambda^k\cdot\mathbb{1}\left\{\underline{n}=OO^c, n\neq\underline{n}\right\} \\
&+ \left(\boldsymbol{\beta}^{k\prime}_d\,\boldsymbol{d}(n,\underline{n}) + \boldsymbol{\beta}^{k\prime}_s\,\boldsymbol{s}(n,\underline{n})\right)\cdot\mathbb{1}\left\{n\neq OO^c, \underline{n}\neq OO^c\right\}
\end{aligned}$$

The right-hand-side of this equation matches the right-hand-side of the Poisson regression in 13. The frequency of moves to neighborhoods within each city's urban core identifies μ^k_{nt}, while the share of households of each aggregate tenure level staying in their home neighborhood exactly identifies $\gamma^k_{nt\bar{\tau}=1}$ and $\gamma^k_{nt\bar{\tau}=2}$. The share of households in each city's outside option moving to its urban core identifies λ^{kc}. Variation in the distance households move within their urban core, conditional on moving, identifies $\boldsymbol{\beta}^{k\prime}_d$. We discuss identifying households' variable moving costs in the main text.

The dependent variable in Equation 13 is the number of sample household moves between tract pairs observed in the data. We detail how we count tract-pair moves and moves between tracts' outside options below. We weight Equation 13 by the number of sample households from each aggregate tenure level in each year and move-origin tract. We use the estimated Poisson regressions to predict the probability each k-type household moves between each neighborhood in their city. Table A7 reports regressions of the predicted conditional choice probabilities for movers on their empirical counterparts constructed from the raw data. As expected, the coefficients from this regression lie below one.[74] In the raw data, many neighborhood choices are never selected by low-income renters from a particular neighborhood, resulting in sparse empirical choice probabilities. The Poisson regression estimates, however, always assign some positive probability to each potential move. The larger coefficient for Non-Black households derives from their larger sample size and subsequent higher share of non-zero empirical choice probabilities.

[74] Additionally including households' home tracts in these regressions yields coefficient estimates very close to 1, but this is a poor description of the smoothing procedure given the highly bimodal distribution of move probabilities.

TABLE A7. Predicted Conditional Choice Probabilities

	Empirical Choice Probabilities	
	Black	Non-Black
Predicted Choice Probabilities	0.486	0.773
	(3.39e-06)	(1.96e-06)
Constant	0.0000912	0.0000418
	(8.09e-09)	(4.92e-09)
PIK-years	9,472,000	25,470,000

Notes: Table A7 reports regressions of sample households' empirical neighborhood choice probabilities on their counterparts predicted from Equation 13. We restrict the sample to empirical choice probabilities involving moves from households' current tracts to another tract within their CBSA's urban core or to their outside option. We weight observations in Equation 13 by the number of sample households in the corresponding move-origin tract, year, and aggregate tenure level. We weight each outside option-year observation equal to the maximum weight of any other observation in the same CBSA, year, and aggregate tenure level. The raw count data is smoothed over three-year bands, as described in the main Appendix B.3 text. We report the total number of PIK-years in the raw data used to construct all empirical choice probabilities.

Constructing Migration Counts. We now describe how we form counts of moves between census tracts within CBSAs' urban cores and between CBSAs' urban cores and their outside options. These counts form our dependent variable in Equation 13, and thus the basis of our structural estimation. We define a CBSA's urban core as the set of census tracts containing the 80% of the CBSA's population that lives closest to its metropolitan divisions' CBDs.[75] We define the outside option separately for each CBSA. For a given CBSA, we consider all sample households residing outside that CBSA's urban core in period t—but within the CBSA—as in that CBSA's outside option during the period (the CBSA's "outer band"). We also consider all sample households who move into a CBSA in period $t+1$—into or outside its urban core—as in that CBSA's outside option during period t.

For each year, sample, and CBSA, we count residential moves between tracts within the urban core, between the urban core and its outside option, and within the outside option. To do so, we record each sample household's tract of residence in period t (move-origin) and period $t+1$ (destination), and classify their moves into one of 12 categories outlined in Table A8. After assigning each sample household to a category, we define their aggregate tenure levels as $\bar{\tau}=1$ if the household lived in a different tract or outside option in $t-5$ and $\bar{\tau}=2$ if it remained in the same tract or outside option in $t-5$. We next sum over moves within each origin-destination location (as defined in Table A8), aggregate tenure level, and year.

Because our structural analysis relies on changes in households' location choices over time, we adjust

[75] Some large CBSAs include multiple metropolitan divisions. We define our CBSA-level urban cores as the tracts closest to metropolitan division CBDs to exclude rural areas lying between metropolitan divisions within a CBSA.

TABLE A8. Classification of Residential Moves

ID	Origin	Destination	Recorded Moves
1	Urban core	Urban core of same CBSA	Tract → Tract
2	Urban core	Outer band of same CBSA	Tract → OO
3	Urban core	Outside any sample CBSA	Tract → OO
4	Urban core	Outer band of another sample CBSA	Tract → OO OO-Dest → OO-Dest
5	Urban core	Urban core of another sample CBSA	Tract → OO OO-Dest → Tract-Dest
6	Outer band	Urban core of same CBSA	OO → Tract
7	Outer band	Outer band of same CBSA	OO → OO
8	Outer band	Outside any sample CBSA	OO → OO
9	Outer band	Outer band of another sample CBSA	OO → OO OO-Dest → OO-Dest
10	Outer band	Urban core of another sample CBSA	OO → OO OO-Dest → Tract-Dest
11	Outside any sample CBSA	Outer band of a sample CBSA	OO-Dest → OO-Dest
12	Outside any sample CBSA	Urban core of a sample CBSA	OO-Dest → Tract-Dest

Notes: Each row in Table A8 row classifies a residential move for a sample household. Moves are defined by (i) the census tract of residence in year t ("origin") and (ii) the census tract in year t+1 ("destination"). "Outer bands" are the tracts in a CBSA that contain the 20% of a CBSA's population living farthest from its metropolitan divisions' CBDs. The right-most column records the type(s) of residential move(s) associated with the sample household's origin and destination locations. "OO" denotes an outside option tract, and the suffix "Dest" indicates that the tract or outside option belongs to the move's destination CBSA. We exclude moves that both start and end outside every sample CBSA because such households are not in our sample during that year. Households that do not move (i.e., remain in their origin tract or in the same outer band) are counted in categories 1 and 7. Moves described by 4, 5, 9, and 10, involve locations in two different CBSA samples; we duplicate these observations and assign one record to each CBSA.

these raw move counts to remove year-to-year noise in baseline move rates introduced by imperfections in the MAF-ARF and LEHD-ICF documented in Table A2. For each sample, we compute the unconditional nationwide, one-year, tract-level move probability for sample households in every year and for the entire 2010-2019 panel. We then form an annual scaling factor as the ratio of a given year's move probability to the pooled move probability. We multiply the yearly counts of households that remain in their origin tract or outside option by this scaling factor. This smoothing keeps baseline move probabilities consistent across years without affecting choice probabilities conditional on moving. Finally, for the Poisson regressions, we further smooth the data by pooling counts within three-year windows centered on each year from 2011 to 2018 (i.e., counts for year t include moves from $t-1$, t, and $t+1$).

B.4. Deriving our Estimating Equation

Consider the set of residential choices detailed in Section 5.2.[76] Given these residential choices, we start our derivation of Equation 12 with an application of the Hotz-Miller inversion, which amounts to differencing Equation 9 across two neighborhood choices in period t, n and OO^c:

$$\begin{aligned} \ln\left(\frac{p^k_{xnt}}{p^k_{xOO^ct}}\right) &= v^k_{xnt} - v^k_{xOO^ct} \\ &= \bar{u}^k_{xnt} - \bar{u}^k_{xOO^ct} + \delta\left(\mathbb{E}\left[\bar{V}^k_{xnt}\right] - \mathbb{E}\left[\bar{V}^k_{xOO^ct}\right]\right), \end{aligned} \tag{A11}$$

where the expectation operator is with respect to both the observable household-level and the full set of city-specific state variables. By assumption (a), we can write these expectations as

$$\begin{aligned} \mathbb{E}\left[\bar{V}^k_{xnt}\right] &= \sum_{x'} \int_{\bar{\omega}'} \bar{V}^k_{xnt}\, dF^{\bar{\omega}}\left(\bar{\omega}'|\bar{\omega}\right) f^x\left(x'|n, x_{nt}\right) \\ &= \sum_{x'} \mathbb{E}_{\bar{\omega}'|\bar{\omega}^{kc}_t}\left[\bar{V}(x', \bar{\omega}')|\bar{\omega}^{kc}_t\right] f^x\left(x'|n, x_{nt}\right) \end{aligned}$$

where x' and $\bar{\omega}'$ denote the next period values for x and $\bar{\omega}$. We can also replace the expectation of the ex-ante continuation values with respect to the city-specific state variables with their realized counterparts and an expectational error defined in Equation 11:

$$\mathbb{E}\left[\bar{V}^k_{xnt}\right] = \sum_{x'} \bar{V}(x', \bar{\omega}^{kc}_{t+1}) f^x\left(x'|n, x_{nt}\right) + e^{\bar{V}}(n, x_{nt}, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}), \tag{A12}$$

with

$$e^{\bar{V}}\left(n, x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}\right) \equiv \sum_{x'} e^{\bar{V}}\left(x', \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}\right) f^x\left(x'|n, x_{nt}\right).$$

These realized continuation values permit minimal assumptions about households' beliefs over the evolution of the city-specific state variables. Substituting our expression for households' expected continuation values conditional on their household-level state variables in A12 to our expression for the difference in conditional choice probabilities in A11 yields

$$\begin{aligned} \ln\left(\frac{p^k_{xnt}}{p^k_{xOO^ct}}\right) &= \bar{u}^k_{xnt} - \bar{u}^k_{xOO^ct} \\ &\quad + \delta\left(\sum_{x'} \bar{V}(x', \bar{\omega}^{kc}_{t+1})\, f^x\left(x'|n, x_t\right) - \sum_{x'} \bar{V}(x', \bar{\omega}^{kc}_{t+1})\, f^x\left(x'|OO^c, x_t\right)\right) \\ &\quad + \delta \cdot \tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}) \end{aligned}$$

where $\tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1})$ is the difference between the expectational errors when residing in neighborhood n relative to OO^c in period t,

$$\tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}) \equiv e^{\bar{V}}(n, x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}) - e^{\bar{V}}(OO^c, x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}).$$

[76] This derivation mostly follows procedures used in the ECCP literature, particularly Diamond, McQuade, and Qian (2018), Davis et al. (2021), and Almagro and Domínguez-Iino (2025).

Next, using Equation 10 to substitute in for $\bar{V}(x', \bar{\omega}^{kc}_{t+1})$, we obtain,

$$\begin{aligned}
&\ln\left(\frac{p^k_{xnt}}{p^k_{xOO^ct}}\right) + \delta\left(\sum_{x'} \ln\left(p^k_{x'\bar{n}t+1}\right) f^x\left(x'|n, x_t\right) - \sum_{x'} \ln\left(p^k_{x'\bar{n}t+1}\right) f^x\left(x'|OO^c, x_t\right)\right) \\
&= \bar{u}^k_{xnt} - \bar{u}^k_{xOO^ct} + \delta\left(\sum_{x'} \nu^k_{\bar{n}}\left(x', \bar{\omega}^{kc}_{t+1}\right)\ f^x\left(x'|n, x_t, \bar{\omega}^{kc}_t\right) - \sum_{x'} \nu^k_{\bar{n}}\left(x', \bar{\omega}^{kc}_{t+1}\right)\ f^x\left(x'|OO^c, x_t, \bar{\omega}^{kc}_t\right)\right) \\
&+ \delta \cdot \tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1})
\end{aligned}$$

Recall that $\bar{n}$ is a renewal action for both households. Therefore, the household-level state variables are set to the same values for both households regardless of their values in period t. This yields identical continuation values in period $t+1$ for both households. The above expression therefore simplifies to,

$$\begin{aligned}
&\ln\left(\frac{p^k_{xnt}}{p^k_{xOO^ct}}\right) + \delta\left(\sum_{x'} \ln\left(p^k_{x'\bar{n}t}\right) f^x\left(x'|n, x_t\right) - \sum_{x'} \ln\left(p^k_{x'\bar{n}t}\right) f^x\left(x'|OO^c, x_t\right)\right) \\
&= \bar{u}^k_{xnt} - \bar{u}^k_{xOO^ct} + \delta\left(MC^k_{t+1}(\bar{n}, n) - MC^k_{t+1}(\bar{n}, OO^c) + \tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1})\right)
\end{aligned}$$

Applying assumption (b) and substituting in for the neighborhoods' flow utilities provides an equation linear in our model parameters,

$$\begin{aligned}
&\ln\left(\frac{p^k_{xnt}}{p^k_{xOO^ct}}\right) + \delta\left(\sum_{x'} \ln\left(p^k_{x'\bar{n}t}\right) f^x\left(x'|n, x_t\right) - \sum_{x'} \ln\left(p^k_{x'\bar{n}t}\right) f^x\left(x'|OO^c, x_t\right)\right) \\
&\quad= \bar{\tilde{\alpha}}^k_n + \tilde{\alpha}^{kc}_t + \tilde{\beta}^k_I \ln(\bar{I}^k_{nt}) + \tilde{\beta}^k_r \ln(r_{\tau nt}) + \tilde{\beta}^k_A \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) \\
&\quad+ \tilde{\beta}^k_\tau \left(\sum_{x'} \ln(\tau_{xt}(x')) f^x\left(x'|n, x_{xt}\right) - \sum_{x'} \ln(\tau_{xt}(x')) f^x\left(x'|OO^c, x_t\right)\right) \\
&\quad+ MC^k_t(n, \underline{n}) - MC^k_t(OO^c, \underline{n}) + \delta\left(MC^k_t(\bar{n}, n) - MC^k_t(\bar{n}, OO^c)\right) \\
&\quad+ \delta \cdot \tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}) + \tilde{\xi}^k_{nt},
\end{aligned}$$

where $\bar{\tilde{\alpha}}^k_n = \tilde{\alpha}^k_n - \alpha^{kc}$.[77] To condense notation, we write this equation as

$$Y^k_{x\underline{n}n\bar{n}t} = \bar{\tilde{\alpha}}^k_n + \tilde{\alpha}^{kc}_t + \tilde{\beta}^k_I \ln(\bar{I}^k_{nt}) + \tilde{\beta}^k_r \ln(r_{\tau nt}) + \tilde{\beta}^k_A \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) + \tilde{\beta}^k_\tau \tilde{\tau}_{xt} + \widetilde{MC}^{kc}_t + \nu^k_{x\underline{n}n\bar{n}t},$$

where

$$\begin{aligned}
Y^k_{x\underline{n}n\bar{n}t} &\equiv \ln\left(\frac{p^k_{xnt}}{p^k_{xOO^ct}}\right) + \delta\left(\sum_{x'} \ln\left(p^k_{x'\bar{n}t+1}\right) f^x\left(x'|n, x_t\right) - \sum_{x'} \ln\left(p^k_{x'\bar{n}t+1}\right) f^x\left(x'|OO^c, x_t\right)\right) \\
\tilde{\tau}_{xt} &\equiv \sum_{x'} \ln(\tau_{xt}(x')) f^x\left(x'|n, x_{xt}\right) - \sum_{x'} \ln(\tau_{xt}(x')) f^x\left(x'|OO^c, x_t\right) \\
\widetilde{MC}^{kc}_t &\equiv MC^k_t(n, \underline{n}) - MC^k_t(OO^c, \underline{n}) + \delta\left(MC^k_{t+1}(\bar{n}, n) - MC^k_{t+1}(\bar{n}, OO^c)\right)
\end{aligned}$$

[77] $\tilde{\beta}^k_r \ln(r_{\tau nt})$ is also formally given by $\tilde{\beta}^k_r \left(\ln(r_{nt}) + \sum_{x'} \ln\left(h_n\left(\tau_{xt}(x'), \{\bar{\omega}^{kc}_t\}^t_{t-\tau}\right)\right) f^x\left(x'|n, x_{xt}\right)\right)$, but we suppress this notation here.

$$\nu^k_{x\underline{n}n\bar{n}t} \equiv \delta \cdot \tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}) + \tilde{\xi}^k_{nt}.$$

This is the same equation reported in 12. Its empirical analogue is given by

$$\hat{Y}^k_{x\underline{n}n\bar{n}t} = \bar{\tilde{\alpha}}^k_n + \tilde{\alpha}^{kc}_t + \tilde{\beta}^k_I \ln(\bar{I}^k_{nt}) + \tilde{\beta}^k_r \ln(\hat{r}_{\tau nt}) + \tilde{\beta}^k_A \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) + \tilde{\beta}^k_\tau \hat{\bar{\tau}}_{xt} + \widehat{\widetilde{MC^F}}^{kc}_t + \nu^k_{x\underline{n}n\bar{n}t},$$

where,

$$\begin{aligned}
\hat{Y}^k_{x\underline{n}n\bar{n}t} &\equiv \ln\left(\frac{\hat{p}^k_{xnt}}{\hat{p}^k_{xOO^ct}}\right) + \delta\left(\sum_{x'} \ln\left(\hat{p}^k_{x'\bar{n}t+1}\right) f^x\left(x'|n, x_t\right)\right. \\
&\qquad \left. - \sum_{x'} \ln\left(\hat{p}^k_{x'\bar{n}t+1}\right) f^x\left(x'|OO^c, x_t\right)\right) - \widehat{\widetilde{MC^V}}^{kc}_t \\
\ln(\hat{r}_{\tau nt}) &\equiv \ln(r_{nt}) + \sum_{x'} \ln\left(h_n\left(\bar{\tau}_{xt}(x'), \{\bar{\omega}^{kc}_t\}\right)\right) f^x\left(x'|n, x_{xt}\right) \\
\hat{\bar{\tau}}_{xt} &\equiv \sum_{x'} \ln(\bar{\tau}_{xt}(x')) f^x\left(x'|n, x_t\right) - \sum_{x'} \ln(\bar{\tau}_{xt}(x')) f^x\left(x'|OO^c, x_t\right) \\
\widehat{\widetilde{MC^\upsilon}}^{kc}_t &\equiv \widehat{MC^\upsilon}^k_t(n, \underline{n}) - \widehat{MC^\upsilon}^k_t(OO^c, \underline{n}) + \delta\left(\widehat{MC^\upsilon}^k_{t+1}(\bar{n}, n) - \widehat{MC^\upsilon}^k_{t+1}(\bar{n}, OO^c)\right) \\
\nu^k_{x\underline{n}n\bar{n}t} &\equiv \delta \cdot \tilde{e}(x_t, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}) + \tilde{\xi}^k_{nt}.
\end{aligned}$$

and where $\hat{\cdot}$ represents estimates from the first step. $\widehat{\widetilde{MC^\upsilon}}^{kc}_t$ is the difference in either the fixed, $\upsilon = F$, or variable, $\upsilon = V$, portion of moving costs.

B.5. Residential Paths

This subsection details the complete set of residential paths we use to estimate our second- and third-step equations and how each type of residential path helps identify our structural parameters.

Complete Set of Residential Paths. Figure A4 plots each pair of residential paths used in the structural estimation. Each pair of residential paths involves two hypothetical type-k households, with one of these households always choosing the outside option in period t. The other household either moves to a different neighborhood within their urban core (paths 1(a), 1(b), 3(a) and 3(b)) or stays in their current neighborhood (paths 2(a) and 2(b)). Residential paths 3(a) and 3(b) both start from within the city's outside option. Residential paths 1(a), 2(a), and 3(a) are initiated by incumbent residents with aggregate tenure length $\bar{\tau} = 1$, while residential paths 1(b), 2(b), and 3(b) are initiated by incumbent residents with aggregate tenure length $\bar{\tau} = 2$. Households with aggregate tenure length $\bar{\tau} = 1$ who stay in their current neighborhood in period t transition to aggregate tenure length $\bar{\tau} = 2$ with probability g^k_{nt}. Every path ends in period $t+1$ with both hypothetical households convening in the same neighborhood within the urban core.

Identification of Fixed Moving Costs and Preference for Tenure. The additional residential paths in Figure A4 beyond those described in Figure 6 help identify households' fixed moving costs and preferences for residential tenure. Table A9 records differences in fixed moving costs and residential tenure utility for each pair of residential paths. The table shows that differences in the likelihood households choose the residential path not passing through the outside option in 1(a)/1(b) relative to in 2(a)/2(b) or in 3(a)/3(b) separately identifies the fixed moving costs for each CBSA given some δ. Variation across neighborhoods in the probability a household with aggregate tenure $\bar{\tau} = 1$ staying in the same path origin transitions to $\bar{\tau} = 2$ in the following period—i.e. g_n^k—identifies $\tilde{\beta}_\tau^k$. Note that only paths 2(a)/2(b) or 3(a)/3(b) are necessary to identify households' fixed moving costs or preferences over residential tenure. We therefore run robustness checks in Section C.1 that exclude residential paths 3(a) and 3(b). These paths identify preferences over residential tenure from households' propensities to transition to $\bar{\tau} = 2$ while in their outside option, a source of variation less attuned to the model framework.

TABLE A9. Fixed Moving Costs and Residential Tenure

Residential Path	$\widetilde{MC^F}^{kc}$	$\tilde{\beta}_\tau^k \cdot \hat{\bar{\tau}}$
Path 1(a)	$(1+\delta)\left(MC^{kc} - MC_{OO}^{kc}\right)$	0
Path 1(b)	$(1+\delta)\left(MC^{kc} - MC_{OO}^{kc}\right)$	0
Path 2(a)	$-MC_{OO}^{kc} + \delta\left(MC^{kc} - MC_{OO}^{kc}\right)$	$\tilde{\beta}_\tau^k \cdot g_n^k \cdot \ln(2)$
Path 2(b)	$-MC_{OO}^{kc} + \delta\left(MC^{kc} - MC_{OO}^{kc}\right)$	$\tilde{\beta}_\tau^k \cdot \ln(2)$
Path 3(a)	$MC_{OO}^{kc} + \delta\left(MC^{kc} - MC_{OO}^{kc}\right)$	$-\tilde{\beta}_\tau^k \cdot g_{OO_c}^k \cdot \ln(2)$
Path 3(b)	$MC_{OO}^{kc} + \delta\left(MC^{kc} - MC_{OO}^{kc}\right)$	$-\tilde{\beta}_\tau^k \cdot \ln(2)$

Notes: Table A9 records differences in fixed moving costs and residential tenure utility for each pair of residential paths displayed in Figure A4 and discussed in text B.5.

Restrictions on Residential Paths. Including all residential path pairs in the second step of the structural estimation is computationally infeasible. For a city with $N^c + 1$ potential path origins in period $t-1$, households can choose among N^c destination census tracts in period t, and $N^c - 1$ renewal actions in period $t+1$. With two individual tenure states, the number of possible residential paths within a single city is $2 \times (N^c + 1) \times N^c \times (N^c - 1)$. For the largest urban core in our sample, New York-Newark-Jersey City, which contains roughly 3,760 census tracts, the total number of potential residential paths rises to over 100 billion. To address this computational challenge and to help credibly identify households' preference parameters, we restrict the set of residential paths used in the second and final step of estimation as follows:

(a) *Restrict set of path origins in period* $t-1$: We restrict the set of path origins within each city to include the city's outside option and the top ten percent of census tracts with the highest share of k-type low-income renters in the city's urban core. Besides easing the computational burden, this restriction helps identify β_τ^k. Since we construct g_n^k separately for each path origin, restricting to path origins with the highest sample shares reduces measurement error in g_n^k. This is important given we estimate β_τ^k via OLS in a third-step conditional on our second-step preference estimates, as described in Footnote 35.

(b) *Restrict set of destination locations in period* t: We drop residential paths from the second estimation step based on the following criteria for destination tracts:

   (i) destination tracts must be above the bottom percentile of the nationwide distribution of tract adult population in 2011;

   (ii) destination tracts must be below the top quartile of the nationwide tract-level adult college-share distribution in 2011;

   (iii) destination tracts must be within the urban core, defined by population weighted distance to metropolitan division CBDs in 2011;

   (iv) destination tracts must be outside the top and bottom percentiles of the distributions for all three instrumental variables and log-differenced endogenous neighborhood characteristics, where the log-differences are between years 2011 and 2018.

   While also easing computation burden, the restrictions prevent the local-average treatment effect that identifies the preference parameters from being driven by outliers or by tracts that households are exceedingly unlikely to choose.

(c) *Restrict set of renewal actions in period* $t+1$: We restrict renewal tracts in period $t+1$ to those most likely to reset households' individual neighborhood tenures. A tract qualifies as a renewal action if it:

   (i) lies inside the urban core, defined by population weighted distance to metropolitan division CBDs in 2011;

   (ii) is 10-15 miles from its destination tracts centroid;

   (iii) is within .2 and .6 2011-population weighted distance from its Metropolitan Division CBD.

   We determined these cutoffs by comparing the subsequent tenure lengths of sample households moving to candidate renewal tracts from destination tracts within the urban core to those arriving from outside the urban core (the outside option). These cutoffs minimize the difference in average future tenure lengths between these groups of households.

Weighting Residential Paths. The number of residential paths selected for estimation scales with the cube of the number of tracts in a CBSA. As a result, unweighted second- and third-step regressions would overweight neighborhoods in larger CBSAs. We therefore weight each residential path according to,

$$w^k_{\bar{\tau},\underline{n}n\bar{n}} = \frac{\left|I^k_{\bar{\tau},n,2011}\right|}{\left|\left\{(\underline{n},\bar{n}) \in \mathbb{N}_{\underline{n}n\bar{n}}\right\}\right|},$$

where $\underline{n}n\bar{n}$ denotes the path origin, destination, and renewal tracts defining a residential path, where $|I^k_{\bar{\tau},n,2011}|$ is the total number of sample households in destination tract n in 2011 with aggregate tenure $\bar{\tau}$, and where $\left|\left\{(\underline{n},\bar{n}) \in \mathbb{N}_{\underline{n}n\bar{n}}\right\}\right|$ is the total number of residential paths among the restricted set of residential paths, $\mathbb{N}_{\underline{n}n\bar{n}}$, that include n as their destination neighborhood. Each destination tract in the second- and third- estimation steps are therefore weighted in proportion to the number of sample households in 2011. These weights also help ensure that our estimation is not driven by large relative changes to initially small move probabilities, which are more susceptible to sampling bias.

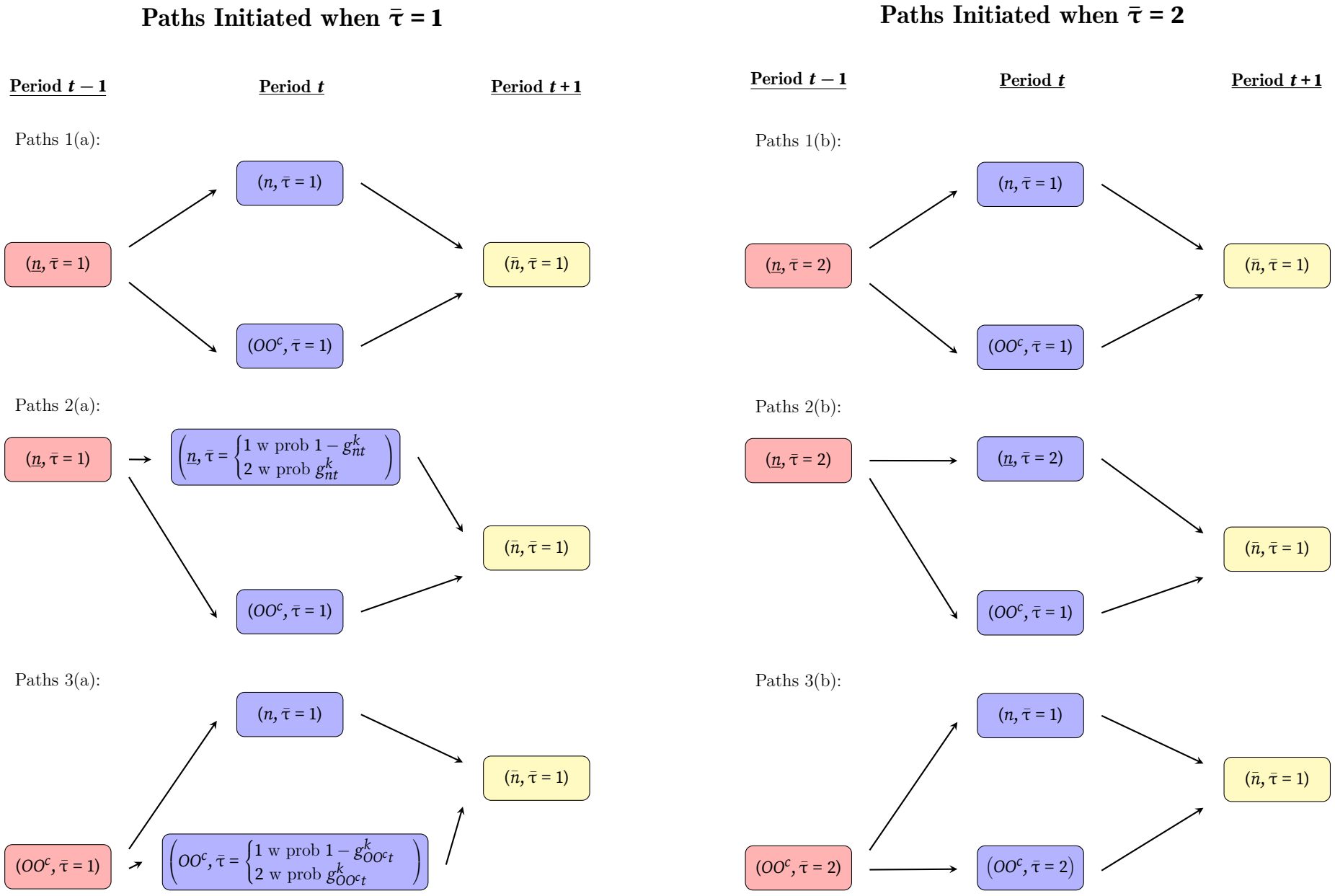


FIGURE A4. Residential Paths

Note: Figure A4 plots the residential paths we use to form estimating equations analogous to Equation 12. Residential paths 1(a), 2(a), and 3(a) are initiated by incumbent residents in neighborhood $\underline{n}$ with aggregate tenure length 1; residential paths 1(b), 2(b), and 3(b) are initiated by incumbent residents with aggregate tenure length 2. We discuss each path's role in identification in the Appendix main text.

Appendix C. Parameter Estimates

C.1. Identifying Endogenous Parameters in Equation 15

Instrument validity implies the following moment restriction,

$$0 = \mathbb{E}\left[\tilde{z}_n \, \Delta\nu^k_{x\underline{n}n\bar{n}}\right],$$

where $\tilde{z}_n = z_n - \mathbb{E}\left[z_n | X_{nt=2010}, \delta_{\underline{n}}\right]$ is our instrument vector residualized on baseline destination tract characteristics and path-origin fixed effects, and where the error term can be decomposed into changes in exogenous neighborhood amenities and changes in households' expectational errors, $\Delta\nu^k_{x\underline{n}n\bar{n}} = \Delta\xi^k_n + \Delta\left(\delta \cdot \tilde{e}(x, \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t'})\right)$. We discussed identification of our endogenous parameters with respect to households' expectational errors in Section 5.3. Here, we more formally discuss instrument exogeneity with respect to changes in neighborhood exogenous amenities, $\Delta\xi^k_n$, and how we support this condition in practice.

Job Market Access Instrument. Identification based on the job market access instrument in Equation 18 requires conditional exogeneity of either the Bartik shares, $\text{EE}^k_{n\iota t_0} / \sum_{\iota'} \text{EE}^k_{n\iota' t_0}$ (Goldsmith-Pinkham, Sorkin, and Swift 2020), or the Bartik shocks, $\ln\left(L^c_{\iota t} / L^c_{\iota t_0}\right)$ (Borusyak, Hull, and Jaravel 2022). The Bartik shares approximate the share of a tract's baseline tradable employment exposure, measured in $t_0 = 2010$, that comes from each industry ι. The Bartik shocks measure national employment growth in each industry, excluding employment growth within CBSA c. For the Bartik shares to be conditionally exogenous, the 2010 locations of establishments in any industry ι must be unrelated to trends in exogenous amenities after conditioning on baseline tract characteristics. This assumption would fail if establishments in particular industries were systematically located near tracts whose exogenous amenities subsequently improved or declined in ways not predicted by our baseline controls.

We view conditional exogeneity of the Bartik shares as a strong assumption in our setting. Even among tradable industries, establishments may choose locations based on local land costs, transportation infrastructure, or other unobserved factors that also predict the evolution of exogenous amenities.[78] We therefore leverage the LEHD's disaggregated 6-digit NAICS industry classifications and rely instead on the conditional exogeneity of the Bartik shocks. Borusyak, Hull, and Jaravel (2022) show that for the Bartik shocks to be conditionally exogenous, it is sufficient that establishments in industries receiving national employment shocks are not concentrated near tracts that share common trends in unobserved exogenous amenities after conditioning on observed tract characteristics. This conditional exogeneity assumption would be violated if, for example, establishments in manufacturing industries that experienced relatively weak national employment growth in 2010–2013 were concentrated near tracts that were observably similar

[78] Our focus on tradable industries nevertheless reduces the most direct concerns about endogenous establishment locations. To see why, consider a non-tradable industry like full-service restaurants. Because these establishments depend directly on local demand, they may be especially attentive to neighborhood-level amenity changes. If full-service restaurants locate in neighborhoods in anticipation of broader improvements in local unobserved amenities, and national restaurant employment grows relatively fast in 2010–2013, including exposure to restaurant employment growth in the instrument could lead us to incorrectly attribute households' migration responses to improved job market access rather than to changes in unobserved amenities that restaurants correctly anticipated.

in 2010 and subsequently experienced worsening unobserved exogenous amenities.[79]

Restricting the job market access instrument to tradable industries introduces a potential incomplete-shares problem, as defined by Borusyak, Hull, and Jaravel (2022). Incomplete shares arise because the sum of tradable-industry shares, $\sum_{\iota \in \mathcal{T}} \left(\text{EE}^k_{n\iota t_0} \Big/ \left[\sum_{\iota'} \text{EE}^k_{n\iota' t_0} \right] \right)$, need not equal one in every tract. The shares would always sum to one if the outer sum were taken over all industries, rather than restricted to the set of tradable industries $\mathcal{T}$. This variation can violate conditional exogeneity if total exposure to tradable-industry shocks is correlated with changes in exogenous neighborhood amenities. For example, if tracts with greater tradable employment exposure also experience improvements in exogenous amenities, positive tradable-industry employment shocks may lead us to overestimate preferences for job market access. Following Borusyak, Hull, and Jaravel (2022), we therefore control for each tract's sum of tradable-industry shares. These are the "Tradable employment share" controls in Tables A10, A11, and A12. We also include them as part of the "BHJ 2022 exposure controls" in Table 2. In addition, we control for baseline total tradable employment exposure, $\sum_{\iota \in \mathcal{T}} \text{EE}^k_{n\iota t_0}$, which we label in the relevant tables as "Tradable employment total". This control absorbs residual within-CBSA differences in the scale of baseline tradable employment exposure that is not fully captured by the sum of tradable-industry shares. Empirically, including this latter control substantially strengthens the first stage. By contrast, additionally conditioning on baseline residential market access substantially weakens the first stage, consistent with the instrument's predictive power operating partly through baseline differences in access. We therefore omit this control.

Even conditional on our rich set of destination-tract controls and path-origin fixed effects, identification through the job market access instrument could fail if tract exposure to sector-wide employment trends is correlated with changes in exogenous amenities. Consider the long-run decline in U.S. manufacturing, NAICS 31–33, and its relatively weak employment growth during 2010–2013 (Sullivan 2014). If employment growth was persistently slower in manufacturing than in other tradable sectors, the instrument may partly capture broad manufacturing exposure rather than idiosyncratic variation across 6-digit industries. This is problematic if manufacturing establishments were concentrated near tracts with common trends in unobserved amenities. For example, Black households continued to suburbanize throughout our analysis period, potentially reflecting rising suburban exogenous amenities for this group (Bartik and Mast 2026). If manufacturing exposure was correlated with rising suburban amenities even after conditioning on our geographic controls, the instrument could attribute moves among Black households toward increasingly attractive suburban locations to apparent insensitivity to declining job market access, biasing preferences for market access downward.

[79] Using the formal representations in Borusyak, Hull, and Jaravel (2022), the conditional exogeneity condition is $\mathbb{E}\left[g_\iota \middle| \left\{\overline{\Delta \nu^k_\iota}\right\}_\iota, \{s^k_\iota\}_\iota\right] = \mu$, where $s^k_\iota = \sum_n s^k_{n,\iota} = \sum_n \frac{\text{EE}^k_{n\iota t_0}}{\sum_{\iota'} \text{EE}^k_{n\iota' t_0}}$, $g_\iota = \ln\left(L^c_{\iota t}/L^c_{\iota t_0}\right)$, and $\overline{\Delta \nu^k_\iota} = \frac{\sum_{\underline{n}\underline{n}\bar{n}} s^k_{\underline{n},\iota} \Delta \nu^k_{\underline{n}\bar{n}}}{s^k_\iota}$. Because the instrument aggregates employment shocks using baseline employment exposure shares, instrument consistency also depends on the structure of the industry-level aggregation of these shares. Given conditional exogeneity of the Bartik shocks and first-stage relevance, Borusyak, Hull, and Jaravel (2022) show that consistency follows if total exposure to industry shocks is not dominated by a small subset of industries and if industries' national employment shocks are mutually uncorrelated conditional on average industry exposure to trends in unobserved amenities and average industry employment exposure shares. These conditions can be formally written as $\mathbb{E}\left[\sum_\iota \left(s^k_\iota\right)^2\right] \to 0$, and $\text{Cov}\left(g_\iota, g_{\iota'} \middle| \left\{\overline{\Delta \nu^k_\iota}\right\}_\iota, \{s^k_\iota\}_\iota\right) = 0 \; \forall \; (\iota, \iota')$.

Following Borusyak, Hull, and Jaravel (2022), we partly address this concern by allowing expected employment growth to differ across broad sectors. Specifically, we control for each tract's employment exposure to broad NAICS sectors, including total manufacturing exposure.[80] These controls absorb the systematic component of the instrument driven by sector-wide employment trends, leaving identification to come from differences in national employment growth across 6-digit industries within broad sectors. The identifying assumption becomes that, conditional on broad-sector exposure and our other destination-tract controls, industries with relatively stronger employment growth than other industries in the same broad sector were not systematically located closer to tracts with shared trends in unobserved exogenous amenities. Thus, manufacturing establishments may be located near tracts with rising suburban amenities, provided that relatively higher-growth manufacturing industries were not differentially located near such tracts.

While sector-exposure controls reduce reliance on conditionally exogenous employment-growth differences across broad sectors, they naturally make identification more reliant on residual variation across 6-digit industries within sectors. If within-sector residual shocks are highly correlated, or if tracts do not differ enough in their exposure to those residual shocks, consistency conditions analogous to those in Footnote 79 become harder to satisfy and the relevance of the job market access instrument declines. Conditioning on broad-sector employment exposure therefore creates a trade-off between the conditional exogeneity of industry shocks and the instrument's relevance. Indeed, we find that controlling for exposure to all sectors substantially weakens the first stage when the instrument is weighted by non-Black sample households' baseline residential locations. By contrast, controlling only for exposure to one or both of the two largest tradable-industry employers of low-income sample households, Health Care & Social Assistance and Manufacturing, leaves enough residual employment growth for a sufficiently strong first stage. Table A11 reports how the parameter estimates vary across specifications with different sets of employment exposure controls. Once we control for tracts' incomplete tradable shares, the coefficient estimates are quite stable whether or not we additionally control for baseline healthcare or manufacturing sector exposure. By contrast, controlling for every sector's exposure substantially raises non-Black households' estimated preference for job market access, though this change is hard to interpret given the accompanying weak first stage.

The source of identifying variation for job market access informs how we interpret the corresponding preference estimates. Because the main identifying variation comes from shocks to employment opportunities in tradable industries, which employ roughly 45 percent of low-income Black renters and 40 percent of low-income non-Black renters in our sample (cf. Table A21), the estimates should not be interpreted narrowly as preferences for proximity to one's own job. Instead, they capture the value households place on living in neighborhoods with greater access to employment opportunities more broadly, including both direct access to jobs and neighborhood characteristics responsive to stronger job market access, such as consequent non-tradable retail activity, public service investments like transportation infrastructure, and the general economic vitality of a neighborhood.[81] We interpret these estimates in the same spirit as the

[80] Baseline employment exposure to sector M is given by $\sum_{\iota \in \mathcal{T}_M} \left(\mathrm{EE}^k_{n\iota t_0} \Big/ \left[\sum_{\iota'} \mathrm{EE}^k_{n\iota' t_0} \right] \right)$, where $\mathcal{T}_M$ is the set of industries in sector M.

[81] This contrasts with approaches to identification that more directly isolate preferences for proximity to one's own workplace. For example, Su (2022) identifies the value workers place on shorter commutes from whether workers in occupations with larger

estimates for college-graduate shares, which capture the value of neighborhood characteristics responsive to changes in the local college-graduate share, rather than the value of college-graduate neighbors per se. In both cases, we believe our holistic interpretations of the corresponding preference parameters accords with the value that low-income renters place on these neighborhood characteristics when choosing where to live.

Proximity Instrument. Recall from Section 6 that the primary concern with the proximity instrument is that a tract's distance-weighted exposure to other tracts' college-graduate shares, given by $\sum_{m \in \mathcal{N}^c \setminus n} e^{-\rho \tau_{n,m}} \left(\text{Coll}_{m,t_0} / \text{Pop}_{m,t_0} \right)$, may predict changes in unobserved exogenous amenities, even conditional on our rich set of destination-tract controls. This concern threatens identification only if these amenity changes are distinct from the gentrification process captured by changes in a tract's own share of college graduates. For example, non-tradable establishments may increasingly locate in tracts with high distance-weighted exposure to college-graduate shares, which we refer to as high-proximity tracts, before those tracts experience measured increases in their college-graduate shares. If these amenity changes reliably predict future increases in a tract's share of college graduates, then they are best interpreted as leading components of gentrification. Like amenities that respond contemporaneously to changes in college-graduate shares, they are part of the endogenous amenity bundle we aim to capture through $\tilde{\beta}_A^k$.

The more serious concern is that high-proximity tracts may experience amenity changes for reasons unrelated to their own gentrification process. For example, public infrastructure investments may be more likely in high-proximity tracts, even when those investments do not reliably predict future increases in those tracts' shares of college graduates. In that case, the proximity instrument would affect household choices through changes in exogenous amenities rather than through changes in a tract's college-graduate share or the endogenous amenities causally tied to those changes. Thus, the relevant threat to identification is that the instrument predicts amenity changes that are not causally tied to expected or realized gentrification. Under this interpretation, the scope for violations of the exclusion restriction is substantially narrower.

We next describe the source of identifying variation for the proximity instrument. Variation comes largely from differences across tracts within the same CBSA. We found in exploratory regressions that additionally controlling directly for a tract's distance-weighted exposure to other tracts' college-graduate shares substantially weakens the first stage. This suggests that much of the instrument's predictive power reflects within-CBSA re-sorting. At the same time, scaling this exposure by CBSA-wide growth in the number of college graduates, $\ln\left(\text{Coll}_t^c \Big/ \text{Coll}_{t_0}^c \right)$, reduces reliance on potential defier neighborhoods in stagnant or declining CBSAs, where high exposure to initially high-college-graduate-share tracts need not predict gentrification. In Table A12, we replace observed CBSA-wide growth in college graduates with predicted growth from a CBSA-level Bartik instrument, constructed using baseline CBSA-wide employment shares and national employment growth among college graduates using the LEHD and LBD. This specification has a meaningfully weaker first stage, suggesting that the proximity measure predicts tract-level gentrification most strongly in CBSAs that actually experienced growth in their college-graduate populations.

increases in the value of time increasingly sort into neighborhoods with shorter expected commutes to occupation-specific job locations.

Recall also from Section 6 that, to generate variation in tract rents, we interact the proximity measure with baseline tract-level shares of urban land development. Because these development measures also enter the estimating equation directly as controls, identification compares how differences in rent growth between heavily and lightly developed tracts vary with the proximity instrument. The tight connection between the rent and college-graduate-share instruments mirrors the tight correlation between changes in tract-level rents and college-graduate shares. As is well known in the literature, this correlation makes it difficult to separately identify preferences over endogenous amenities and rents. For example, despite using a separate price instrument to help estimate households' disutility from housing costs and preferences over endogenous amenities, Diamond (2016)'s preferred specification externally calibrates households' housing expenditure shares so that it is over-identified.

Because both the rent and college-graduate share instruments are functions of the same proximity measure, the spatial decay parameter, ρ, cannot be calibrated separately for their respective first stages. Moreover, the two first stages need not be strongest at the same value of the spatial decay parameter. College-graduate-share growth may respond most strongly to high values of ρ if gentrification expands outward from nearby high-college-share neighborhoods. Rent growth, by contrast, may reflect demand pressure across a broader housing submarket, and therefore respond more strongly to lower values of ρ. In exploratory analyses, we find that the choice of ρ meaningfully affects the instruments' relative first-stage strengths. We therefore calibrate ρ using k-fold cross-validation on tract-level first-stage regressions for changes in college-graduate shares, and then verify that the implied rent first stage remains sufficiently strong. When weighting tracts by their baseline shares of Black sample households, the selected value of ρ is stable and yields robust first stages for both endogenous variables. When the cross-validation procedure uses non-Black sample household weights, the value of ρ selected from the college-graduate-share first stage yields a relatively weak rent first stage. By contrast, the value selected using Black-household weights yields robust first stages for both endogenous variables when using either Black- or non-Black sample household weights. We therefore use this value of ρ for both samples. For each specification and endogenous variable, we report Sanderson-Windmeijer first-stage F-statistics (SW F-stat) to assess whether the excluded instruments provide sufficient independent variation for each endogenous variable (Sanderson and Windmeijer 2016).

Increases in college-graduate shares are accompanied by heterogeneous combinations of rent growth and changes in job market access. Using the proximity instrument in the descriptive analysis would therefore identify a difficult-to-interpret local average treatment effect for the tracts whose college shares respond to the proximity IV. Moreover, a strong first stage in the structural analysis does not require the instrument to induce gentrification in low-college-graduate-share origin tracts, since identification comes from variation across the destination tracts renters choose among.

C.2. Robustness Checks

Table A12 examines the sensitivity of our estimates to several modeling decisions. Column (1) addresses whether our job market access instrument captures variation specific to low-income renters. If so, controlling for contemporaneous changes in college-educated workers' job market access, while correlated with overall

changes in local labor demand, should not meaningfully affect low-income renters' estimated preferences for job market access. Consistent with this prediction, the job market access coefficient estimates in column (1) are nearly unchanged from our baseline estimates.

Column (2) controls for changes in a tract's proximity to nearby tracts' college-graduate shares. Our proximity instrument is almost surely correlated with such changes, and conditioning on them helps isolate the value low-income renters place on a tract's own college-graduate share and its corresponding amenities, holding the college shares of surrounding neighborhoods fixed. The point estimates are quite similar to our baseline estimates, suggesting that the college-share coefficient captures preferences primarily over amenities tied to the neighborhood itself rather than to its surroundings. This control absorbs much of the instruments' identifying variation, however, so the estimates are far less precise, particularly for Black households. Column (3) excludes residential paths 3(a) and 3(b), described in Figure A4, which begin in each CBSA's outside option. Variation from these paths helps identify preferences over neighborhood characteristics but is less useful for identifying the value of high neighborhood tenure, as described in Appendix B.5. Our estimates are quantitatively similar when we exclude these paths, though the estimated value of neighborhood tenure rises modestly for non-Black households, as reported in Table A13.

Column (4) modifies the proximity instrument by replacing the log change in each CBSA's total college-educated adult population with a Bartik shift that interacts baseline industry employment shares of college-educated adults with log changes in national industry employment. While this formulation may make conditional exogeneity more plausible, predicted changes in CBSA-wide college employment during 2010-2013 translate only loosely into changes in the college-educated population over 2010-2019. The instrument consequently loses considerable relevance, and the coefficient estimates become quite unstable.

Column (5) controls for changes in neighborhood racial composition, specifically the log changes in the shares of adult residents who are white, Black, and of another race. We obtain similar coefficient estimates when including these controls, which is reassuring given our job market access measures are sample-specific. One concern might otherwise be that racial homophily draws low-income renters toward neighborhoods with more same-race residents, which could be correlated with measured job market access. Note, however, that job market access does not mechanically increase with the number of same-race neighbors and may in fact decline, because market access is an equilibrium object that measures the supply of relevant jobs relative to the supply of relevant workers. Holding the number of jobs fixed and all else equal, an inflow of relevant workers reduces measured market access.

Finally, column (6) replaces our proxy for endogenous neighborhood amenities, the log change in the adult college-graduate share, with the log change in the white adult college-graduate share. A concern with our baseline proxy is that it does not directly capture racial change, even though gentrification is often closely associated with changes in neighborhood racial composition (Freeman 2006; Pattillo 2010). The resulting parameter estimates are quantitatively similar to our baseline estimates. This similarity suggests that our race-neutral proxy captures much of the variation in neighborhood change that is relevant for our estimated welfare effects.

TABLE A10. Complete Parameter Estimates

	OLS		IV	
	Black	Non-Black	Black	Non-Black
Endogenous variables				
Rent $\left(\tilde{\beta}_r^k\right)$	-0.0422	.197	-2.633	-3.133
	(0.0247)	(.00707)	(0.383)	(0.280)
College share $\left(\tilde{\beta}_A^k\right)$	-0.187	-.203	0.485	1.293
	(0.00669)	(0.00628)	(0.253)	(0.390)
Job market access $\left(\tilde{\beta}_w^k\right)$	-0.0434	.0216	4.226	1.681
	(0.0235)	(.00860)	(0.684)	(0.684)
Continuous controls				
High tenure	-0.535	-1.095	-1.201	-0.952
	(1.363)	(0.643)	(2.062)	(0.789)
Population-weighted CBD distance	-0.331	-0.357	-0.622	0.882
	(0.053)	(0.0556)	(0.379)	(0.343)
Population-weighted CBD distance squared	0.415	0.406	0.317	-0.851
	(0.071)	(0.0629)	(0.358)	(0.319)
Log college share	-0.0222	-0.0147	0.317	0.427
	(0.0108)	(0.00711)	(0.0605)	(0.0505)
Log college share squared	0.00169	-0.0307	0.0247	0.00653
	(0.00243)	(0.00181)	(0.00717)	(0.00408)
Log hedonic rents	-1.193	-5.713	-3.89	-11.12
	(0.122)	(0.154)	(0.49)	(0.541)
Log hedonic rents squared	0.0964	0.463	0.237	0.795
	(0.00933)	(0.0117)	(0.0298)	(0.0351)
White share	-0.691	-0.502	-0.608	0.0447
	(0.0294)	(0.0227)	(0.0975)	(0.0394)
White share squared	0.928	0.149	0.474	-0.508
	(0.0254)	(0.0191)	(0.0786)	(0.0811)
Black share	-0.232	-0.582	-0.957	-0.542
	(0.0245)	(0.0238)	(0.116)	(0.0586)
Black share squared	0.205	0.746	0.62	0.633
	(0.0192)	(0.0219)	(0.119)	(0.0302)
Urban development	-0.0621	0.078	0.0717	0.302
	(0.0104)	(0.00914)	(0.0182)	(0.0191)
Urban development square	0.0557	-7.94e-5	0.0209	-0.145
	(0.00762)	(0.00607)	(0.0155)	(0.016)
BHJ 2022 exposure controls				
Tradable employment total	2.09e-5	3.79e-7	1.49e-4	2.09e-5
	(2.41e-6)	(8.61e-7)	(3.93e-5)	(1.82e-5)
Tradable employment share	-0.0829	1.236	-0.281	-0.495
	(0.095)	(0.0819)	(0.222)	(0.408)
Tradable healthcare employment share	-0.204	-0.696	-0.871	-0.307
	(0.0788)	(0.0768)	(0.126)	(0.101)
Tradable manufacturing employment share	0.302	-0.589	0.144	0.297
	(0.144)	(0.0659)	(0.295)	(0.382)
Concentric population-weighted CBD-distance rings	✓	✓	✓	✓
Path-origin fixed effects	✓	✓	✓	✓
SW F-stat (Rent)	—	—	93.69	42.66
SW F-stat (College share)	—	—	78.31	53.67
SW F-stat (Job market access)	—	—	83.26	45.57
Kleibergen-Paap F-stat	—	—	22.86	9.29
Residential paths (000s)	58,320	71,610	58,320	71,610

Notes: Table A10 extends the 2SLS estimates from Table 2 by including estimated coefficients for all continuous control variables and all corresponding OLS estimates. See notes to Table 2 for more details.

TABLE A11. Parameter Estimates and Sector Exposure

	(1)		(2)		(3)		(4)		(5)		(6)	
	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black
Rent $\left(\tilde{\beta}_r^k\right)$	-2.42	-2.824	-2.125	-3.181	-2.079	-3.117	-2.423	-3.083	-2.633	-3.133	-2.592	-2.527
	(0.6)	(0.44)	(0.457)	(0.176)	(0.463)	(0.187)	(0.413)	(0.263)	(0.383)	(0.28)	(0.354)	(0.481)
College share $\left(\tilde{\beta}_A^k\right)$	1.101	1.451	0.138	1.562	0.133	1.457	0.344	1.247	0.485	1.293	0.363	0.0488
	(0.158)	(0.122)	(0.325)	(0.207)	(0.333)	(0.219)	(0.269)	(0.363)	(0.253)	(0.39)	(0.182)	(0.603)
Job market access $\left(\tilde{\beta}_w^k\right)$	0.606	0.543	5.374	0.987	5.304	1.265	4.776	1.683	4.226	1.681	3.739	4.46
	(0.33)	(0.268)	(0.946)	(0.246)	(0.958)	(0.298)	(0.758)	(0.646)	(0.684)	(0.684)	(0.532)	(1.114)
Path-origin fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Baseline destination-tract controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
BHJ 2022 exposure controls												
Tradable employment total	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Tradable employment share			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Tradable healthcare employment share					✓	✓			✓	✓	✓	✓
Tradable manufacturing employment share							✓	✓	✓	✓	✓	✓
Each employment sector share											✓	✓
SW F-stat (Rent)	79.68	108.4	84.8	339.2	80.17	221.2	107.4	48.03	93.69	42.66	116.7	26.89
SW F-stat (College share)	76.21	115.7	58.83	260.8	56.5	204.5	78.92	57.26	78.31	53.67	131.9	31.65
SW F-stat (Job market access)	172.1	111.6	60.47	303.3	58.43	209.3	80.95	48.86	83.26	45.57	131.8	26.32
Kleibergen-Paap F-stat	17.94	35.8	18.65	48.61	17.87	39.32	24.27	10.38	22.86	9.29	31.89	5.741
Residential paths (000s)	58,320	71,610	58,320	71,610	58,320	71,610	58,320	71,610	58,320	71,610	58,320	71,610

Notes: Table A11 reports 2SLS estimates of the endogenous variables in Equation 15 for different market-access exposure controls following Borusyak, Hull, and Jaravel (2022). We discuss our choice of these controls in the main text of Appendix C.1. We describe the residential paths used in estimation and their weights in Appendix B.5. Standard errors, in parentheses, are clustered by path origin. "SW F-stat" denotes the Sanderson–Windmeijer conditional F-statistic.

TABLE A12. Parameter Estimates Robustness Checks

	(1)		(2)		(3)		(4)		(5)		(6)	
	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black
Rent $\left(\tilde{\beta}_r^k\right)$	-1.846	-2.991	-2.548	-2.778	-2.675	-3.212	-3.977	-4.397	-2.271	-2.881	-2.681	-3.342
	(0.399)	(0.294)	(0.618)	(0.17)	(0.37)	(0.264)	(0.391)	(0.161)	(0.354)	(0.248)	(0.442)	(0.389)
College share $\left(\tilde{\beta}_A^k\right)$	0.444	1.342	0.391	1.181	0.501	1.306	-0.515	2.32	0.387	1.169	0.112	0.993
	(0.247)	(0.386)	(0.579)	(0.264)	(0.253)	(0.384)	(0.435)	(0.463)	(0.309)	(0.37)	(0.0606)	(0.318)
Job market access $\left(\tilde{\beta}_w^k\right)$	4.201	1.638	5.008	1.298	4.214	1.715	7.612	1.493	4.072	1.839	4.77	1.886
	(0.672)	(0.719)	(3.32)	(0.926)	(0.679)	(0.691)	(1.236)	(1.137)	(0.723)	(0.607)	(0.460)	(0.651)
Path-origin fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Baseline destination-tract controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
BHJ 2022 exposure controls												
Tradable employment total	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Tradable employment share	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Tradable healthcare employment share	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Tradable manufacturing employment share	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Robustness Checks												
Change in college-educated job market access	✓	✓										
Change in proximity to neighborhood college shares			✓	✓								
Exclude residential paths 3(a) and 3(b)					✓	✓						
Interact Bartik shift-share with proximity measure							✓	✓				
Change in racial shares									✓	✓		
White college share as the endogenous variable											✓	✓
SW F-stat (Rent)	83.07	38.84	5.357	80.96	90.73	47.13	172.5	502.8	91.85	53.19	189.2	37.99
SW F-stat (College share)	75.29	51.71	4.709	37.78	76.65	56.87	53.25	41.68	60.1	63.15	239.2	51.68
SW F-stat (Job market access)	81.49	43.42	4.578	31.99	81.24	47.26	55.26	32.47	64.39	51.83	219.8	47.55
Kleibergen-Paap F-stat	22.12	9.109	1.466	6.436	22.46	9.546	11.79	5.66	17.61	11.22	56.47	8.606
Residential paths (000s)	58,320	71,610	58,320	71,610	58,030	71,270	58,320	71,610	58,320	71,610	58,320	71,610

Notes: Table A12 reports 2SLS estimates of the endogenous variables in Equation 15 for six robustness checks. We discuss these robustness checks in Appendix C.2. We describe the residential paths used in estimation and their weights in Appendix B.5. Standard errors, in parentheses, are clustered by path origin. "SW F-stat" denotes the Sanderson–Windmeijer conditional F-statistic.

TABLE A13. Fixed Moving Cost Pseudo-Percentiles and Value of Neighborhood Tenure

	(1)						(2)					
	Black			Non-Black			Black			Non-Black		
Pseudo-percentile	25th	50th	75th	25th	50th	75th	25th	50th	75th	25th	50th	75th
Fixed tract moving cost $\left(MC^{kc}\right)$	-3.736	-3.319	-2.922	-3.587	-3.221	-2.948	-3.719	-3.308	-2.919	-3.526	-3.175	-2.907
Fixed CBSA moving cost $\left(MC^{kc}_{OO}\right)$	-3.784	-3.368	-2.938	-3.664	-3.348	-3.118	-3.707	-3.226	-2.853	-3.500	-3.147	-2.86
High neighborhood tenure $\left(\tilde{\beta}^{k}_{\tau}\right)$		0.351			0.255			0.389			0.413	
		(0.073)			(0.0329)			(0.0169)			(0.0222)	
Exclude residential paths 3(a) and 3(b)							✓	✓	✓	✓	✓	✓
Residential paths (000s)		423,300			506,400			421,300			504,000	

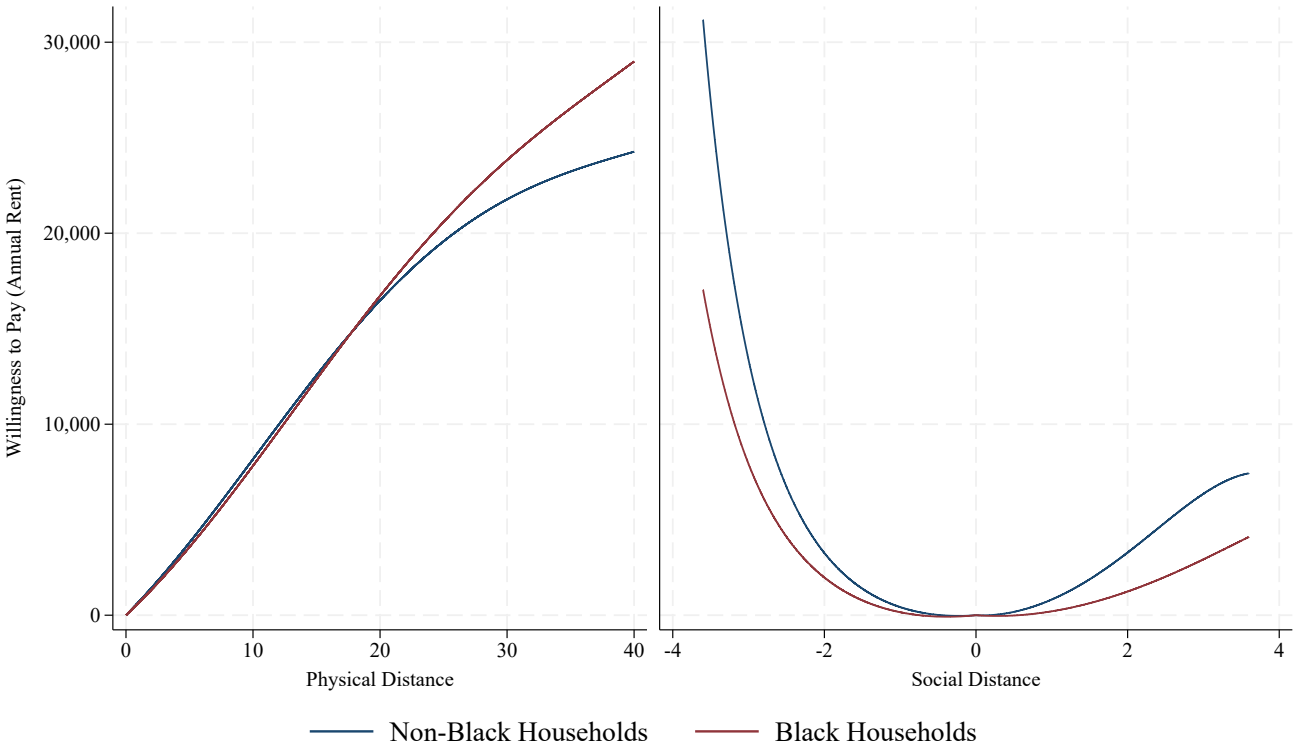
Notes: Table A13 reports the pseudo-percentiles of the city-specific fixed moving cost parameters from Equation 14 as well as the value of high neighborhood tenure. Pseudo-percentiles are constructed as the mean of the five CBSA-specific parameter estimates whose ranks are centered on the indicated percentile of the CBSA distribution. We pool all cross-sections from 2010–2019 and use the same residential paths and weights as when estimating Equation 15. Column (2) drops residential paths that originate in a city's outside option (paths 3(a) and 3(b) in Figure A4). The value of high neighborhood tenure is constrained to be constant across CBSAs. We could not compute standard errors for the fixed moving cost parameters due to computational constraints. We report the total number of residential paths used to estimate Equation 14.

Table A14. Urban-Core Structure and Fixed Moving Costs

	Fixed tract moving cost $\left(MC^{kc}\right)$		Fixed CBSA moving cost $\left(MC^{kc}_{OO}\right)$	
	Non-Black	Black	Non-Black	Black
Log mean tract density in CBSA urban core	-0.838 (0.0566)	-0.923 (0.0617)	-0.857 (0.0573)	-0.908 (0.0624)
Log number of tracts in CBSA urban core	-0.326 (0.041)	-0.333 (0.0427)	-0.274 (0.0415)	-0.333 (0.0432)
Constant	4.784 (0.300)	5.544 (0.356)	4.559 (0.304)	5.402 (0.360)
Adjusted R-squared	0.938	0.931	0.931	0.928
N	90	90	90	90

Notes: Table A14 reports estimates from CBSA-level regressions of fixed moving costs on urban-core characteristics. Tract density is measured as population per square mile in 2010. Mean tract density is the average density across tracts in the CBSA urban core, weighted by the number of low-income renters in the sample in each tract. Regressions are weighted by the number of tracts in each CBSA urban core. Outliers are excluded separately from each renter sample. Standard errors are reported in parentheses. N is the rounded number of CBSAs with valid fixed moving cost estimates.

Figure A5. Variable Moving Cost Willingness to Pay Measures



Notes: Figure A5 translates the variable moving cost coefficients in Table 3 into households' willingness to pay, over and above any fixed moving costs, to avoid a move of a given distance within the same city. Physical distance is in miles; social distance is the log-difference between tracts' college-graduate shares. For reference, a social-distance value of 2 represents a move from a tract with a 9 percent college-graduate share (the sample average) to one with 66 percent.

Appendix D. Descriptive Analyses

D.1. Baseline Residential Mobility

Table A15 reports additional baseline residential mobility statistics not shown in Figure 1. Column (2) reports these statistics for the 2010 sample of low-income renters used to construct the empirical neighborhood choice probabilities, defined after applying restriction (g) in Table A1. Comparing Columns (1) and (2) shows similar baseline mobility patterns, though low-income renters outside low-college-share tracts near the metropolitan division CBD tend to move slightly less frequently, but over longer distances. The smaller average differences among Black renters partly reflect the fact that a larger share of Black low-income renters already lived in low-college-share tracts near the metropolitan division CBD.

Panel C shows that the median non-Black and Black low-income renter moved once between 2010 and 2019. The mean number of moves is only slightly higher, at 1.145 for non-Black renters and 1.395 for Black renters, while the positive skew indicates a right tail of more frequent movers. Together, these statistics suggest that most households made few moves over the decade, while a smaller group of renters moved more often. This pattern is consistent with duration dependence among low-income renters, and therefore helps motivate allowing the value of remaining in place to vary with residential tenure in the structural model.

We find evidence of lower mobility among low-income renters in census tracts where a larger share of sample households are predicted to live in housing units covered by a binding rent control ordinance. Figure A6 plots conditional means from a regression of tract out-migration between 2010 and 2019 on our tract-level measure of predicted binding rent control coverage. The specification is analogous to Equation 1, but excludes the low-income housing unit controls (which include our rent control measures) and replaces the gentrification measure with our measure of binding rent control coverage. Low-income renters predicted to be more likely to live in binding rent-controlled units are significantly less likely to leave their census tract between 2010 and 2019. This relationship is more pronounced for non-Black households, consistent with evidence that racial minorities are less likely to receive the protections afforded by rent control or rent stabilization policies (Ambrose et al. 2026). We incorporate this evidence of lower mobility under binding rent control in the structural model by allowing rents within tracts covered by binding rent control ordinances to vary with the length of renters' residential tenure.

D.2. Gentrification and Residential Mobility

Table A17 reports the coefficient estimates presented in Figure 3 from Equation 1, along with additional estimates for renters' 2019 distance from their origin census tract and from their metropolitan division CBD. Panel A also reports OLS estimates from regressions of tract out-migration on our gentrification measure, without instrumenting for it using the change in the number of adult college graduates normalized by the tract's 2010 population. The OLS estimates are marginally smaller than the corresponding IV estimates, even though the mechanical link between out-migration and measured gentrification raises the OLS estimates, all else equal. Panels D and E show that when a tract's share of college graduates increases

TABLE A15. Baseline Residential Mobility

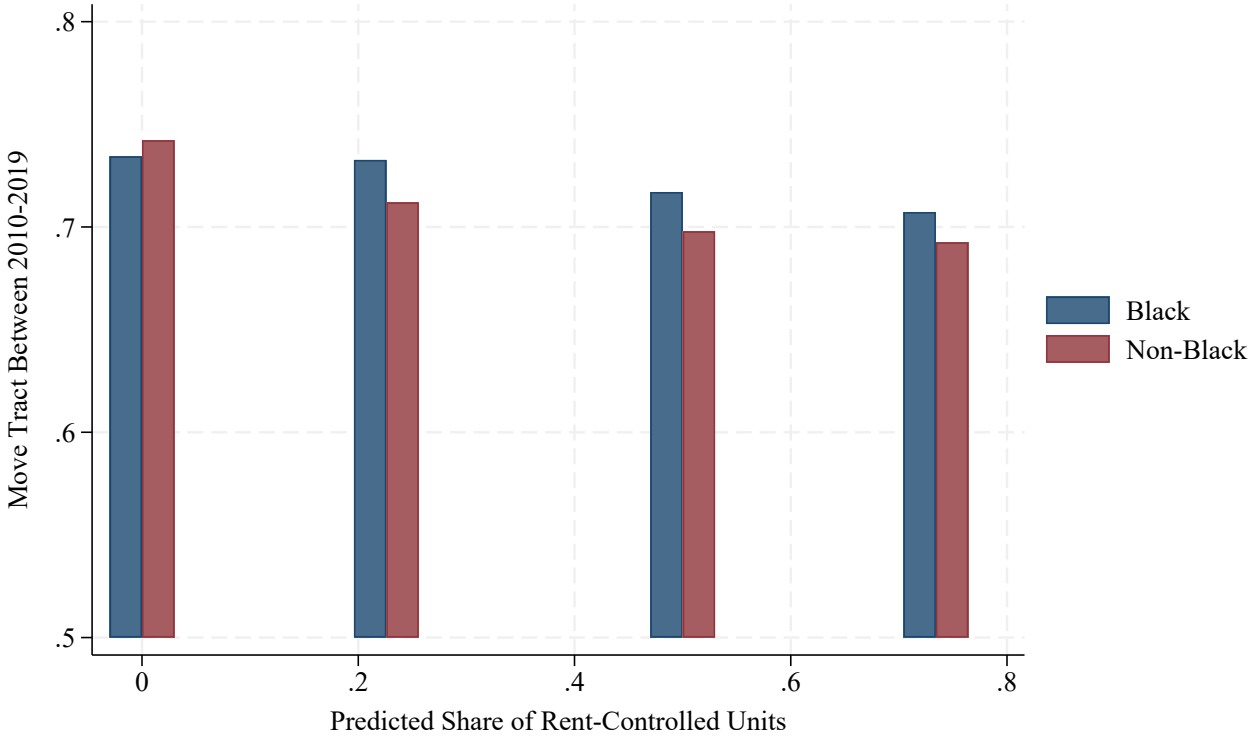
	(1) Urban Low-Inc Renters		(2) Low-Inc Renters		(3) All Low-Inc HHs		(4) All HHs
	Non-Black	Black	Non-Black	Black	Non-Black	Black	
Panel A							
Move tract in 2010	0.1849	0.1869	0.1808	0.1844	0.1139	0.1457	0.1025
Move CBSA in 2010	0.03401	0.02230	0.04221	0.02715	0.02800	0.02251	0.02473
Move > 1 mile in 2010	0.1750	0.1770	0.1742	0.1770	0.1105	0.1400	0.09972
Move > 5 miles in 2010	0.1060	0.1014	0.1180	0.1087	0.07758	0.08832	0.07212
First-move distance	7.734	7.189	9.118	7.970	9.748	8.254	10.31
Standard deviation	(9.316)	(8.021)	(10.42)	(8.639)	(10.90)	(8.870)	(10.89)
Pseudo-median	[4.702]	[4.768]	[5.691]	[5.244]	[6.192]	[5.457]	[6.756]
Skew	⟨3.67⟩	⟨4.675⟩	⟨3.459⟩	⟨3.649⟩	⟨3.330⟩	⟨3.458⟩	⟨2.884⟩
Panel B							
Move tract between 2010-2019	0.6984	0.7298	0.7068	0.7269	0.5409	0.6307	0.5207
Distance from 2010 tract in 2019	8.687	7.854	10.07	8.726	10.88	9.107	11.77
Standard deviation	(10.50)	(8.920)	(11.23)	(9.522)	(11.69)	(9.771)	(12.00)
Pseudo-median	[5.299]	[5.158]	[6.429]	[5.724]	[7.135]	[6.061]	[7.956]
Skew	⟨3.601⟩	⟨3.901⟩	⟨2.894⟩	⟨3.179⟩	⟨2.769⟩	⟨3.088⟩	⟨2.495⟩
Panel C							
Number moves in 2010–2019	1.145	1.395	1.040	1.309	0.7246	1.084	0.6345
Standard deviation	(1.289)	(1.437)	(1.214)	(1.393)	(1.072)	(1.329)	(0.9839)
Pseudo-median	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]
Skew	⟨1.315⟩	⟨1.131⟩	⟨1.374⟩	⟨1.180⟩	⟨1.806⟩	⟨1.397⟩	⟨1.940⟩
N	517,000	351,000	1,994,000	799,000	5,866,000	1,594,000	24,830,000

Notes: Table A15 summarizes residential mobility for households in 2010. Panel A compares each household head's 2010 residential tract to their 2011 residential tract; Panel B compares their 2010 residential tract to their 2019 residential tract; Panel C reports the number of residential tract moves between 2010 and 2019. Columns (1), (2), (3), and (4) use the samples defined after applying restrictions (h.i), (g), (f), and (d) in Table A1, respectively. Household counts correspond to Panel A (counts for Panels B and C are similar). We compute residential mobility statistics only when the household head's census tract is observed in both years relevant to that panel (2010 and 2011 for Panel A; 2010 and 2019 for Panels B and C). The move-distance statistics are computed only for households we observe in the same CBSA in the end year of our comparisons. "First-move distance" and "Distance from 2010 tract in 2019" are measured in miles and are computed only for households who leave their origin census tract.

by one log point, non-Black and Black low-income renters live only 0.078 and 0.081 miles farther from their origin tract in 2019, respectively, than comparable renters in non-gentrifying neighborhoods. They also live only 0.012 and 0.035 miles farther from their CBD, respectively. The estimates are quantitatively similar when restricting the sample to gentrifying tracts, to tracts not covered by any rent control ordinance, or to long-term renters.

Table A18 reports coefficient estimates from variants of the baseline specification in Equation 1. We choose these variants to assess the robustness of the baseline results, further examine whether the small average mobility responses mask heterogeneity across relevant subsamples, and evaluate how much the baseline estimates are attenuated when using ACS-based gentrification measures. Panel A in Table A18 first replaces our baseline gentrification measure, constructed from the de-biased tract estimates, with two alternative measures. The first uses our 5-year ACS tract estimates, and the second uses the log change in the share of white adult college graduates. Although our de-biased neighborhood change estimates remain imperfect (cf. Figure A2), they appear to meaningfully reduce attenuation from classical measurement error. Using the 5-year ACS tract estimates alone produces coefficients that are only 33 and 30 percent

FIGURE A6. Baseline Residential Mobility and Rent Control



Notes: Figure A6 shows the probability incumbent renters move census tracts between 2010-2019 as a function of their origin tract's predicted share of binding rent-controlled units. The bars plot conditional means estimated from a regression that includes the same baseline fixed effects and individual- and tract-level controls as in Equation 1. The number of bins is chosen to minimize integrated mean squared error, though seven bins are clustered at 0 (Cattaneo et al. 2024). Construction of the predicted rent-controlled shares is described in Section A.1. The sample comprises all urban low-income renter households observed in both 2010 and 2019, defined after applying restriction (h.ii) in Table A1.

as large as the corresponding estimates using the de-biased tract estimates for non-Black and Black households, respectively.[82] Measuring gentrification as the log change in the share of white adult college graduates yields point estimates close to our baseline estimates.

Panel B of Table A18 restricts the descriptive sample to subsamples that are useful for assessing whether the small average estimates mask meaningful heterogeneity. We first limit the sample to low-income renters initially residing within 0.25 units of 2010-population-weighted distance from their metropolitan division CBD, compared with 0.5 units in the full descriptive sample. The estimates are somewhat smaller (by 0.004 and 0.009) to those from the baseline sample. We next drop all household heads whom we observe to have at least a Bachelor's degree, as well as those whom the LEHD imputes as likely to have at least a Bachelor's degree.[83] This restriction serves two purposes. First, it limits the analysis to renters who are more likely to be socioeconomically disadvantaged, conditional on income. Second, it avoids a mechanical

[82]The standard deviation of the de-biased gentrification measure is 0.2816, while the standard deviation of the ACS 5-year estimate centered at 2010 and 2019 is 0.5953. These statistics are computed using all low-income renters in our descriptive sample as weights. The weighted standard deviation of the difference between these tract-level gentrification estimates is 0.5575.

[83]Excluding renters whom the LEHD imputes as likely to have at least a Bachelor's degree drops more households than our estimated number of college graduates in the sample, as reported in Table 1. We therefore view this restriction as conservative.

TABLE A16. Baseline Controls

Control	Description	Source
Date of birth	Household head's year of birth.	LEHD-ICF, Numident
Adjusted household income	Quadratic in baseline adjusted household income. See footnote 59 for the definition of adjusted income.	LEHD-EHF, LEHD-JHF
Residential tenure	Quadratic in length of residential tenure as of 2010.	MAF-ARF
Left-censored panel indicator	Indicator equal to one if the household head is left-censored in the residential panel, defined as residing in their origin tract since at least 2001.	MAF-ARF
CBD proximity	Quadratic in 2010-population-weighted proximity to the metropolitan division CBD.	Gazetteer, Fee and Hartley (2013) CBD definitions
CBD distance bands	Indicators for deciles of 2010-population-weighted distance from the metropolitan division CBD.	Gazetteer, Fee and Hartley (2013) CBD definitions
Baseline racial composition	Quadratics in the tract's baseline shares of adult residents who are white and Black.	De-biased tract aggregates (see Section 2)
Baseline college-graduate share	Quadratic in the tract's baseline share of adult residents with a college degree.	De-biased tract aggregates (see Section 2)
LIHTC units	Total number of LIHTC units in the tract, normalized by the total number of sample households in the tract.	HUD LIHTC database, MAF-ARF, LEHD-ICF
HUD-subsidized units	Total number of HUD-subsidized housing units in the tract, normalized by the total number of sample households in the tract.	HUD POSH, MAF-ARF, LEHD-ICF
Rent control ordinance coverage	Indicator equal to one if the tract is covered by any rent control ordinance.	Compiled rent control ordinances
Predicted rent control unit shares	Share of units in which sample households reside that are predicted to be covered by a binding rent control ordinance.	Compiled rent control ordinances, MAF-ARF, LEHD-ICF, ACS
Residential market access	Quadratic in the natural log of baseline residential market access (see Section B.1)	MAF-ARF, LEHD
Baseline rents	Quadratic in baseline hedonically adjusted rents, measured in 2010 dollars. See Section A.3.	ACS
Baseline development	Quadratic in the tract's baseline level of development.	National Land Cover Database prepared by Baum-Snow and Han (2024)

Notes: Table A16 lists the baseline controls included in the descriptive regressions unless otherwise specified. Tract-level sociodemographic controls are constructed from the de-biased tract aggregates. These controls, as well as the relevant renter-level controls, exclude sociodemographic characteristics imputed by the LEHD. We therefore do not control for renters' educational attainment because observed measures of educational attainment are too sparse. Residential market access enters in logs so that scale differences across CBSAs are absorbed by the metropolitan division fixed effects. Our choice of control variables broadly follows those used in the literature (e.g. Baum-Snow, Hartley, and Lee 2019; Brummet and Reed 2021) that have been shown to shape residential mobility in low-income neighborhoods (Frenette, Picot, and Sceviour 2004), though we include additional controls especially relevant to low-income renters, such as predicted rent control coverage. We did not find our coefficient estimates to be sensitive to any particular combination of controls.

relationship between incumbents' migration choices and measured gentrification, since out-migration among incumbent college graduates mechanically lowers the measured change in a tract's college-graduate share, artificially lowering the coefficient estimate. Excluding these renters has a negligible effect on the point estimates.

One concern with our gentrification measure is that, in tracts with very low initial college-graduate shares, modest absolute increases in the number of college graduates can generate large log changes in the measured college-graduate share. This concern is partly mitigated by excluding low-income renters in tracts in the bottom percentile of the total population distribution. It is also unlikely that these tracts heavily influence the measure, given the strong relationship between measured gentrification and changes in tract rents shown in Figure A7. Nevertheless, we assess this possibility directly by dropping incumbent renters in tracts in the bottom and top vintiles of the gentrification-measure distribution. Dropping these outliers substantially raises the first-stage F statistics but again has a negligible effect on the point estimates.

TABLE A17. Gentrification and Residential Mobility

	(1) Baseline		(2) Gentrifying Tracts		(3) No Rent Control		(4) Long-Term Renters		(5) Movers	
	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black
Panel A: Moved Tract OLS										
Coefficient estimate	0.011	0.017	0.021	0.029	0.007	0.011	0.003	0.023	—	—
Standard error	(0.005)	(0.005)	(0.006)	(0.007)	(0.006)	(0.005)	(0.007)	(0.008)	—	—
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓		
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓		
Panel B: Moved Tract IV										
Coefficient estimate	0.015	0.027	0.020	0.033	0.019	0.017	0.009	0.027	—	—
Standard error	(0.007)	(0.007)	(0.009)	(0.009)	(0.007)	(0.006)	(0.009)	(0.010)	—	—
Kleibergen-Paap F-stat	[101.4]	[413.3]	[96.80]	[437.7]	[38.98]	[190.7]	[142.5]	[466.0]	—	—
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓		
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓		
Panel C: Moved CBSA IV										
Coefficient estimate	0.013	0.006	0.015	0.004	0.027	0.011	0.000	-0.001	—	—
Standard error	(0.004)	(0.004)	(0.005)	(0.005)	(0.007)	(0.005)	(0.005)	(0.005)	—	—
Kleibergen-Paap F-stat	[101.4]	[413.3]	[96.8]	[437.7]	[38.98]	[190.7]	[142.5]	[466]	—	—
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓		
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓		
N	472,000	324,000	389,000	255,000	260,000	219,000	201,000	127,000	—	—
Panel D: Distance from Origin Tract IV										
Coefficient estimate	0.078	0.081	0.100	0.101	0.088	0.057	0.049	0.055	0.099	0.043
Standard error	(0.015)	(0.015)	(0.019)	(0.020)	(0.020)	(0.017)	(0.019)	(0.021)	(0.014)	(0.015)
Kleibergen-Paap F-stat	[94.46]	[406.1]	[90.06]	[432.9]	[33.34]	[184.5]	[149.3]	[466.8]	[88.67]	[395.1]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel E: Distance from CBD IV										
Coefficient estimate	0.012	0.035	0.033	0.057	0.052	0.027	-0.019	0.012	0.012	0.039
Standard error	(0.019)	(0.015)	(0.023)	(0.018)	(0.030)	(0.020)	(0.015)	(0.013)	(0.027)	(0.021)
Kleibergen-Paap F-stat	[94.46]	[406.1]	[90.06]	[432.9]	[33.34]	[184.5]	[149.3]	[466.8]	[88.67]	[395.1]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	392,000	278,000	324,000	219,000	212,000	190,000	177,000	114,000	249,000	190,000

Notes: Table A17 reports coefficient estimates for our gentrification measure from regression Equation 1. In Panels A and B, the dependent variable is an indicator equal to one if the household resides in a different census tract in 2019 than in 2010. In Panel C, the dependent variable is an indicator equal to one if the household resides in a different CBSA in 2019 than in 2010. In Panel D, the dependent variable is log of the straight-line distance in miles from the centroid of the household's 2019 tract to the centroid of its 2010 origin tract. In Panel E, the dependent variable is the log change in the household's distance in miles from its metropolitan division CBD to its tract centroid between 2010 and 2019. In Panels B-E, we instrument for gentrification using changes in the adult college-educated population, as described in Section 3. All specifications include CBSA fixed effects and the full set of baseline controls listed in Table A16. We define gentrifying tracts as those whose college share rose from 2010 to 2019; tracts without rent control as those not covered by any type of rent-control ordinance in 2010; long-term renters as households residing in the same tract since at least 2005; and movers as households observed in a different census tract in 2019 than in 2010. Robust standard errors clustered at the origin census-tract level are in parentheses (Abadie et al. 2023). Kleibergen-Paap F-statistics are in square brackets. The sample comprises urban low-income renter households observed in both 2010 and 2019, defined after applying restriction (h.ii) in Table A1. Panels D and E further subset to households who are observed to reside in the same CBSA in 2019 as in 2010. All coefficient estimates and standard errors are rounded to three decimal places, while Kleibergen-Paap F-statistics are rounded to four significant figures.

Another potential concern is that the association between gentrification and incumbent residential mobility may be larger in tracts with somewhat higher initial college-graduate shares.[84] To assess this

[84] Freeman (2005) considers tracts to be potentially gentrifiable if, among other criteria, median household income is below the 40th percentile of the metro-area. Brummet and Reed (2021) have a similar criterion.

possibility, we expand the analysis to include tracts below the median of their CBSA's tract-level college-graduate share distribution. The corresponding point estimates are 0.003 and 0.007 smaller than the baseline estimates for non-Black and Black households, respectively.

The small average mobility responses to gentrification may mask heterogeneity across tracts with different housing supply elasticities. In less elastic tracts, gentrification may be associated with larger rent increases, making incumbent renters less likely to remain as neighborhood characteristics change. In more elastic tracts, rents may rise less, and incumbent renters may be more likely to stay if they value the accompanying changes in amenities or access to employment opportunities, masking heterogeneous responses in the average point estimates. We find, however, mixed evidence of heterogeneity by housing supply elasticity. Relative to the baseline estimates, the association between gentrification and residential mobility is larger for non-Black but smaller for Black renters in less elastic tracts.[85] Conversely, in more elastic tracts, the association is smaller for non-Black renters but larger for Black renters. In all cases, however, the estimates remain economically small.

The last set of estimates in Panel B examines whether older household heads, who may be more residentially stable, are more likely to leave gentrifying neighborhoods. We restrict the sample to household heads born before 1975, who were at least 45 years old in 2010. The resulting point estimates differ only modestly from the baseline estimates, by -0.7 percentage points for non-Black household heads and 0.5 percentage points for Black household heads.

Panel C of Table A18 adds controls that help clarify the interpretation of the descriptive association between gentrification and residential mobility. We first control for gentrification between 2005 and 2009, so the coefficient on gentrification during 2010–2019 is identified from variation that is orthogonal to neighborhood change immediately before 2010. Although the analysis remains descriptive, this specification makes the comparison less sensitive to households having selected into neighborhoods based partly on pre-existing gentrification trends. The estimates change by only 0.2 and 0.1 percentage points relative to the baseline estimates for non-Black and Black households, respectively.

Building on this specification, we then restrict the sample to long-term low-income incumbent renters. This restriction further narrows the scope for selection into gentrifying tracts to shape the descriptive comparison, since these renters were already living in their origin tract during 2005. In this specification, selection into gentrifying tracts would have to operate through pre-2005 residential choices that predict variation in 2010–2019 gentrification that is orthogonal to 2005–2009 gentrification. The point estimate is 0.4 percentage points lower to the baseline estimate for non-Black households. For Black households, the point estimate increases by 0.2 percentage-points relative to the baseline estimate.

Our coefficient estimates remain similar when we control for the log change in the adult population over 2010–2019, suggesting that the small mobility responses persist when measured gentrification more directly reflects compositional change among adult residents. The same is true when we control for the log changes in the shares of Black, white, and non-Black, non-white adult residents.

[85] We partition tracts into housing-supply-elastic and housing-supply-inelastic tracts based on the unweighted tract-level distribution of housing supply elasticities among all census tracts in our descriptive sample. We define tracts as inelastic if they are below the median tract-elasticity estimate and elastic otherwise. We use the housing supply elasticities generated from Baum-Snow and Han (2024)'s quadratic finite mixture model for total units in 2011.

We also separately control for the log change in hedonically adjusted rents between 2010 and 2019. If incumbent renters positively value the non-price neighborhood changes associated with gentrification, then conditioning on rent changes should move the coefficient on gentrification toward zero. For non-Black households, the coefficient becomes close to zero after adding this control. For Black households, however, the coefficient remains nearly identical to the baseline estimate. Although the differences are modest, this heterogeneity foreshadows the structural estimates in Table 2, where Black household heads have a lower willingness to pay for neighborhood amenity changes causally attributed to gentrification.

TABLE A18. Gentrification and Residential Mobility Heterogeneity

	Non-Black			Black			Baseline Controls	CBSA FEs
	Coefficient	F-Stat	N	Coefficient	F-Stat	N		
Panel A: Alternative Gentrification Measures								
ACS 5-year Aggregates	0.005 (0.003)	23.57	472,000	0.008 (0.003)	206.8	324,000	✓	✓
White College Share	0.011 (0.005)	85.40	472,000	0.014 (0.004)	245.2	324,000	✓	✓
Panel B: Additional Sample Restrictions								
Inner City	0.011 (0.009)	64.27	293,000	0.018 (0.008)	221.5	209,000	✓	✓
No College Educated	0.014 (0.007)	95.98	425,000	0.030 (0.007)	432.1	293,000	✓	✓
Drop Outliers	0.009 (0.008)	1457	427,000	0.023 (0.009)	927.6	289,000	✓	✓
Below Median College Share	0.012 (0.007)	278.6	795,000	0.020 (0.007)	532.7	483,000	✓	✓
Inelastic Tracts	0.024 (0.010)	258.1	246,000	0.021 (0.009)	202.1	169,000	✓	✓
Elastic Tracts	0.008 (0.010)	43.22	225,000	0.032 (0.010)	217.3	155,000	✓	✓
Older Households	0.008 (0.011)	107.5	243,000	0.032 (0.009)	393.4	156,000	✓	✓
Panel C: Additional Controls								
Gentrification 2005-2009	0.017 (0.007)	101.6	472,000	0.028 (0.007)	415.4	324,000	✓	✓
Gentrification 2005-2009, Long-Term Renters	0.011 (0.009)	143.7	201,000	0.029 (0.010)	470.5	127,000	✓	✓
Log Change in Adult Population	0.012 (0.007)	92.84	472,000	0.028 (0.007)	399.4	324,000	✓	✓
Log Change in Racial Shares	0.015 (0.008)	90.55	472,000	0.023 (0.007)	363.5	324,000	✓	✓
Log Change in Hedonic Rents	0.002 (0.007)	96.63	472,000	0.018 (0.007)	399.1	324,000	✓	✓

Notes: Table A18 reports coefficient estimates for our gentrification measure from Equation 1. The dependent variable is an indicator equal to one if the household resides in a different census tract in 2019 than in 2010. Panel A replaces the gentrification measure with either (i) its counterpart constructed using only 5-year ACS aggregates, or (ii) the log-change in a tract's share of adults who are both white and hold at least a bachelor's degree. Panel B reports estimates from regression specifications with additional sample selection criteria discussed in the main text of Appendix D.2. Panel C reports estimates from regression specifications with additional regression controls also discussed in the main text of Appendix D.2. Otherwise, the coefficient estimates are analogous to those in Table A17.

D.3. Gentrification and Experienced Neighborhood Characteristics

Table A19 reports the coefficient estimates from Figure 4 alongside their corresponding standard errors and F-statistics. We discuss the key point estimates in Section 3.3, and expand on the job market access estimates in Section D.5. Here we note one further point of consideration for the rent-control heterogeneity results. Among incumbent renters whose origin tracts were not covered by any rent control ordinance, the difference in 2019 tract-level median hedonic rents between those from gentrifying and non-gentrifying tracts is smaller than the corresponding difference among renters whose origin tracts were covered by rent control. These results should not be interpreted as evidence that rent control does not protect incumbent renters from rent growth in gentrifying neighborhoods, however. Three facts are important for interpreting these results. First, as shown in Figure 4, baseline mobility rates are higher in tracts without rent control coverage, so the 2019 tract-level hedonic rents experienced by incumbent renters may reflect more residential re-sorting across neighborhoods. Second, rent control coverage is itself endogenous to rent growth. The sample of tracts without rent control coverage excludes all California tracts and most New York tracts, two states that experienced some of the largest rent increases between 2010 and 2019. Third, the estimates measure median hedonic rents in the tracts where incumbent renters lived in 2010 and 2019, not the rents actually paid by those renters. We account for this nuance in our structural analysis by allowing rents to vary with residential tenure duration in tracts covered by binding rent control ordinances.

TABLE A19. Gentrification and Experienced Neighborhood Characteristics

	(1) Baseline		(2) Gentrifying Tracts		(3) No Rent Control		(4) Long-Term Renters		(5) Movers		(6) Stayers	
	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black
Panel A: College Share												
Coefficient estimate	0.508	0.427	0.542	0.462	0.340	0.311	0.595	0.526	0.160	0.093	1.000	1.000
Standard error	(0.017)	(0.015)	(0.021)	(0.018)	(0.015)	(0.016)	(0.014)	(0.016)	(0.016)	(0.014)	(0.000)	(0.000)
Kleibergen-Paap F-stat	[94.46]	[406.1]	[90.06]	[432.9]	[33.34]	[184.5]	[149.3]	[466.8]	[88.67]	[395.1]	[75.46]	[269.9]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: Hedonic Rents												
Coefficient estimate	0.135	0.121	0.158	0.146	0.058	0.058	0.161	0.144	0.033	0.031	0.258	0.239
Standard error	(0.012)	(0.011)	(0.014)	(0.014)	(0.008)	(0.009)	(0.014)	(0.014)	(0.005)	(0.006)	(0.024)	(0.022)
Kleibergen-Paap F-stat	[94.46]	[406.1]	[90.06]	[432.9]	[33.34]	[184.5]	[149.3]	[466.8]	[88.67]	[395.1]	[75.46]	[269.9]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel C: Job Market Access												
Coefficient estimate	0.034	0.023	0.036	0.026	0.012	0.011	0.039	0.021	0.030	0.024	0.050	0.021
Standard error	(0.006)	(0.006)	(0.007)	(0.007)	(0.005)	(0.005)	(0.007)	(0.007)	(0.005)	(0.004)	(0.009)	(0.010)
Kleibergen-Paap F-stat	[94.46]	[406.1]	[90.06]	[432.9]	[33.34]	[184.5]	[149.3]	[466.8]	[88.67]	[395.1]	[75.46]	[269.9]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel D: White Population Share												
Coefficient estimate	0.069	0.098	0.078	0.097	0.041	0.044	0.082	0.185	0.002	-0.099	0.182	0.494
Standard error	(0.011)	(0.025)	(0.013)	(0.030)	(0.009)	(0.034)	(0.013)	(0.030)	(0.011)	(0.028)	(0.019)	(0.038)
Kleibergen-Paap F-stat	[94.46]	[406.1]	[90.06]	[432.9]	[33.34]	[184.5]	[149.3]	[466.8]	[88.67]	[395.1]	[75.46]	[269.9]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	392,000	278,000	324,000	219,000	212,000	190,000	177,000	114,000	249,000	190,000	143,000	87,500

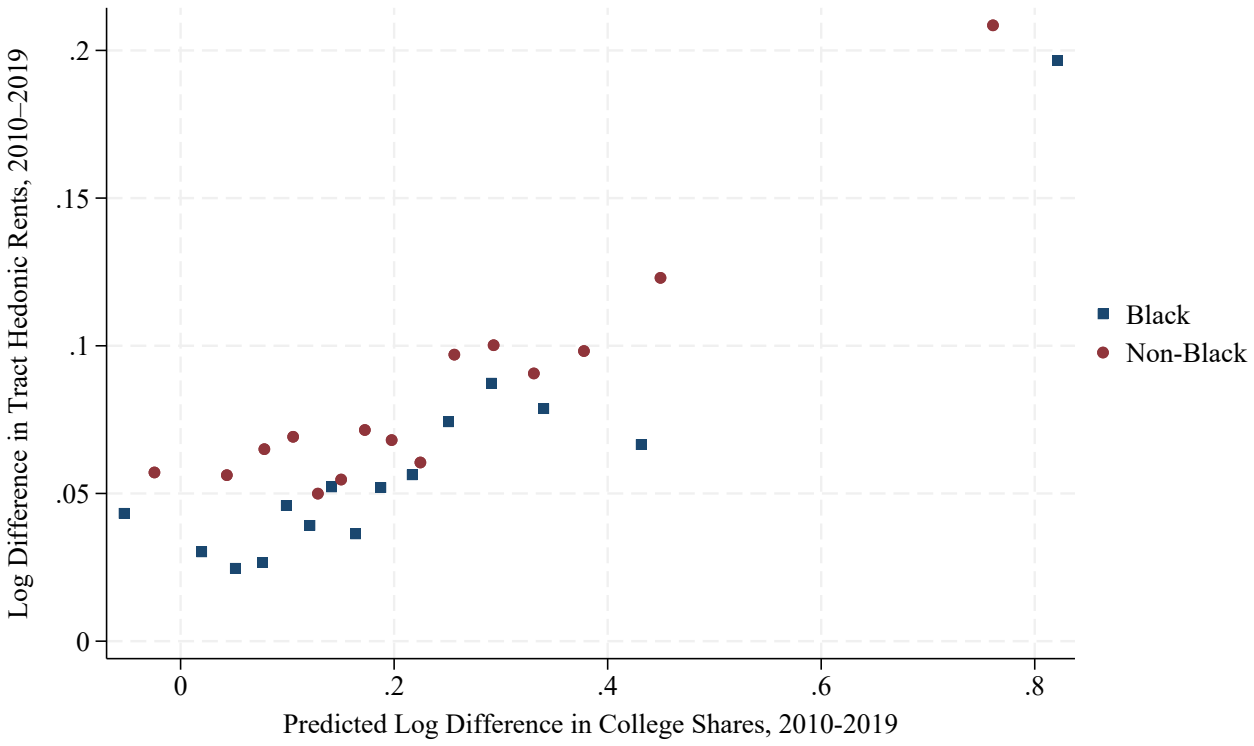
Notes: Table A19 reports coefficient estimates for our gentrification measure from regression equation 1. In Panel A, the dependent variable is the log difference in the share of adults with a college degree between each household's 2019 and 2010 census tracts. In Panel B, the dependent variable is the log difference in hedonic rents between each household's 2019 and 2010 census tracts. In Panel C, the dependent variable is the log difference in job market access, measured using the sample-specific residential market access index defined in Section 4.2. In Panel D, the dependent variable is the log difference in the share of white adults between each household's 2019 and 2010 census tracts. The sample comprises urban low-income renter households observed in both 2010 and 2019 to live in the same CBSA, defined after applying restriction (h.ii) in Table A1. Otherwise, the coefficient estimates are analogous to those in Table A17.

D.4. Rent and Residential Mobility

We show in Section 3 that there was only a small association between gentrification and residential mobility among incumbent low-income renters. This finding holds in supply-inelastic census tracts, where gentrification might have been expected to generate larger rent increases. We therefore investigate two related questions. First, is gentrification associated with meaningful increases in rents? Second, do rent increases predict residential mobility, regardless of whether they occur in gentrifying neighborhoods?

Figure A7 plots conditional means of tract-level hedonic rents against our gentrification measure after residualizing both variables on CBSA fixed effects. We estimate the conditional means separately for the Black and non-Black samples, weighting tracts by the number of sample households from each group living there in 2010. The figure shows a strong positive relationship between our gentrification measure and log changes in hedonic tract rents. This confirms that gentrifying neighborhoods experienced meaningful rent growth, even if incumbent renters were partly insulated from those increases.

FIGURE A7. Relationship Between Rents and Gentrification

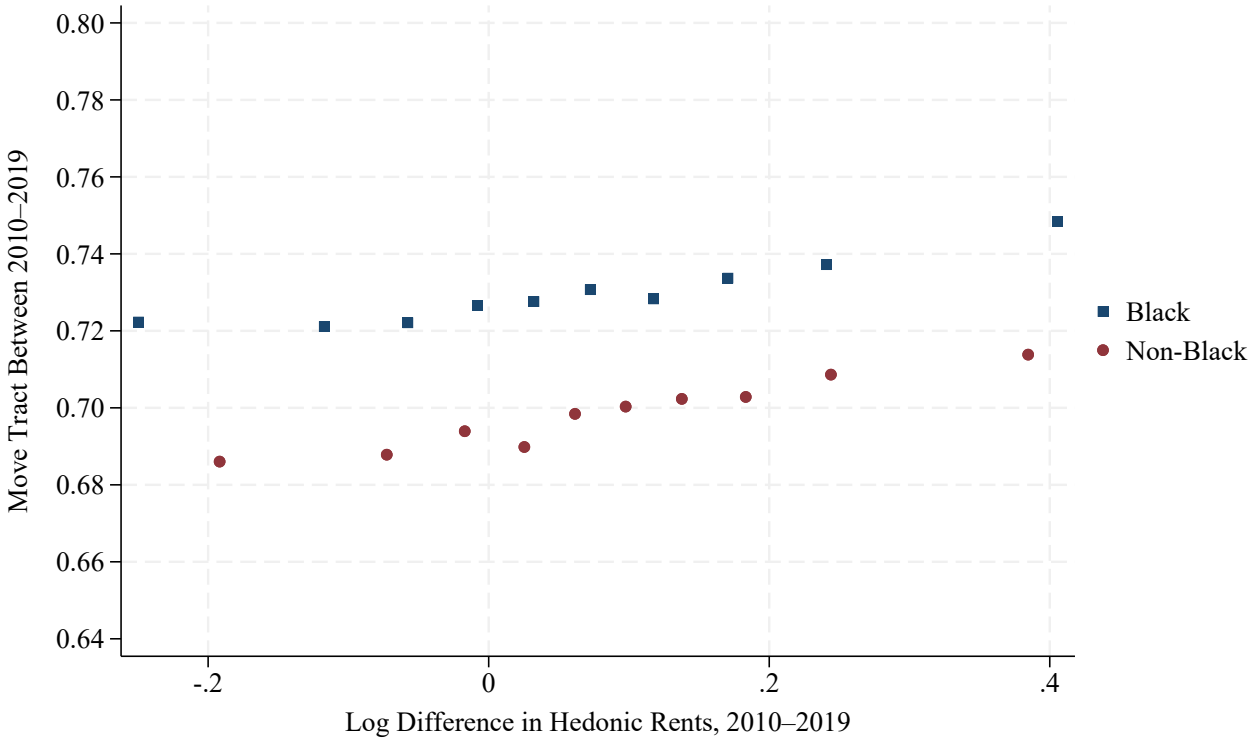


Notes: Figure A7 shows the log change in tract-level hedonic rents as a function of the predicted log change in the share of college-educated adults in sample households' 2010 residential tracts. The points plot conditional means estimated from a regression that includes CBSA fixed effects (Cattaneo et al. 2024). Tracts are weighted separately by the number of Black and non-Black sample household residing in each tract in 2010. The samples comprise urban low-income renter households in 2010, defined after applying restriction (h.i) in Table A1.

One possible explanation for the muted mobility response to gentrification is that incumbent renters in our sample are relatively insensitive to tract-level rent growth. To test this, we re-estimate Equation 1 after replacing our gentrification measure with the log change in tract-level hedonic rents. Figure A8 shows that the relationship between rent growth and residential mobility is somewhat stronger than the

relationship between gentrification and residential mobility, especially given that the OLS estimates are likely attenuated by measurement error in the ACS records. A one-log-point increase in tract-level hedonic rents is associated with a 5.4 percentage-point increase in the probability of leaving one's origin tract for non-Black incumbent renters and a 4.4 percentage-point increase for Black incumbent renters.

FIGURE A8. Changing Rents and Residential Mobility



Notes: Figure A8 shows the probability incumbent renters move census tracts between 2010-2019 as a function of their origin tracts' log change in hedonic rents. The points plot conditional means estimated from regressions analogous to Equation 1, which include CBSA fixed effects and the full set of baseline controls listed in Table A16 (Cattaneo et al. 2024). The sample comprises urban low-income renter households observed in both 2010 and 2019, defined after applying restriction (h.ii) in Table A1. The corresponding point estimates from the two regressions are 0.054 (0.008) for non-Black-headed households and 0.044 (0.008) for Black-headed households, where the corresponding standard errors are in parentheses.

How should we interpret modest mobility responses to rent increases but not to gentrification? First, there is substantial variation in rent growth conditional on gentrification, so our gentrification measure does not mechanically capture the rent increases renters face. This can be seen from the range of the log changes in hedonic rents in Figure A7 and Figure A8. In Figure A7, the conditional means of log changes in tract-level hedonic rents range from 0.025 in the bottom decile of gentrification to 0.21 in the top decile. By contrast, Figure A8 shows that the full rent-growth measure ranges from -0.25 in the bottom decile of rent growth to 0.41 in the top decile. Variation in rent growth conditional on gentrification may meaningfully attenuate incumbents' mobility responses to gentrification relative to rent growth.

Second, the disutility from higher rents may be partly offset by improvements in amenities or by the modest gains in access to employment opportunities that accompany gentrification, as shown in Section 3 and emphasized by Vigdor (2010). We quantify incumbent renters' willingness to pay for changes

in these characteristics in our structural analysis. In Table 2, we estimate that non-Black incumbent low-income renters are willing to pay $267 in annual rents to live in a neighborhood with a 10 percent higher share of adult college graduates, while Black incumbent low-income renters are willing to pay $107. This explanation is also consistent with the heterogeneity analysis in Table A18, where mobility responses to gentrification become even more muted after we control for contemporaneous changes in log rents. Consistent with our structural estimates, this attenuation is larger for non-Black low-income renters than for Black low-income renters, reflecting a higher estimated willingness to pay for neighborhood characteristics associated with a larger college-graduate share.

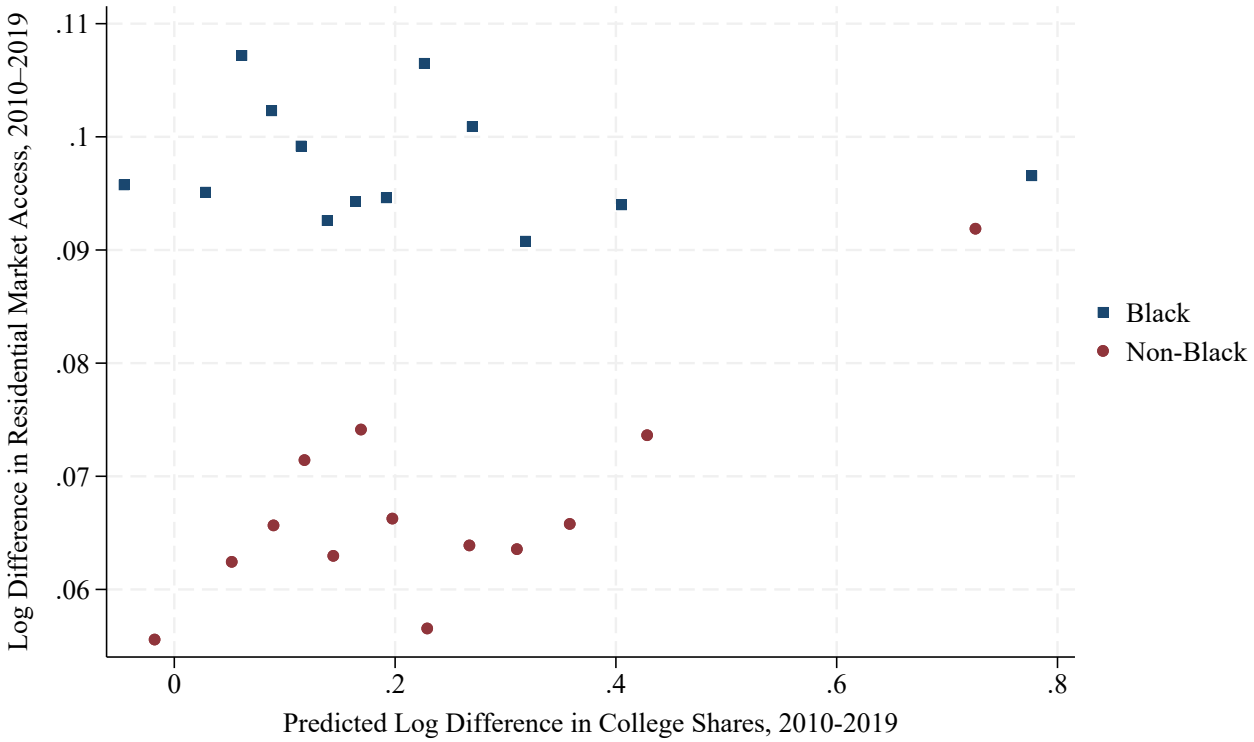
Low-income renters might be more responsive to tract-level rent changes that are less directly tied to gentrification, consistent with our structural estimates showing positive valuations of neighborhood characteristics associated with gentrification. Even so, the estimated effects remain modest. Our point estimates imply that moving from the bottom decile to the top decile of log changes in hedonic rents increases the ten-year probability of leaving one's origin tract from 69 percent to 72 percent for Black households and from 67 percent to 70 percent for non-Black households. These changes correspond to roughly 4.3 percent and 4.4 percent increases in residential mobility, respectively. The large majority of incumbent renters move regardless of trends in their origin tract's hedonically adjusted rents.

D.5. Gentrification, Economic Outcomes, and Job Mobility

Throughout the paper and as noted in Table A1, we construct race-specific job market access measures using all low-income household heads, not only renters, so that the measures better capture employment opportunities available to low-income workers more generally. This assumes less labor-market segmentation by homeownership status than by race in low-income workers' local labor markets. Consistent with this assumption, tradable job shares among low-income workers are much more similar across homeownership status than across racial groups. This can be seen in Table A21 by comparing columns (2) and (3) with the corresponding differences between racial groups within each column. Additionally, Figure A11 shows substantial differences in industry composition between low-income non-Black and Black renters.

Figure A9 shows how job market access varied with gentrification before accounting for sample households' subsequent residential choices. The figure plots conditional means of both job market access measures against our gentrification measure, weighting each tract by the number of low-income renters of the corresponding sample living there in 2010. For non-Black renters, the positive association between gentrification and job market access is driven largely by the top decile of gentrifying tracts. For Black renters, the average relationship between origin-tract gentrification and changes in job market access is negligible. This contrasts with the positive association between gentrification and the experienced job market access of both Black stayers and Black movers discussed in Section 3.4. One plausible explanation is that Black stayers and movers selected into tracts with improving job market access. This explanation does not rely on a mechanical link between greater local demand from Black households and higher measured job market access. Holding fixed the number of jobs, the measure falls as the local supply of low-income workers rises. The selection explanation is also consistent with our structural estimates, which show that Black renters have a relatively high willingness to pay for job market access.

FIGURE A9. Gentrification and Job Market Access

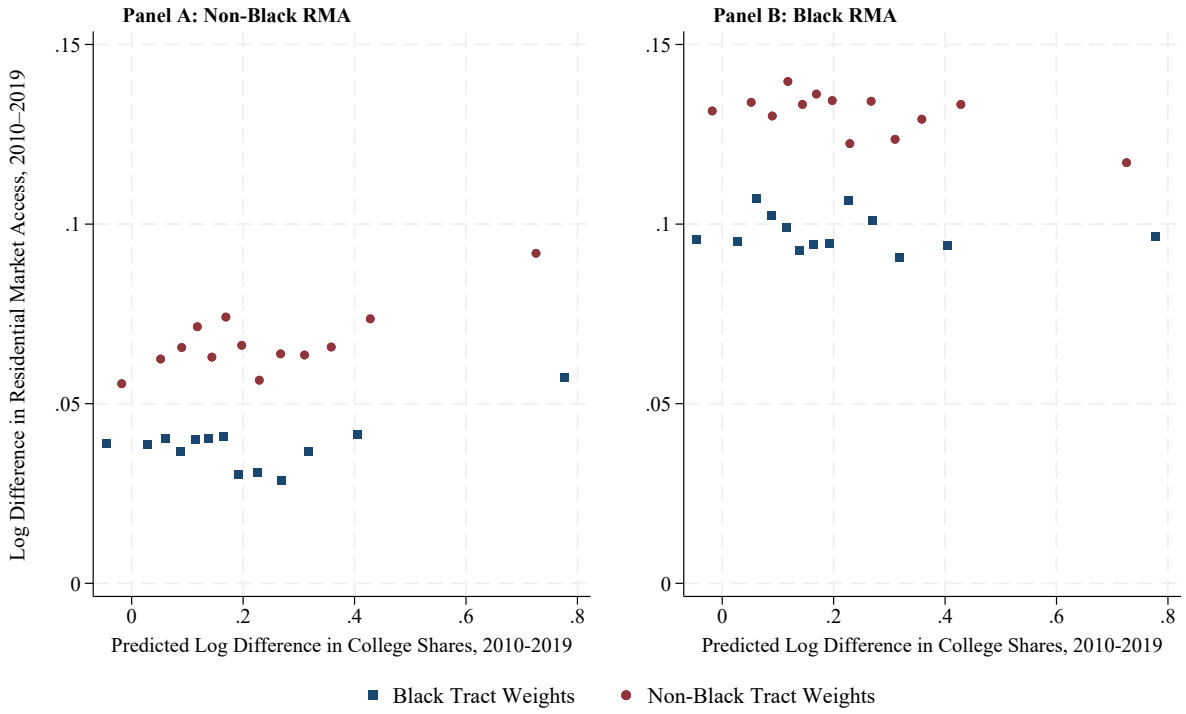


Notes: Figure A9 shows the log change in job market access, measured using the sample-specific residential market access index defined in Section 4.2, as a function of the predicted log change in the share of college-educated adults in sample households' 2010 residential tracts. The points plot conditional means estimated from regressions that include CBSA fixed effects (Cattaneo et al. 2024). The residential market access measures are weighted by the number sample households from the corresponding race residing in each tract in 2010. The weights comprise urban low-income renter households in 2010, defined after applying restriction (h.i) in Table A1. The residential market access measures are derived from all low-income households in 2010 and 2019, defined after applying restriction (f) in Table A1.

One explanation for the racial heterogeneity in changes in low-income workers' job market access is that gentrification was associated with larger improvements in non-Black than Black job market access within the same tracts. Another is that non-Black incumbent renters in gentrifying neighborhoods were initially concentrated in tracts where job market access rose more for both groups. To distinguish between these explanations, we re-plot conditional means of log changes in job market access against our gentrification measure, weighting each race-specific job market access measure by each tract's 2010 count of Black or non-Black incumbent low-income renters. Figure A10 shows that the tract-level weights have little effect on the slope of the relationship. If anything, Black job market access appears negatively associated with gentrification when weighted by non-Black incumbent renters' 2010 residential locations. This suggests that the racial heterogeneity reflects race-specific changes in job market access within gentrifying neighborhoods, rather than differences in where Black and non-Black incumbent renters initially lived.

Improvements in experienced job market access need not translate directly into better realized economic outcomes. Table A20 shows that gentrification is associated with slightly worse employment outcomes for Black households and negligible changes for non-Black households, despite positive job market access estimates among both stayers and movers in both groups. One possible explanation is

FIGURE A10. Gentrification and Job Market Access Weighted by Incumbent Shares



Notes: Figure A10 shows the log change in job market access, measured using the sample-specific residential market access index defined in Section 4.2, as a function of the predicted log change in the share of college-educated adults in sample households' 2010 residential tracts. Panel A plots log changes in residential market access derived from only non-Black low-income households in 2010 and 2019. Panel B plots log changes in residential market access derived from only Black low-income households in 2010 and 2019. The series within each panel are differentiated by the race of households comprising the tract-level weights. Otherwise, the figures are analogous to Figure A9.

that renters sort into areas with high job market access after adverse employment shocks. The income estimates closely follow the non-participation estimates, primarily because non-participants are coded as having zero income, or as $\log(0+1)$ in our regression specifications. Finally, the commute-time estimates are economically small. Since commute time is measured in minutes, a one-log-point increase in a tract's share of adult college graduates is associated with average increases of only 1.6 percent for non-Black households and 0.3 percent for Black households, or roughly 0.4 and 0.1 minutes at the sample means. These calculations exclude household heads who are not observed in the LEHD in 2019.

TABLE A20. Gentrification and Economic Outcomes

	(1) Baseline		(2) Gentrifying Tracts		(3) No Rent Control		(4) Long-Term Renters		(5) Movers		(6) Stayers	
	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black
Panel A: Non-Participation												
Coefficient estimate	-0.000	0.007	0.001	0.010	-0.000	0.007	-0.006	0.011	0.002	0.007	-0.002	0.009
Standard error	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)	(0.004)	(0.005)	(0.006)
Kleibergen-Paap F-stat	[94.46]	[406.1]	[90.06]	[432.9]	[33.34]	[184.5]	[149.3]	[466.8]	[88.67]	[395.1]	[75.46]	[269.9]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: Income												
Coefficient estimate	0.028	-0.081	0.011	-0.109	0.006	-0.084	0.092	-0.151	-0.018	-0.100	0.073	-0.075
Standard error	(0.035)	(0.035)	(0.042)	(0.044)	(0.045)	(0.043)	(0.049)	(0.051)	(0.040)	(0.040)	(0.057)	(0.065)
Kleibergen-Paap F-stat	[94.46]	[406.1]	[90.06]	[432.9]	[33.34]	[184.5]	[149.3]	[466.8]	[88.67]	[395.1]	[75.46]	[269.9]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	392,000	278,000	324,000	219,000	212,000	190,000	177,000	114,000	249,000	190,000	143,000	87,500
Panel C: Commute Time												
Coefficient estimate	0.016	0.003	0.028	0.011	-0.005	-0.015	0.021	0.016	0.018	-0.001	0.011	0.009
Standard error	(0.007)	(0.007)	(0.008)	(0.009)	(0.009)	(0.008)	(0.009)	(0.009)	(0.010)	(0.008)	(0.010)	(0.011)
Kleibergen-Paap F-stat	[84.27]	[377.1]	[81.87]	[390.2]	[36.24]	[179.0]	[125.4]	[455.7]	[113.5]	[412.6]	[57.11]	[193.1]
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	210,000	165,000	176,000	131,000	111,000	111,000	96,500	68,500	131,000	112,000	78,500	53,000

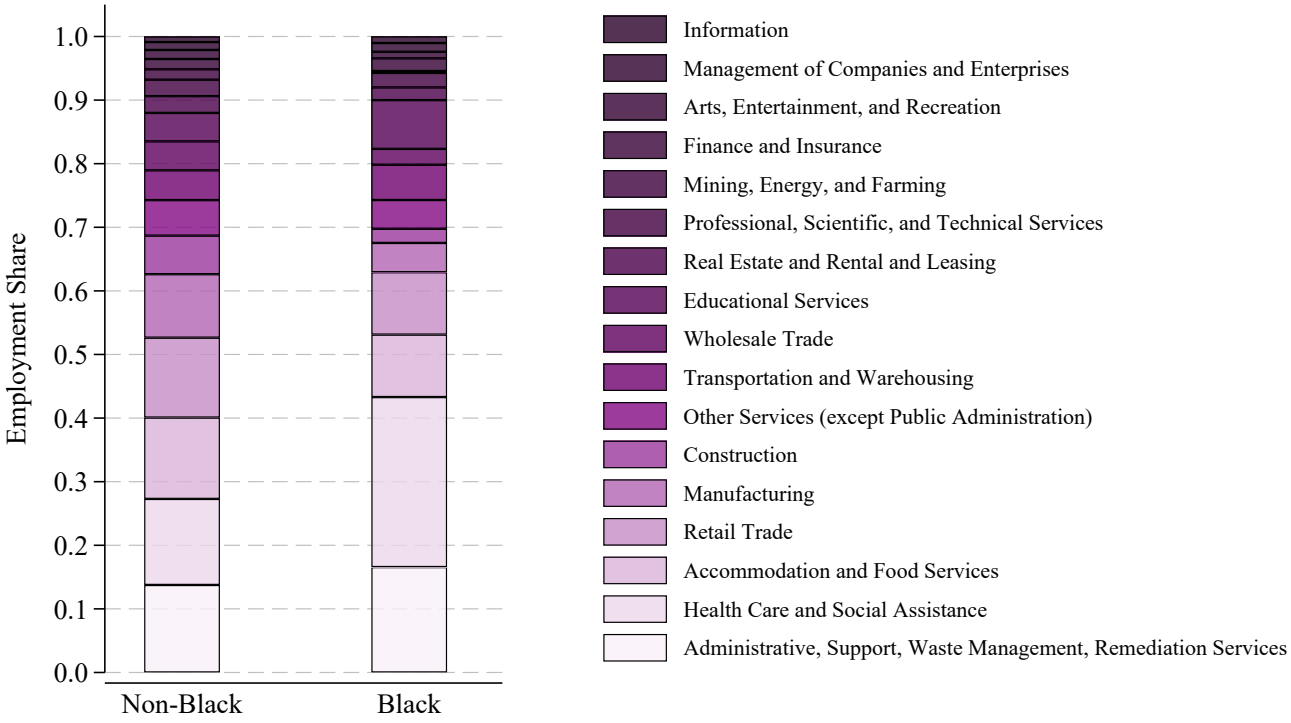
Notes: Table A20 reports coefficient estimates for our gentrification measure from regression Equation 1. In Panel A, the dependent variable is an indicator equal to one if we do not observe the household head with positive income in the LEHD in 2019. In Panel B, the dependent variable is the log difference in each household head's income (2010 dollars, plus one) between 2019 and 2010. In Panel C, the dependent variable is the log difference in each household head's commute time between 2019 and 2010. We instrument for gentrification using changes in the adult college-educated population, as described in Section 3. In Panels A and B, the sample comprises urban low-income renter households we observe in both 2010 and 2019 to live in the same CBSA, defined after applying restriction (h.ii) in Table A1. In Panel C, the sample is further restricted to household heads we observe with positive income in the LEHD in both 2010 and 2019. Otherwise, the coefficient estimates are analogous to those in Table A17 and Table A19.

TABLE A21. Baseline Job Characteristics and Mobility

	(1) Urban Low-Inc Renters		(2) Low-Inc Renters		(3) All Low-Inc HHs		(4) All HHs
	Non-Black	Black	Non-Black	Black	Non-Black	Black	
Panel A							
Switch job in 2010	0.2338	0.2492	0.2329	0.2477	0.2025	0.2325	0.1505
Number job switches 2010-2019	2.076	2.355	2.039	2.314	1.835	2.205	1.502
Standard deviation	(1.613)	(1.749)	(1.590)	(1.731)	(1.579)	(1.734)	(1.501)
Pseudo-median	[2.000]	[2.000]	[2.000]	[2.000]	[2.000]	[2.000]	[1.000]
Skew	⟨0.8165⟩	⟨0.6668⟩	⟨0.8419⟩	⟨0.6899⟩	⟨0.9657⟩	⟨0.7513⟩	⟨1.141⟩
Panel B							
Tradable job in 2010	0.4044	0.4512	0.3879	0.4461	0.3785	0.438	0.3892
Commute time in 2010	25.12	30.12	24.04	30.34	23.05	29.01	25.6
Standard deviation	(21.13)	(24.16)	(19.70)	(24.86)	(19.04)	(23.53)	(19.83)
Pseudo-median	[20.00]	[20.00]	[20.00]	[20.00]	[20.00]	[20.00]	[20.00]
Skew	⟨2.904⟩	⟨2.225⟩	⟨2.869⟩	⟨2.309⟩	⟨3.398⟩	⟨2.542⟩	⟨3.061⟩
Commute distance in 2010	9.783	9.611	10.05	10.09	10.96	10.86	13.36
Standard deviation	(9.551)	(9.185)	(9.829)	(9.380)	(10.21)	(9.812)	(11.45)
Pseudo-median	[7.252]	[7.309]	[7.384]	[7.798]	[8.287]	[8.311]	[10.51]
Skew	⟨2.626⟩	⟨3.715⟩	⟨2.435⟩	⟨2.807⟩	⟨2.617⟩	⟨2.368⟩	⟨2.084⟩
N	517,000	351,000	1,994,000	799,000	5,866,000	1,594,000	24,830,000

Notes: Table A21 summarizes household heads' job mobility and baseline job characteristics in 2010. To align with our annual residential-tract assignments, each household head is assigned one firm ID per calendar year, determined by the establishment from which they earned the highest wage income. Panel A reports job switching either between 2010 and 2011 or year-to-year over 2010-2019. By construction, a household head can switch at most once per year, and job-mobility statistics are computed only when a household head is assigned a job in both the beginning- and end-year associated with the statistic. Panel B summarizes characteristics of household heads' assigned jobs in 2010. A job is classified as tradable if its industry's trade costs fall in the bottom 75% of industries, as defined in Gervais and Jensen (2019) (see Footnote 39 for details). Commute-time and commute-distance statistics are computed for household heads matched to the 2010 person-level ACS. Commute time is measured in minutes rounded to the nearest five, and distance is measured in miles. Columns (1)–(4) use the samples defined after applying restrictions (h.i), (g), (f), and (d) in Table A1. Household counts correspond to Panel A.

FIGURE A11. Sectoral Employment Shares



Notes: Figure A11 plots sectoral employment shares for urban low-income renter households in 2010, defined after applying restriction (h.i) in Table A1. Industries are stacked in ascending order of the non-Black sample's employment shares. Sectors aggregate 2-digit NAICS (2012) industries as follows: 23–Construction; 11, 21, and 22–Mining, Energy, and Farming; 31, 32, 33–Manufacturing; 42–Wholesale Trade; 44, 45–Retail Trade; 48, 49–Transportation and Warehousing; 52–Finance and Insurance; 51–Information; 53–Real Estate and Rental and Leasing; 54–Professional, Scientific, and Technical Services; 55–Management of Companies and Enterprises; 56–Administrative and Support and Waste Management and Remediation Services; 61–Educational Services; 62–Health Care and Social Assistance; 71–Arts, Entertainment, and Recreation; 72–Accommodation and Food Services; and 81–Other Services (except Public Administration).

Appendix E. Welfare Simulations

E.1. Empirical Setup

Data Construction. We conduct our counterfactual welfare simulations using data procured outside the Census restricted IRE. This lets us report tract-level estimates of expected welfare that would otherwise fail the Census Bureau's disclosure requirements for small-area geographic estimates. We construct tract-level aggregates in a way that closely matches those we construct inside the Census restricted IRE. We compute raw tract-level measures of median gross rent paid as well as the share of adult residents over 25 years old with and without a college degree from the 5-year ACS tract estimates (Manson et al. 2022). The US Census Bureau does not publicly release tract-level counts of low-income Black and non-Black renters separately by low ($\bar{\tau} = 1$) and high ($\bar{\tau} = 2$) tenure. We therefore approximate these counts in several steps. First, we construct tract-level counts of Black and non-Black renters in each tract from the 5-year ACS tract estimates. Second, we approximate tract-level counts of low-income Black and non-Black households from 5-year ACS income-bin tabulations. We define low-income households as those earning among the bottom tercile of their respective CBSA's income distribution for cohorts of 35 to 45 year olds, estimated from the Integrated Public Use Microdata Series (IPUMs) with person-level weights (Ruggles et al. 2025). Third, we construct Public Use Microdata Area (PUMA) estimates of counts of low-income renter households by race and length of tenure from IPUMS using household weights. We finally combine these estimates by scaling the tract-level counts of low-income Black and non-Black households by the share of Black and non-Black households who are renters in each tract, and then apportioning the resulting counts into low- and high-tenure groups according to the share of low-income renters of each race in the corresponding PUMA who are low- and high-tenure.[86]

We similarly construct tract-level and race-specific measures of residential market access for low-income households using publicly available data to approximate our estimates inside the Census IRE. We estimate tract-level jobs belonging to low-income household heads by race using the LEHD Origin-Destination Employment Statistics (LODES) workplace-area job counts by tract, year, and industry sector, excluding public administration jobs. We allocate these sector jobs to Black and non-Black low-income household heads using baseline race-specific sector shares from our analysis sample recorded in Figure A11, rescaling each job count by the ratio of the sector's baseline share of low-income renter employment to its share of total LODES employment in each year. We then sum across sectors to approximate tract-year race-specific low-income employment counts up to a common scale factor. We combine these job counts with the tract-level counts of low-income Black and non-Black households and solve the corresponding unique (to scale) fixed point in Equation A5 to obtain values for RMA_{nt}^{k} in each tract and year for which the LODES job counts are available. When constructing these measures, we use the 50th percentile 2010 parameter cluster estimates (η_{2010}^{kc}) reported in Table A6. We also calibrate tract-pair commute times based off tract pairs' physical distance and their distances to the metropolitan division CBD, ensuring they are consistent with households' mean commute times reported in Table 1 and estimates from our commute-time forecasts

[86] We experimented with apportioning the PUMA-level counts of low-income renters by tenure level to tracts based on the tract estimates, but we found the overall count of low-income renter households in each PUMA to be particularly variable across years. We therefore only rely on the ratio of high- and low-tenure low-income renter households in each PUMA-year.

in Table A4.

We crosswalk all our tract-level estimates to 2010 tabulation Census tract boundaries using household population weights, and deflate gross rents and low-income cutoffs to real 2010 dollars. We further deflate gross rents by the ratio of mean hedonic tract rents to mean gross tract rents, separately for each sample using the corresponding estimates in Table 1. For tracts covered by a binding rent-control ordinance, we impute rents for high-tenure households using the ratio of hedonically adjusted predicted rents for low- and high-aggregate-tenure households, as described in Appendix A.3. We approximate aggregate-tenure transition probabilities using the mean tenure transition probabilities for all households in each sample. Finally, we smooth each tract-level estimate using a kernel-weighted average of estimates from nearby tracts in the same metropolitan division and from nearby years (inclusive of the base year), where the weights decay with both geographic distance and time.[87]

Tract Aggregate Forecasting. We compute our expected welfare measures by backward induction from a terminal period in which we assume the economy is in steady state. Ideally, this terminal period would be far enough in the future that discounting makes the steady-state assumption largely irrelevant for our expected welfare measures. Because we estimate tract-level aggregates only through 2021, setting a later terminal year requires forecasting tract characteristics forward. We use 2030 as the terminal year because our forecasts of tract characteristics appear credible through that horizon, and because the present-discounted contribution of the terminal steady-state assumption becomes quite small. After some experimentation, we used the following tract-fixed-effects panel vector error correction model (VECM) to forecast neighborhood characteristics through 2030.

We define the log and the log-first-difference of each endogenous neighborhood characteristic Y_{nt}, as $y_{nt} \equiv \log(Y_{nt})$ and $\Delta y_{nt} \equiv \ln\left(\frac{Y_{nt}}{Y_{nt-1}}\right)$, respectively.[88] We estimate the cointegrating equation using all endogenous tract characteristics in 2011-2021 and tract fixed effects (ζ_n), defining the error-correction term (ECT) as the within-tract residual,

$$\text{ECT}_{nt} = y_{nt}^{\text{rent}} - \hat{\theta}' \boldsymbol{y}_{nt}^{-\text{rent}} - \hat{\zeta}_n,$$

where y_{nt}^{rent} is the log of median gross rents in tract n and year t, and $\mathbf{y}_{nt}^{-\text{rent}}$ stacks the logs of the remaining endogenous tract characteristics. We use ECT_{nt} to estimate the following VECM,

$$\Delta \boldsymbol{y}_{nt} = \mathbf{A}\Delta \boldsymbol{y}_{nt-1} + \boldsymbol{\phi}\text{ECT}_{nt-1} + \upsilon_n + \boldsymbol{u}_{nt},$$

where $\mathbf{A}$ is the one-period VAR coefficient matrix and $\boldsymbol{\phi}$ collects the error-correction coefficients. We forecast through 2022-2030 recursively by generating one-step-ahead predictions for the log of each

[87] Our counterfactual welfare analyses require smooth tract-level estimates. Otherwise, households in our simulations could arbitrage measurement noise by repeatedly moving to neighborhoods with artificially low rents or spuriously high market access and amenities.

[88] The endogenous neighborhood characteristics we forecast forward are counts of adults over 25 years with and without a college degree, median gross rents, counts of both Black and non-Black low-income renter households with low tenure ($\bar{\tau} = 1$), and separate residential market access measures for Black and non-Black low-income households.

endogenous characteristic and corresponding deviations from the cointegrating relationship,[89]

$$\widehat{\boldsymbol{y}}_{nt} = \widehat{\boldsymbol{y}}_{nt-1} + \widehat{\Delta \boldsymbol{y}_{nt}}, \qquad \widehat{\text{ECT}}_{nt} = \hat{y}_{nt}^{\text{rent}} - \hat{\theta}' \widehat{\boldsymbol{y}}_{nt}^{-\text{rent}} - \hat{\zeta}_n.$$

When transforming back to levels, we replace any negative endogenous values with the smallest positive estimate within the same CBSA-year.

E.2. Counterfactual Simulation Details

For each census tract in our analysis, we compute and compare two tract-level measures of expected welfare for low-income Black and non-Black renter households. Both tract-level measures are constructed from the perspective of the year 2010, separately for each tenure length $\bar{\tau}$. The first expected welfare measure assumes the city's economy is in a steady-state in the year 2010. We denote this measure $\bar{V}^k_{SS}\left(x(n,\bar{\tau}), \bar{\omega}^{kc}_{2010}\right)$. We compare it to an alternative expected welfare measure, $\bar{V}^k_{Gent}\left(x(n,\bar{\tau}), \bar{\omega}^{kc}_{2010}\right)$, that assumes households believe their city's tract characteristics will evolve just as they do in the data.[90] We quantify the welfare effects of gentrification for incumbent low-income renter households by comparing these tract-level measures across tracts that gentrify and those that do not.

Steady-State Expected Welfare. We compute $\bar{V}^k_{SS}\left(x(n,\bar{\tau}), \bar{\omega}^{kc}_{2010}\right)$ by holding observed city-level state variables fixed at their 2010 values and then solving for time-invariant continuation values and exogenous amenities $\left\{\tilde{\xi}^k_{n,2010}\right\}_n$ that generate a stationary distribution of type-k low-income renter households. To define a stationary distribution, we denote $q^k_{nt}(\bar{\tau})$ as the model-implied share of each city's type-k households with tenure length $\bar{\tau}$ living in neighborhood n in period t. Stacking $q^k_{nt}(\bar{\tau})$ yields a $2(N^c+1)$-dimensional vector $\boldsymbol{q}^k_t$. We define a stationary distribution in year t among our sample of low-income renter households as $\boldsymbol{q}^k_t = \boldsymbol{\Lambda}^k_t \boldsymbol{q}^k_t$, where $\boldsymbol{\Lambda}^k_t$ is a transition probability matrix constructed so the distribution of type-k household tract shares evolve such that,

$$q^k_{nt}(\bar{\tau}) = \sum_{\bar{\tau}'} \sum_{n'} q^k_{n't-1}(\bar{\tau}') \cdot p^k_n\left(x(n',\bar{\tau}'), \bar{\omega}^{kc}_t\right) \cdot f^x(x(n,\bar{\tau})|n, x(n',\bar{\tau}')) \quad \forall\, n, \bar{\tau}.$$

We define a steady-state equilibrium in 2010 as a stationary distribution among type-k low-income renter households in which continuation values are time-invariant and all city-specific state variables are held fixed at their 2010 levels for all $t > 2010$. The resulting time-invariant continuation values are our steady-state measures of expected welfare, $\bar{V}^k_{SS}\left(x(n,\bar{\tau}), \bar{\omega}^{kc}_{2010}\right)$. Algorithm 1 describes how we compute these welfare measures in practice, along with the corresponding vector of exogenous amenities.[91]

[89] Since many tracts have very low values for Black low-income renter counts, we winsorize that variable's one-step-ahead log-difference forecasts at the 35th and 55th percentiles of its in-sample log-difference distribution. For all other variables, we winsorize one-step-ahead log-difference forecasts at the corresponding 5th and 95th percentiles.

[90] This perfect-foresight assumption implies that households' expected ex-ante continuation values equal their realized counterparts in the simulations, or equivalently that the expectational errors equal zero, $e^{\bar{V}}(x', \bar{\omega}^{kc}_t, \bar{\omega}^{kc}_{t+1}) = 0$. The assumption is the degenerate special case of Assumption (c).

[91] In step 6 of Algorithm 1, we exploit that exogenous amenities do not vary with households' length of tenure. Although Algorithm 1 need not generate an exactly stationary distribution for type-k households with high neighborhood tenure ($\bar{\tau} = 2$), we find that it closely approximates stationarity in practice. We define $\hat{q}^k_{nt}(\bar{\tau} = 1)$ as the empirical counterpart to $q^k_{nt}(\bar{\tau} = 1)$.

Algorithm 1 Compute Steady State Expected Welfare

1: Guess value of unobserved neighborhood amenities: $\left\{\tilde{\xi}_n^k\right\}_n$

Compute continuation values:

2: Guess $\left\{\bar{V}_{SS}^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)\right\}_{n,\bar{\tau}}$

3: Compute for all n and $\bar{\tau}$,

$$\bar{V}_{SS}'^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)=\ln\left(\sum_{n'\in\mathcal{N}^c}\exp\left(\bar{u}_{n'}^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)+\delta\sum_{\bar{\tau}'}\bar{V}_{SS}^k\left(x(n',\bar{\tau}'),\bar{\omega}_t^{kc}\right)\cdot f^x(x(n',\bar{\tau}')|n',x(n,\bar{\tau}))\right)\right)+\gamma$$

4: Set $\bar{V}_{SS}^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)=\bar{V}_{SS}'^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right),\quad\forall\, n,\bar{\tau}$

Repeat steps 3 - 4 until $\max_{n,\bar{\tau}}\left\{\left|\bar{V}_{SS}^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)-\bar{V}_{SS}'^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)\right|\right\}<\epsilon_V$ for some $\epsilon_V>0$.

Compute exogenous amenities:

5: Compute the probability a type-k household chooses n given $\left\{\bar{V}_{SS}^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)\right\}_{n,\bar{\tau}}$ and $\left\{\tilde{\xi}_n^k\right\}_n$

$$p_n^k\left(x(\underline{n},\bar{\tau}),\bar{\omega}_t^{kc}\right)=\frac{\exp\left(v_n^k\left(x(\underline{n},\bar{\tau}),\bar{\omega}_t^{kc}\right)\right)}{\sum_{n'\in\mathcal{N}^c}\exp\left(v_{n'}^k\left(x(\underline{n},\bar{\tau}),\bar{\omega}_t^{kc}\right)\right)},\quad\forall\, n,\underline{n},\bar{\tau}$$

6: Update exogenous amenities using observed neighborhood shares of low-income Black and non-Black households with low tenure, $q_n^k(\bar{\tau}=1)$:

$$\tilde{\xi}_n'^k=\tilde{\xi}_n^k+\ln\left(\hat{q}_n^k(\bar{\tau}=1)\right)-\ln\left(\sum_{\bar{\tau}'}\sum_{\underline{n}}q_{\underline{n}}^k(\bar{\tau}')\cdot p_n^k\left(x(\underline{n},\bar{\tau}'),\bar{\omega}_t^{kc}\right)\cdot f^x(x(n,\bar{\tau}=1)|n,x(\underline{n},\bar{\tau}'))\right)\tag{A13}$$

Set $\tilde{\xi}_n^k=\tilde{\xi}_n'^k$.

Repeat steps 2 - 6 until $\max_n\left\{\left|\tilde{\xi}_n^k-\tilde{\xi}_n'^k\right|\right\}<\epsilon_\xi$ for some $\epsilon_\xi>0$.

Take $\bar{V}_{SS}^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)$ as the measures of steady-state expected welfare with the city-specific state variables fixed at their period t levels.

Expected Welfare Change. We compute $\bar{V}_{Gent}^k\left(x(n,\bar{\tau}),\bar{\omega}_{2010}^{kc}\right)$ by first computing steady-state expected welfare measures in the year 2030, $\bar{V}_{SS}^k\left(x(n,\bar{\tau}),\bar{\omega}_{2030}^{kc}\right)$. We compute these and 2030 exogenous amenities analogously to how we compute their 2010 counterparts, but hold observed city-level state variables fixed at their 2030 forecast values. We then solve for $\bar{V}_{Gent}^k\left(x(n,\bar{\tau}),\bar{\omega}_{2010}^{kc}\right)$ via backward induction given $\bar{V}_{SS}^k\left(x(n,\bar{\tau}),\bar{\omega}_{2030}^{kc}\right)$ until we obtain $2N^c$ tract-level measures of expected welfare from the perspective of 2010, one for each aggregate tenure level in each tract. During each step of the backward induction procedure, we solve for the levels of exogenous amenities that rationalize the observed share of type-k low-income renter households with low aggregate tenure across census tracts in that year. To do so, we repeat steps 5 and 6 in Algorithm 1, but substitute $\left\{\bar{V}_{SS}^k\left(x(n,\bar{\tau}),\bar{\omega}_t^{kc}\right)\right\}_{n,\bar{\tau}}$ for the measures of expected welfare computed in the previous year's iteration.

With $\bar{V}^k_{Gent}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$ and $\bar{V}^k_{SS}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$ in hand, we define expected welfare change as,

$$\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) \equiv \bar{V}^k_{Gent}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) - \bar{V}^k_{SS}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right). \tag{A14}$$

Welfare Effect Decomposition. We decompose $\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$ into one component capturing incumbent renters' own-tract flow utilities and another capturing their option value. To do so, we apply the Hotz and Miller (1993) inversion to $\bar{V}^k_{Gent}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$ and $\bar{V}^k_{SS}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$, taking the household's origin tract as the reference option in the inversion:

$$\begin{aligned}
\bar{V}^k_{\mathcal{U}}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) &= \ln\left(\sum_{n'\in\mathcal{N}^c}\exp\left(v^k_{\mathcal{U}n'}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)\right)\right)+\gamma \\
&= v^k_{\mathcal{U}n}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)+\ln\left(1+\sum_{n'\neq n}\exp\left(v^k_{\mathcal{U}n'}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)-v^k_{\mathcal{U}n}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)\right)\right)+\gamma \\
&= \underbrace{v^k_{\mathcal{U}n}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)}_{\text{payoff from staying}} - \underbrace{\ln\left(p^k_{\mathcal{U}n}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)\right)}_{\text{option value of moving}}+\gamma,
\end{aligned}$$

where $\mathcal{U}\in\{SS, Gent\}$ indicates that the respective object can be computed within the steady-state environment or under the observed evolution of tract characteristics. The second equality highlights how the probability of staying in one's origin neighborhood reflects the relative desirability of a household's broader choice set. When other tracts are less desirable relative to the origin tract, households are less likely to leave, and expected welfare lies closer to the value of remaining in the origin tract. In the limiting case in which the household stays in its origin tract with probability one, the option value of moving is zero, so expected welfare is equal to the origin tract's conditional value function, $\bar{V}^k_{\mathcal{U}}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) = v^k_{\mathcal{U}n}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$.

We can express expected welfare change under this decomposition as,

$$\begin{aligned}
\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) &\equiv \bar{V}^k_{Gent}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) - \bar{V}^k_{SS}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) \\
&= \Delta\underbrace{v^k_n\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)}_{\Delta\text{ payoff from staying}} - \Delta\underbrace{\ln\left(p^k_n\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)\right)}_{\Delta\text{ option value of moving}}, \qquad \forall\, n,\ \bar{\tau}.
\end{aligned} \tag{A15}$$

where $\Delta v^k_n\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$ and $\Delta\ln\left(p^k_n\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)\right)$ are the differences in origin tracts' conditional value functions and option values computed under the observed evolution of tract characteristics and under the steady-state environment. Recursively substituting the conditional value function (8) into A15 yields,

$$\begin{aligned}
\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right) = &\sum_{s=0}^{20}\delta^s\sum_{\bar{\tau}_{2010+s}}\Pi_{2010+s}(\bar{\tau}_{2010+s}|\bar{\tau})\left[\Delta\bar{u}^k_n\left(x(n,\bar{\tau}_{2010+s}),\bar{\omega}^{kc}_{2010+s}\right)-\Delta\ln\left(p^k_n\left(x(n,\bar{\tau}_{2010+s}),\bar{\omega}^{kc}_{2010+s}\right)\right)\right] \\
&+\frac{\delta^{21}}{1-\delta}\sum_{\bar{\tau}_{2031}}\Pi_{2031}(\bar{\tau}_{2031}|\bar{\tau})\left[\Delta\bar{u}^k_n\left(x(n,\bar{\tau}_{2031}),\bar{\omega}^{kc}_{2031}\right)-\Delta\ln\left(p^k_n\left(x(n,\bar{\tau}_{2031}),\bar{\omega}^{kc}_{2031}\right)\right)\right],
\end{aligned}$$

where $\Pi_t(\bar{\tau}_t|\bar{\tau})$ is the probability of being in tenure state $\bar{\tau}_t$ after staying in tract n until period t given

the initial state $\bar{\tau}$,

$$\Pi_{t+1}(\bar{\tau}_{t+1}|\bar{\tau}) = \sum_{\bar{\tau}_t} f^x(x(n,\bar{\tau}_{t+1})|n, x(n,\bar{\tau}_t)) \cdot \Pi_t(\bar{\tau}_t|\bar{\tau}).$$

Because flow utility is linear in tracts' characteristics, we can additionally decompose differences in the payoffs from staying in n during any period t,

$$\Delta\bar{u}_n^k\left(x(n,\bar{\tau}_t), \bar{\omega}_t^{kc}\right) = \tilde{\beta}_I^k \Delta \ln\left(\bar{I}_{nt}^k\right) + \tilde{\beta}_r^k \Delta \log\left(r_{\bar{\tau}_t nt}\right) + \tilde{\beta}_A^k \Delta \ln\left(\frac{\text{Coll}_{nt}}{\text{Pop}_{nt}}\right) + \Delta\xi_{nt}^k,$$

where Δ now denotes the difference between the period-t components of the payoff from staying in tract n under the observed evolution of tract characteristics and under the steady-state environment. Finally, we can denote the present discounted value for any component of the expected welfare change decomposition as,

$$\text{PDV}\Delta\chi = \sum_{s=0}^{20}\delta^s \sum_{\bar{\tau}_{2010+s}} \Pi_{2010+s}(\bar{\tau}_{2010+s}|\bar{\tau})\Delta\chi_s + \frac{\delta^{21}}{1-\delta}\sum_{\bar{\tau}_{2031}} \Pi_{2031}(\bar{\tau}_{2031}|\bar{\tau})\Delta\chi_{2031},$$

for all $\chi_s \in \left\{ \ln\left(p_n^k\left(x(n,\bar{\tau}_{2010+s}), \bar{\omega}_{2010+s}^{kc}\right)\right),\ \bar{u}_n^k\left(x(n,\bar{\tau}_{2010+s}), \bar{\omega}_{2010+s}^{kc}\right),\ \tilde{\beta}_r^k \ln\left(r_{\bar{\tau}_s ns}\right),\ \tilde{\beta}_A^k \ln\left(\frac{\text{Coll}_{ns}}{\text{Pop}_{ns}}\right),\ \tilde{\beta}_I^k \ln\left(\bar{I}_{ns}^k\right),\ \xi_{ns}^k \right\}$ and analogously for χ_{2031}. This setup lets us write the final decompositions we present in Section 8:

$$\Delta\bar{V}^k\left(x(n,\bar{\tau}), \bar{\omega}_{2010}^{kc}\right) = \underbrace{\text{PDV}\Delta\bar{u}_n^k\left(x(n,\bar{\tau}), \bar{\omega}^{kc}\right)}_{\text{origin-tract flow utility changes}} - \underbrace{\text{PDV}\Delta\ln\left(p_n^k\left(x(n,\bar{\tau}), \bar{\omega}^{kc}\right)\right)}_{\text{option value changes}} \tag{A16}$$

and,

$$\begin{aligned}\Delta\bar{V}^k\left(x(n,\bar{\tau}), \bar{\omega}_{2010}^{kc}\right) = {} & \underbrace{\text{PDV}\tilde{\beta}_I^k\Delta\ln\left(\bar{I}_n^k\right)}_{\text{market access changes}} + \underbrace{\text{PDV}\tilde{\beta}_r^k\Delta\ln\left(r_{\bar{\tau}n}\right)}_{\text{rent changes}} + \underbrace{\text{PDV}\tilde{\beta}_A^k\Delta\ln\left(\frac{\text{Coll}_n}{\text{Pop}_n}\right)}_{\text{amenity changes}} \\ & + \underbrace{\text{PDV}\Delta\xi_n^k}_{\text{exogenous changes}} - \underbrace{\text{PDV}\Delta\ln\left(p_n^k\left(x(n,\bar{\tau}), \bar{\omega}^{kc}\right)\right)}_{\text{option value changes}}.\end{aligned} \tag{A17}$$

Calibrating Model Parameters. Discrepancies between the publicly available and internal tract-level estimates induce differences between our model-implied mobility rates and their empirical counterparts. In particular, imputing tract-level counts of low-income Black and non-Black renters by tenure level yields noisy estimates in the public data, and this noise implies greater variation in tract-level mean utilities than in the internal microdata we use to estimate the model parameters.[92] In addition, because we cannot disclose the CBSA-specific moving cost parameters, we instead predict them from the estimates in Table A14, and any prediction error in the largest CBSAs—which carry the most weight in our simulations—passes directly into the simulated mobility rates. Both of these factors could generate higher

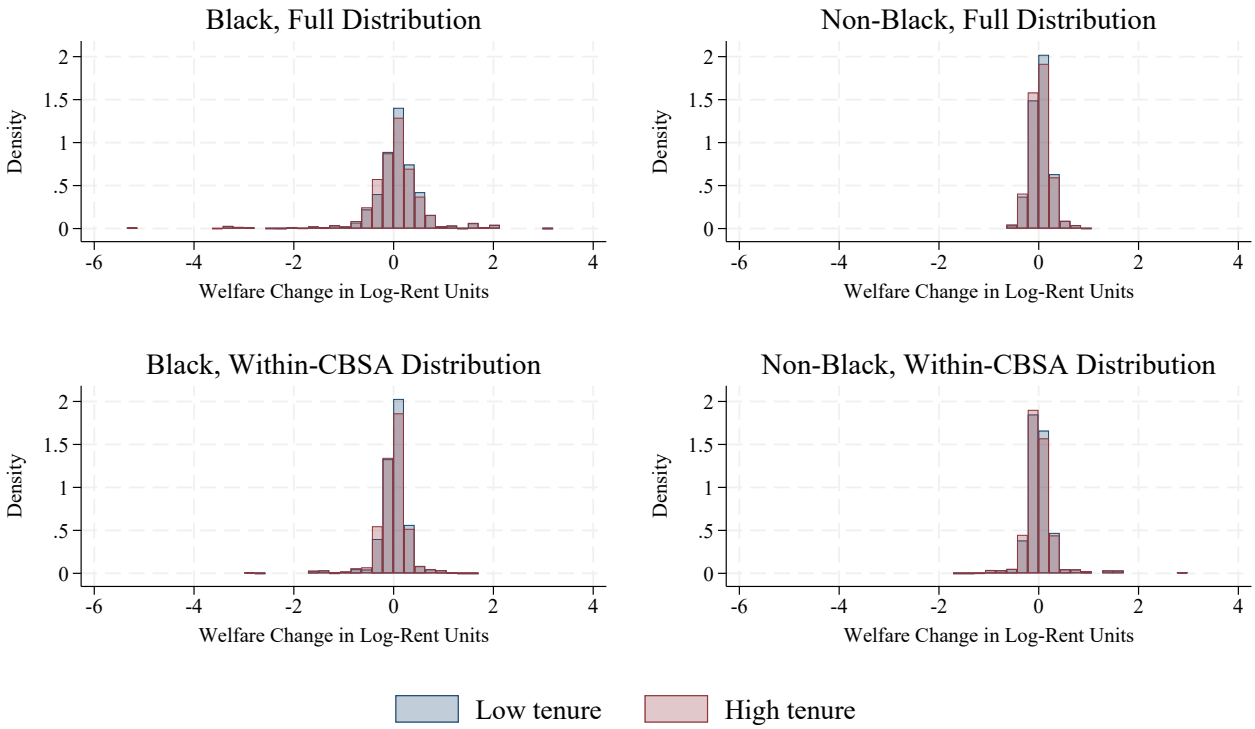
[92] To see why this increases model-implied baseline mobility, note that the simulations invert exogenous amenities to rationalize observed tract shares of low-tenure low-income renters (cf. Equation A13 in Algorithm 1), so classical measurement error in the log shares of these renters passes directly into conditional values, with $\hat{v}_n^k = v_n^k + e_n$, where e_n is i.i.d., mean zero. The model-implied move probability for a household in origin tract n is approximately $1 - \hat{p}_n^k \approx \sum_{n' \neq n} \exp\left(v_{n'}^k - v_n^k\right)\exp\left(e_{n'} - e_n\right)$, and so by Jensen's inequality, noisier tract shares will mechanically inflate model-implied move rates.

mobility in our simulations than we observe in the data, and inferring welfare differences from artificially high mobility would bias us toward finding limited cross-tract differences in welfare changes.

To correct for this, we calibrate the scale parameter so that model-implied mobility matches the empirical 2010 one-year tract move rates reported in Table A15. Specifically, we match the model-implied 2010 tract move rates to their empirical counterparts when households believe tract characteristics evolve as they do in the data (i.e., using the same backward induction process that we use to compute $\bar{V}^k_{Gent}\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$), weighting each tract and aggregate tenure group by its share of low-income renters, separately for each k-type. By lowering model-implied mobility rates, we view this calibration as a conservative check on our analysis. The calibrated scale parameters are 0.783 and 0.789 for Black and non-Black households, respectively. Nevertheless, for robustness, Table A23 replicates the specifications underlying Figure 8 both without calibrating the scale parameter and calibrating it to the lower baseline mobility in the ACS, which we report in Figure A1. Table A23 reports quantitatively similar results for these specifications.

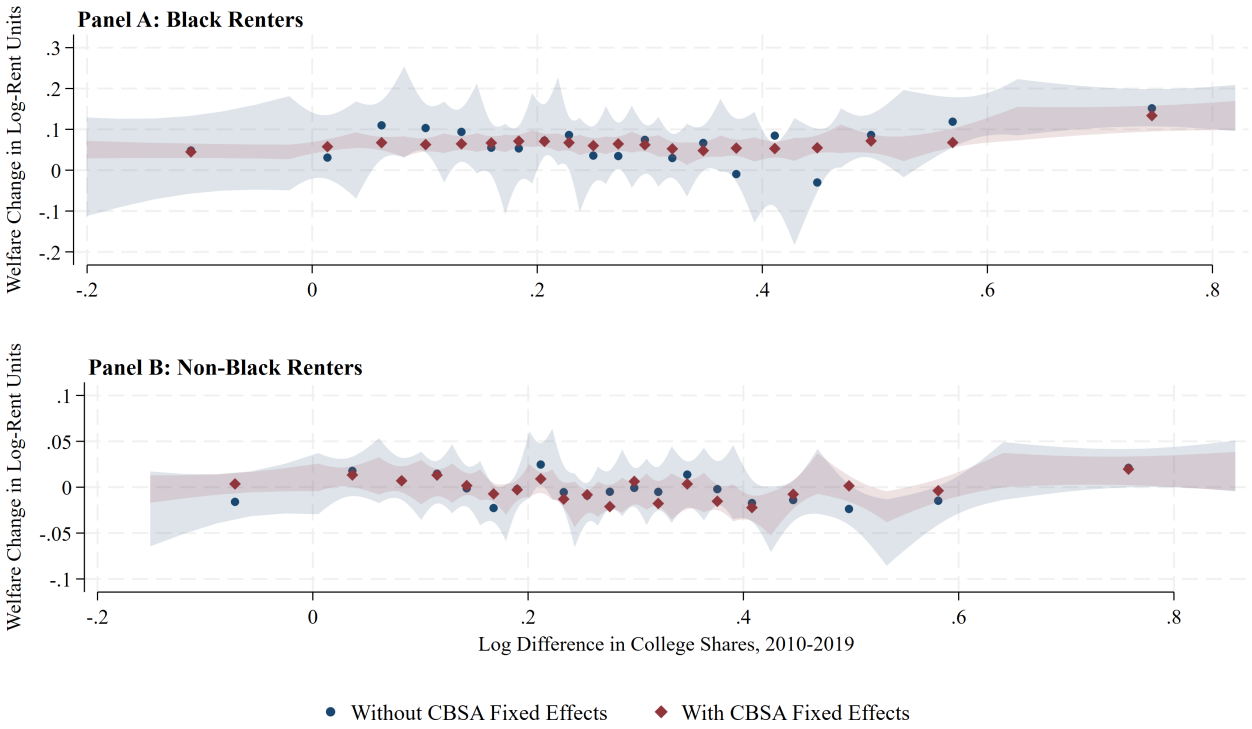
E.3. Additional Counterfactual Simulation Results

FIGURE A12. Distribution of Expected Welfare Changes



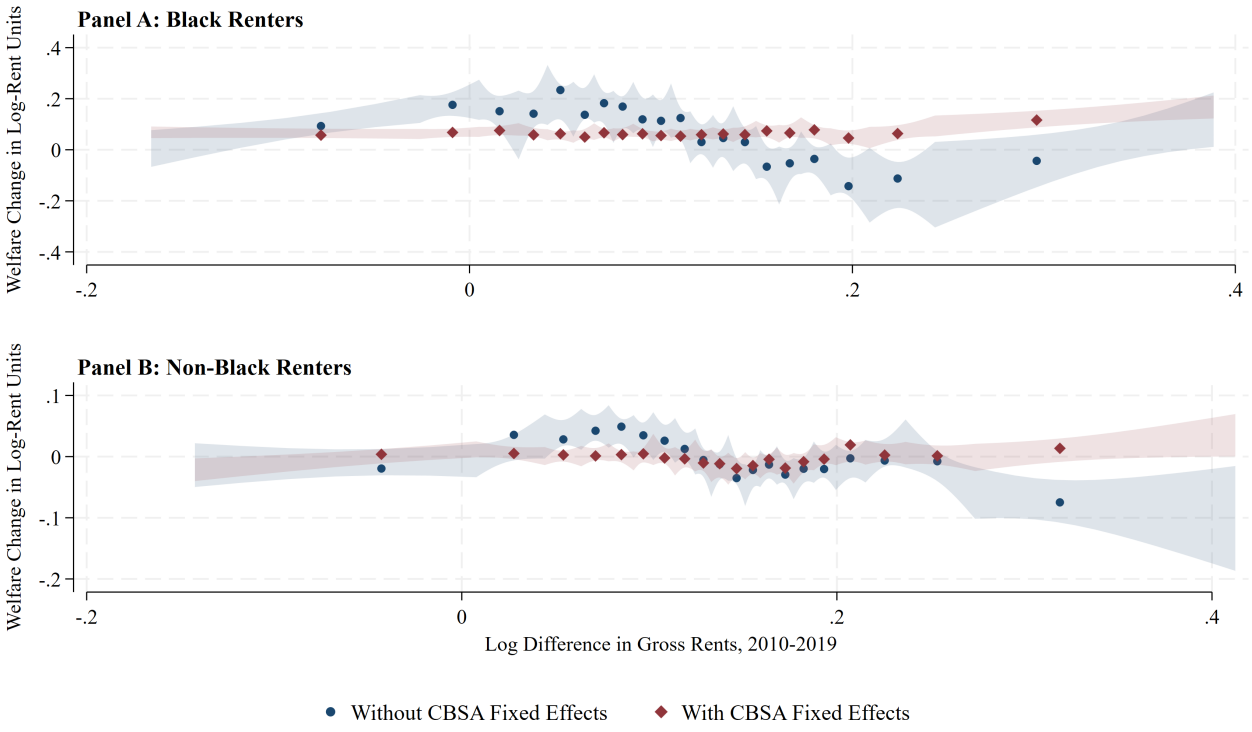
Notes: Figure A12 plots the distribution of expected welfare differences, $\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$. The bottom two plots show within-CBSA distributions and are identical to those in Figure 7. The top two plots instead show the corresponding distributions before demeaning by CBSA average expected welfare changes. All other plot details match Figure 7.

FIGURE A13. Gentrification and Changes in Expected Welfare Within and Across CBSAs



Notes: Figure A13 plots the conditional means from Figure 8 alongside analogous series estimated without absorbing CBSA fixed effects. All other details match those reported in the notes to Figure 8.

FIGURE A14. Rents and Changes in Expected Welfare Within and Across CBSAs



Notes: Figure A14 is identical to Figure A13, except that the independent variable is the log difference in median gross tract rents in over 2010–2019.

TABLE A22. Gentrification and Changes in Expected Welfare

	College Share				Rent			
	Low Tenure		High Tenure		Low Tenure		High Tenure	
	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black
Panel A: Baseline controls and CBSA FEs								
Coefficient estimate	0.019	0.098	0.000	0.084	0.105	0.101	0.091	0.050
Standard error	(0.009)	(0.015)	(0.010)	(0.016)	(0.023)	(0.035)	(0.025)	(0.037)
R-squared	[0.803]	[0.971]	[0.775]	[0.964]	[0.804]	[0.970]	[0.776]	[0.963]
N	7,285	7,283	7,285	7,283	7,285	7,283	7,285	7,283
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: Baseline controls only								
Coefficient estimate	0.013	-0.023	-0.007	-0.046	-0.247	-0.849	-0.267	-0.959
Standard error	(0.016)	(0.048)	(0.016)	(0.049)	(0.038)	(0.103)	(0.038)	(0.104)
R-squared	[0.005]	[0.019]	[0.014]	[0.018]	[0.015]	[0.033]	[0.025]	[0.035]
N	7,285	7,283	7,285	7,283	7,285	7,283	7,285	7,283
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects								
Panel C: CBSA FEs only								
Coefficient estimate	0.006	0.053	-0.012	0.040	0.038	0.113	0.024	0.073
Standard error	(0.008)	(0.013)	(0.009)	(0.014)	(0.023)	(0.032)	(0.025)	(0.033)
R-squared	[0.794]	[0.969]	[0.763]	[0.962]	[0.794]	[0.969]	[0.763]	[0.962]
N	7,285	7,283	7,285	7,283	7,285	7,283	7,285	7,283
Baseline controls								
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Panel D: No baseline controls or CBSA FEs								
Coefficient estimate	0.015	0.071	-0.025	0.022	-0.198	-0.808	-0.260	-0.964
Standard error	(0.013)	(0.041)	(0.013)	(0.041)	(0.036)	(0.100)	(0.035)	(0.100)
R-squared	[0.000]	[0.001]	[0.001]	[0.000]	[0.007]	[0.013]	[0.012]	[0.019]
N	7,285	7,283	7,285	7,283	7,285	7,283	7,285	7,283
Baseline controls								
CBSA fixed effects								

Notes: Table A22 reports the regression estimates corresponding to Figures 8, A13, and A14. The columns split expected welfare measures by initial low- and high-aggregate-tenure levels and the panels alternate between specifications with and without baseline controls or CBSA fixed effects. Otherwise, the specifications match those described in the notes to Figure 8.

TABLE A23. Gentrification and Changes in Expected Welfare Robustness

	College Share				Rent			
	Low Tenure		High Tenure		Low Tenure		High Tenure	
	Non-Black	Black	Non-Black	Black	Non-Black	Black	Non-Black	Black
Panel A: No calibration								
Coefficient estimate	0.006	0.030	0.001	0.041	0.023	0.032	0.018	0.040
Standard error	(0.003)	(0.005)	(0.004)	(0.007)	(0.007)	(0.012)	(0.009)	(0.016)
R-squared	[0.995]	[0.999]	[0.994]	[0.998]	[0.995]	[0.999]	[0.994]	[0.998]
N	7,285	7,283	7,285	7,283	7,285	7,283	7,285	7,283
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: Calibrated to ACS mobility								
Coefficient estimate	0.023	0.110	0.005	0.094	0.128	0.112	0.118	0.061
Standard error	(0.011)	(0.017)	(0.011)	(0.018)	(0.027)	(0.040)	(0.029)	(0.041)
R-squared	[0.688]	[0.956]	[0.659]	[0.948]	[0.689]	[0.955]	[0.661]	[0.948]
N	7,285	7,283	7,285	7,283	7,285	7,283	7,285	7,283
Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Table A23 reports estimates from the same specifications used in Panel A of Table A22, but computes $\Delta\bar{V}^k\left(x(n,\bar{\tau}),\bar{\omega}^{kc}_{2010}\right)$ under alternative calibrations of the scale parameter governing the variance of households' idiosyncratic tract utility shocks, σ^k. Panel A uses the uncalibrated values of σ^k, while Panel B calibrates σ^k to match baseline mobility rates for the corresponding samples in the ACS, displayed in Figure A1.

TABLE A24. Gentrification and Welfare-Contribution Estimates

	Non-Black Renters				Black Renters			
	Coef.	S.E.	R-squared	N	Coef.	S.E.	R-squared	N
Standardized welfare-contribution coefficients								
Rents	-3.569	(0.296)	[0.579]	14,654	-3.295	(0.267)	[0.576]	14,650
Job market access	-0.985	(0.190)	[0.734]	14,654	0.210	(0.249)	[0.948]	14,650
College graduate share	5.270	(0.100)	[0.854]	14,654	5.144	(0.107)	[0.856]	14,650
Exogenous amenities	-0.395	(0.348)	[0.680]	14,654	1.458	(0.353)	[0.954]	14,650
Staying welfare	-1.892	(0.475)	[0.535]	14,654	-0.972	(0.398)	[0.795]	14,650
Option value	2.094	(0.442)	[0.253]	14,654	1.245	(0.289)	[0.130]	14,650
Total welfare	-0.035	(0.261)	[0.773]	14,654	0.427	(0.485)	[0.964]	14,650
CBSA fixed effects	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Table A24 reports the regression coefficient estimates and standard errors shown in Figure 10, alongside each regression's R-squared value and tract count.

Appendix F. Experienced Welfare

The main welfare analysis in Section 8 uses the model to decompose welfare changes into the components affecting the payoff from staying in a gentrifying tract and the option value of moving from it. Because the analysis relies on publicly available census data, it also allows us to document heterogeneity across CBSAs in these components. At the same time, with only aggregate tract data, we infer household exposure to tract-level changes and mobility responses from the model. Discrepancies between the publicly available data and the restricted-access data, especially in our job market access measures, further require us to calibrate several welfare-relevant parameters described in Section E.2, including the variance of the scale parameter, σ^k. The complementary analysis below therefore uses internal census microdata to construct a household-level measure of experienced welfare along observed residential histories.

We use the internal census microdata to construct a measure of experienced welfare, which we define as the present discounted value of the model-implied flow utilities associated with the neighborhoods in which households resided from 2010 to 2019, inclusive of moving costs and conditional expectations over idiosyncratic preference shocks. This measure combines households' realized residential histories with our parameter estimates, and therefore approximates the ex-ante value function in Equation 7 using observed neighborhood choices rather than simulated future choices. To derive this measure, we substitute households' expectational errors into the ex-ante value function for a household observed in neighborhood n at time t:

$$\begin{aligned}
\bar{V}^k\left(x_{it}, \bar{\omega}_t^{kc}\right) &= v_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right) - \ln\left(p_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right)\right) + \gamma \\
&= \bar{u}_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right) + \delta \mathbb{E}_t\left[\bar{V}^k\left(x_{i,t+1}, \bar{\omega}_{t+1}^{kc}\right) \mid n, x_{it}, \bar{\omega}_t^{kc}\right] - \ln\left(p_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right)\right) + \gamma \\
&= \bar{u}_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right) - \ln\left(p_n^k\left(x_{it}, \bar{\omega}_t^{kc}\right)\right) + \gamma + \delta\left[\bar{V}^k\left(x_{i,t+1}, \bar{\omega}_{t+1}^{kc}\right) - e^{\bar{V}}\left(x_{i,t+1}, \bar{\omega}_t^{kc}, \bar{\omega}_{t+1}^{kc}\right)\right].
\end{aligned}$$

Applying this expression recursively from 2010 through 2019 and using n_{is} to denote household i's tract choice in period s yields,

$$\begin{aligned}
\bar{V}^k\left(x_{i,2010}, \bar{\omega}_{2010}^{kc}\right) = &\underbrace{\sum_{s=2010}^{2019} \delta^{s-2010}\left[\bar{u}_{n_{is}}^k\left(x_{is}, \bar{\omega}_s^{kc}\right) - \ln\left(p_{n_{is}}^k\left(x_{is}, \bar{\omega}_s^{kc}\right)\right) + \gamma\right]}_{\text{observed}} \\
&\underbrace{+ \delta^{10}\bar{V}^k\left(x_{i,2020}, \bar{\omega}_{2020}^{kc}\right) - \sum_{s=2011}^{2020} \delta^{s-2010} e^{\bar{V}}\left(x_{is}, \bar{\omega}_{s-1}^{kc}, \bar{\omega}_s^{kc}\right)}_{\text{unobserved}}.
\end{aligned} \tag{A18}$$

We estimate the observed component of the decomposition by evaluating flow utilities and choice probabilities along households' realized residential histories. This gives our measure of experienced welfare,

$$\hat{\bar{V}}^k\left(x_{i,2010}, \bar{\omega}_{2010}^{kc}\right) = \sum_{s=2010}^{2019} \delta^{s-2010}\left[\hat{\bar{u}}_{n_{is}}^k\left(x_{is}, \bar{\omega}_s^{kc}\right) - \ln\left(\hat{p}_{n_{is}}^k\left(x_{is}, \bar{\omega}_s^{kc}\right)\right) + \gamma\right].$$

We estimate regressions analogous to those in Section 3, but replace the dependent variable with

$\widehat{\bar{V}}^k\left(x_{i,2010}, \bar{\omega}^{kc}_{2010}\right)$. These regressions compare experienced welfare across households in the same origin CBSA whose 2010 tracts gentrified to different degrees. Interpreting the coefficients as estimates of changes in the full ex-ante value function requires the unobserved component of the recursive decomposition, namely the remaining continuation value after 2019 and the accumulated expectational errors over 2010–2019, to be orthogonal to measured gentrification in the household's origin tract.[93]

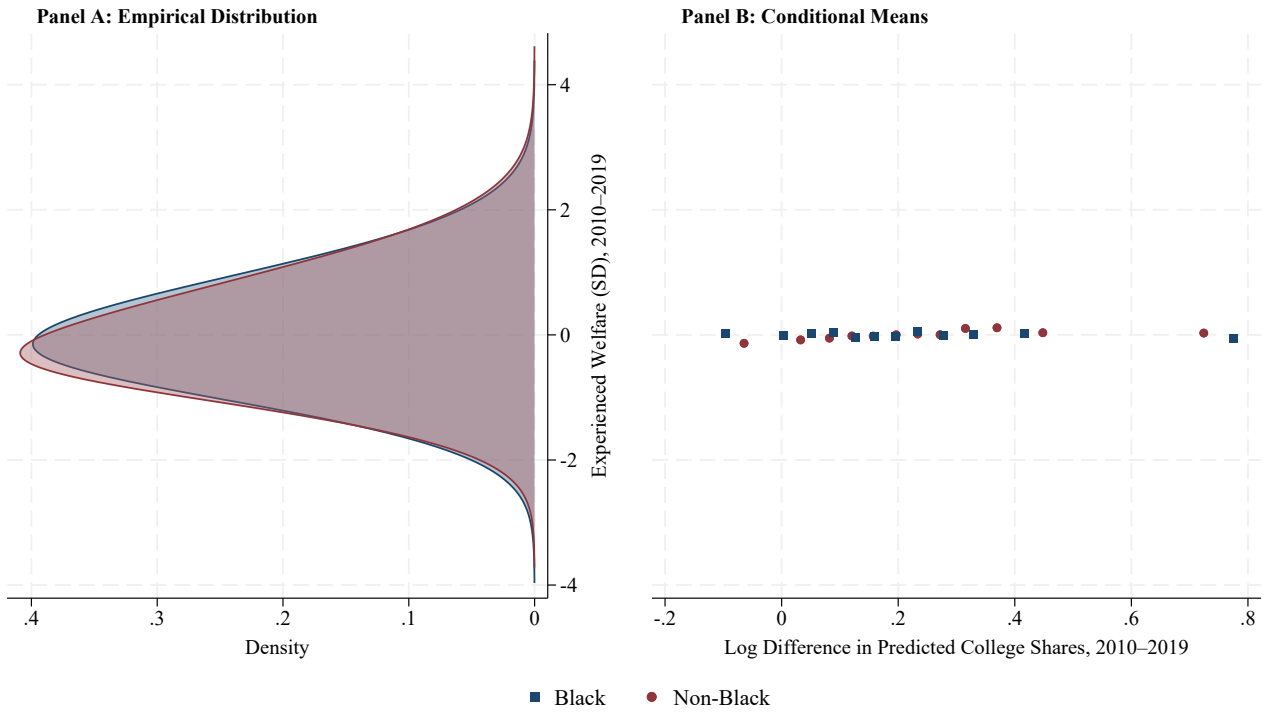
Panel A of Figure A15 plots standardized distributions of experienced welfare by race. The distribution for Black households has a marginally thicker left tail than the distribution for non-Black households. Panel B relates experienced welfare to measured gentrification. It plots conditional means from regressions of experienced welfare on log changes in origin tracts' college-graduate shares over 2010–2019, controlling for the baseline covariates listed in Table A16 and the remaining origin-tract characteristics that comprise flow utility. As with the main welfare analysis in Figure 8, experienced welfare is essentially unrelated to measured gentrification. Panel A of Table A25 reports similarly economically insignificant estimates for incumbent renters initially residing in tracts not covered by any rent-control ordinance and for incumbent renters with high aggregate tenure in 2010.

Figure A16 plots coefficient estimates from regressions analogous to those underlying Figure 8, but recomputes $\widehat{\bar{V}}^k\left(x_{i,2010}, \bar{\omega}^{kc}_{2010}\right)$ for each regression after omitting the referenced component of experienced utility. The estimates show that, while total experienced welfare is largely unrelated to gentrification, omitting rents from the construction of $\widehat{\bar{V}}^k\left(x_{i,2010}, \bar{\omega}^{kc}_{2010}\right)$ yields a strong positive relationship between gentrification and experienced welfare. Omitting market access, and especially college-graduate shares, pushes the estimates downward. These patterns are consistent with the descriptive evidence in Section 3 and the parameter estimates in Table 2. In particular, although low-income Black renters highly value job market access, gentrification produces relatively little change in their realized exposure to it, leading to only modest changes in the estimates when market access is omitted. Omitting either neighborhood tenure or moving costs has little effect on the relationship between gentrification and experienced welfare, consistent with the limited mobility responses to gentrification documented in Section 3.2. Table A25 reports the full set of coefficient estimates.

Overall, the experienced-welfare analysis is consistent with both our descriptive evidence and the main welfare analysis. We find little relationship between gentrification and total experienced welfare, with this null relationship seemingly reflecting offsetting changes in the welfare-relevant characteristics of gentrifying neighborhoods.

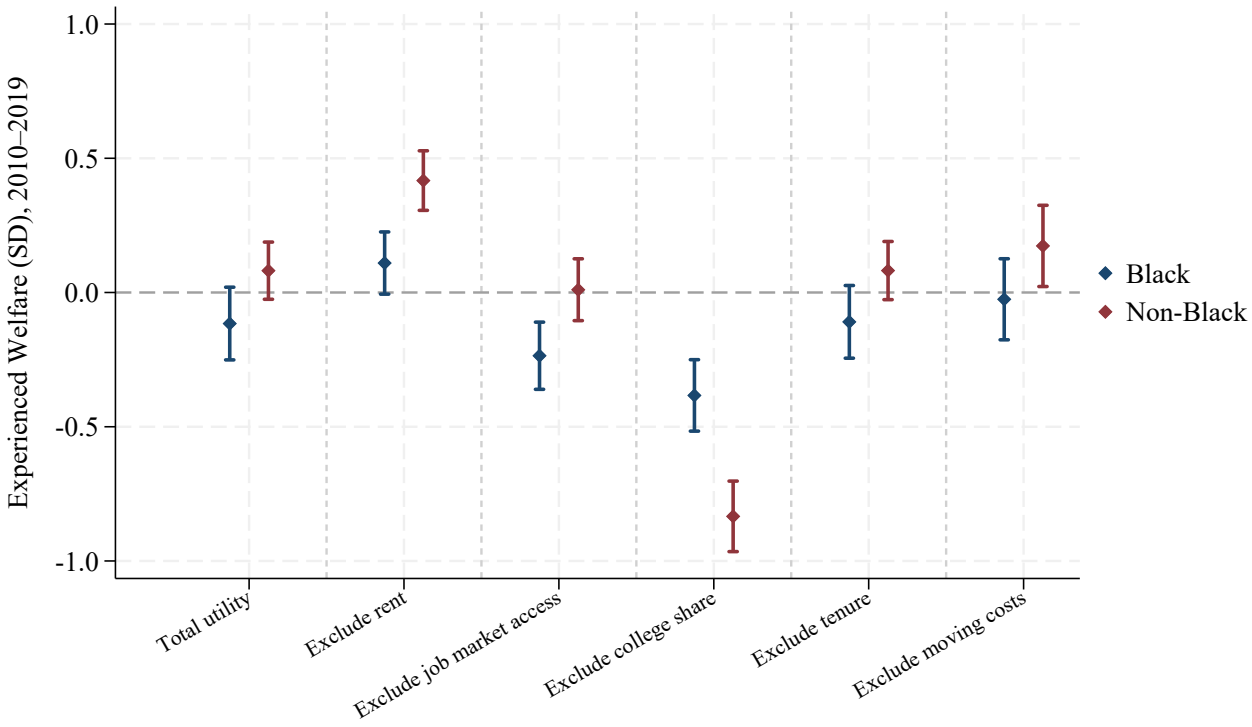
[93] One practical difference from the regressions in Section 3 is that we replace CBSA fixed effects with origin-CBSA-by-final-sample-year fixed effects. The final sample year records the last year a household remains in the analysis sample. Because we do not compute $\widehat{\bar{u}}^k_{n_{is}}\left(x_{is}, \bar{\omega}^{kc}_{s}\right)$ after households move across CBSAs, many households exit the sample before 2019. Interacting households' final sample year with their origin CBSA allows us to retain these households while absorbing both cross-CBSA differences in mean utility levels and CBSA-specific differences in sample-exit timing that may be correlated with gentrification. Another practical difference is that we limit the sample to households in origin tracts that experienced positive changes in their college-graduate shares over 2010–2019.

FIGURE A15. Experienced Welfare



Notes: Figure A15 presents the experienced-welfare measures discussed in the main text. Panel A plots residuals from type-specific regressions of experienced welfare on CBSA-by-final-sample-year fixed effects, the baseline controls listed in Table A16, and the remaining neighborhood characteristics entering households' 2010 flow utilities net of moving costs and the value of residential tenure. These additional controls include the natural logarithms of the 2010 tract college share and hedonic rent and the tract's estimated exogenous amenities in 2010. Panel B shows how experienced welfare varies with a tract's predicted log change in its college share over 2010–2019. The points plot conditional means estimated from regressions with the same fixed effects and controls used in Panel A. The sample comprises urban, low-income renter households, defined after applying restriction (h.i) in Table A1.

FIGURE A16. Experienced Welfare Decomposition



Notes: Figure A16 plots the estimated coefficients on our gentrification measure from regressions analogous to those underlying Panel B of Figure A15. The dependent variable in the baseline specification is experienced welfare. The remaining dependent variables are constructed identically but exclude the indicated component of experienced utility by setting its welfare coefficient to zero. The regressions otherwise use the same fixed effects and controls described in Figure A15. We display 95% confidence intervals based on robust standard errors clustered at the origin census-tract level (Abadie et al. 2023). The sample comprises urban, low-income renter households initially residing in gentrifying tracts, defined after applying restriction (h.iii) in Table A1.

TABLE A25. Gentrification and Experienced Welfare

	(1) Baseline		(2) No Rent Control		(3) Long-Term Renters	
	Non-Black	Black	Non-Black	Black	Non-Black	Black
Panel A: Baseline						
Coefficient estimate	0.081	-0.116	0.055	-0.205	0.125	-0.125
Standard error	(0.054)	(0.069)	(0.067)	(0.075)	(0.069)	(0.091)
Kleibergen-Paap F-stat	[98.21]	[445.2]	[37.41]	[224.0]	[143.5]	[515.4]
Baseline controls	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓
Panel B: Exclude Hedonic Rents						
Coefficient estimate	0.417	0.110	0.311	0.024	0.502	0.105
Standard error	(0.057)	(0.059)	(0.074)	(0.056)	(0.075)	(0.078)
Kleibergen-Paap F-stat	[98.21]	[445.2]	[37.41]	[224.0]	[143.5]	[515.4]
Baseline controls	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓
Panel C: Exclude Market Access						
Coefficient estimate	0.010	-0.236	0.012	-0.255	0.054	-0.222
Standard error	(0.059)	(0.064)	(0.071)	(0.070)	(0.073)	(0.084)
Kleibergen-Paap F-stat	[98.21]	[445.2]	[37.41]	[224.0]	[143.5]	[515.4]
Baseline controls	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓
Panel D: Exclude College Shares						
Coefficient estimate	-0.834	-0.383	-0.602	-0.431	-0.894	-0.433
Standard error	(0.067)	(0.068)	(0.069)	(0.083)	(0.077)	(0.089)
Kleibergen-Paap F-stat	[98.21]	[445.2]	[37.41]	[224.0]	[143.5]	[515.4]
Baseline controls	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓
Panel E: Exclude Tenure Length						
Coefficient estimate	0.082	-0.109	0.060	-0.199	0.122	-0.120
Standard error	(0.055)	(0.069)	(0.068)	(0.075)	(0.070)	(0.091)
Kleibergen-Paap F-stat	[98.21]	[445.2]	[37.41]	[224.0]	[143.5]	[515.4]
Baseline controls	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓
Panel F: Exclude Moving Costs						
Coefficient estimate	0.174	-0.025	0.202	-0.123	0.128	-0.050
Standard error	(0.077)	(0.077)	(0.095)	(0.086)	(0.088)	(0.097)
Kleibergen-Paap F-stat	[98.21]	[445.2]	[37.41]	[224.0]	[143.5]	[515.4]
Baseline controls	✓	✓	✓	✓	✓	✓
CBSA fixed effects	✓	✓	✓	✓	✓	✓
N	408,000	269,000	214,000	171,000	178,000	109,000

Notes: Table A25 reports the estimated coefficients on our gentrification measure from the specifications described in Figure A16. Panel A uses experienced welfare as the dependent variable. Panels B–F reconstruct experienced welfare after excluding the indicated component of experienced utility. Column (1) reports the baseline estimates plotted in Figure A16. Columns (2) and (3) repeat these specifications after restricting the sample to census tracts not covered by any rent control ordinance and to incumbent renters who had lived in their 2010 origin tract since at least 2005, respectively. Standard errors clustered at the origin census-tract level are reported in parentheses, and Kleibergen–Paap first-stage F-statistics are reported in brackets.